Nov 2026 attempt

AFM Last-Day Revision — Master Pack

84 pages · all 15 chapters · ordered by ICAI marks weightage · each sheet checked against the ICAI AFM Study Material (and Saransh where noted)

#	Chapter	Weight	Pages	Done
1	Ch 9 · Derivatives Analysis & Valuation	12.7%	2–8	☐
2	Ch 10 · Foreign Exchange Exposure & Risk Mgmt	11.6%	9–14	☐
3	Ch 6 · Portfolio Management	10.7%	15–20	☐
4	Ch 8 · Mutual Funds	10.5%	21–25	☐
5	Ch 5 · Security Valuation	9.6%	26–32	☐
6	Ch 14 · Mergers, Acquisitions & Restructuring	8.2%	33–39	☐
7	Ch 11 · International Financial Management	7.1%	40–45	☐
8	Ch 12 · Interest Rate Risk Management	5.9%	46–50	☐
9	Ch 2 · Risk Management	4.0%	51–53	☐
10	Ch 1 · Financial Policy & Corporate Strategy	3.4%	54–57	☐
11	Ch 15 · Startup Finance	3.4%	58–62	☐
12	Ch 3 · Advanced Capital Budgeting	3.4%	63–67	☐
13	Ch 4 · Security Analysis	3.4%	68–72	☐
14	Ch 7 · Securitization	3.4%	73–77	☐
15	Ch 13 · Business Valuation	2.8%	78–84	☐

How to read: **SM text = what to write.** Anything marked "app-written", "app-computed", "exam aid" or "not in SM" is a study aid. In the exam, write the SM version. Every chapter ends with an actual-link correlation table.

#1 · 12.7% of marks

Ch 9 — Derivatives Analysis & Valuation

Tested every attempt: strangle / straddle payoffs, index-futures beta hedge, Black-Scholes, real options (risk-neutral), futures + options combos. Theory: real vs financial options.

① Users & Forward vs Futures (SM)

Users: *corporations — hedge currency & inventory risk · individual investors — speculation, hedging, yield enhancement · institutional investors — hedging asset allocation, yield enhancement, arbitrage · dealers — hedging, position taking, exploiting inefficiencies, dealer spreads.*

Feature	Forward	Futures
Trading	Personal basis / telephone	Competitive arena
Size	Individually tailored	Standardised
Exchange	OTC	Organised exchange
Settlement	On agreed date	Daily via clearing house
Delivery date	As agreed; actual delivery	Cyclical; delivery rarely takes place
Transaction cost	Bid-ask spread	Brokerage on buy & sell
Marking to market	No	Yes — daily in margin account
Margins	Not required	Required
Credit risk	Borne by each party	Not a concern (clearing house)

Basis = Spot − Futures. *Normal (**contango**) market: futures > spot (full cost of carry) → basis negative. **Backwardation / inverted**: spot > futures (factors other than carry) → basis positive. Basis → 0 at expiry = **convergence**.*

② Futures pricing — cost of carry

Situation	Formula
Basic (discrete)	$F = S + \text{cost of carry} - \text{dividend} = S(1+r)^t - D$
Continuous compounding	$F = S \cdot e^{rt}$
Known dividend	$F = (S - \text{PV of dividend}) \cdot e^{rt}$
Index with dividend yield y	$F = S \cdot e^{(r-y)t}$
Commodity with storage cost	$F = (S + \text{PV storage}) \cdot e^{rt}$ (convenience yield reduces F)

Actual F > Theoretical F

*Futures overpriced → **cash-and-carry**: borrow, **buy spot, sell futures**; deliver at expiry.*

Actual F < Theoretical F

*Futures underpriced → **reverse cash-and-carry**: **short-sell spot, invest, buy futures**.*

*SM: Future price = Spot + carrying cost − returns (dividends etc.). RTP Jan 2026: S = 125, r 8%, D = 4 → F = 125 + 10 − 4 = **₹131**; contract (1,000 shares) ₹1,31,000. Price falls 6% → S = 117.50 → F = 122.90 → long loses ₹8,100.*

REPEAT R17

③ Margins & mark-to-market

- → **Initial margin = μ + 3σ** of daily changes in contract value (SM illustration: μ ₹10,000 + 3 × ₹2,000 = ₹16,000; maintenance 75% = ₹12,000)
- → **Maintenance margin** (usually lower); balance falls below it → **margin call** to restore to **initial** margin
- → Daily MTM: long gains when price rises; losses debited to margin account. Cumulative MTM = (final − entry price) × lot

④ Beta hedge — formulas

No. of contracts to fully hedge = (Portfolio value × β_p) ÷ (Futures price × lot size)

Change beta from β_1 to β_2: contracts = ($\beta_2 - \beta_1$) × Portfolio value ÷ (Futures price × lot) — negative = **sell**, positive = **buy**

Long stock + short stock: hedge each leg separately (sell futures for long leg, buy for short leg) or net the beta-weighted values

Example	Working
Portfolio ₹1 cr, β 1.2; Nifty futures 22,000; lot 75	1,00,00,000 × 1.2 ÷ (22,000 × 75) = 7.27 → **sell 7 contracts**
Reduce β 1.2 → 0.6 (same data)	(0.6 − 1.2) × 1 cr ÷ 16,50,000 = −3.64 → sell **4**
RTP Jan 2026: portfolio 9,94,450, 5 Nifty contracts sold at 8,767.07 × 25 for full hedge	Implied β = 5 × 8,767.07 × 25 ÷ 9,94,450 = **1.102**; 1.102 = 0.9 × 1.10 + 0.1 × β_{10} → β_{10} = **1.12**

⚠ TRAP

Use the **futures** price (not spot index) in the denominator. Lot size is whatever the question gives — don't assume. Profit on hedge = (sell price − buy-back price) × lot × contracts; overall P/L = cash position change + futures P/L.

⑤ Options vs single stock futures (SM 6.7)

Point	Options	Stock futures
Nature	Buyer has right, not obligation	Long must square off at / before expiry
Movement of market	Premium moves with **delta**; may move independently of stock	Follows the underlying much more closely
Price of investing	Premium = **sunk cost**; lost if not ITM at expiry	Initial margin + maintenance (MTM) margin

7 Basics

	Call (right to BUY at K)	Put (right to SELL at K)
In the money	$S > K$	$S < K$
At the money	$S = K$	$S = K$
Out of the money	$S < K$	$S > K$
Intrinsic value	$\max(S - K, 0)$	$\max(K - S, 0)$
Time value	Premium − intrinsic value (all premium is time value if OTM / ATM)	
Buyer's net pay-off	$\max(S_T - K, 0)$ − premium	$\max(K - S_T, 0)$ − premium
Writer's pay-off	Exact mirror image of buyer (zero-sum)	

⚠ TRAP

Exercise decision ignores premium (sunk) — exercise whenever ITM, even if net result is still a loss.

Premium & brokerage are per share × **lot**. European = exercise only on expiry; American = any time.

8 Strategies & break-evens

exam aid — not described in SM; PYQs test the pay-offs

Strategy	Position	Break-even / view
Long straddle	Buy call + buy put, **same K**	K ± total premium; expects big move either way
Long strangle	Buy OTM call (K_c) + OTM put (K_p), $K_c > K_p$	Upper = K_c + total premium; lower = K_p − total premium
Strip	1 call + 2 puts (same K)	Bearish volatility bet
Strap	2 calls + 1 put	Bullish volatility bet
Bull spread	Buy low-K call, sell high-K call	Limited gain / loss; moderately bullish
Bear spread	Buy high-K put, sell low-K put (or calls reversed)	Moderately bearish
Butterfly	Buy 1 low-K + 1 high-K call, sell 2 middle-K calls	Expects price to stay near middle K
Covered call	Long stock / long futures + **sell call**	Earns premium; caps upside (Jan 2026 Q3(a))
Protective put	Long stock + buy put	Floor on loss (insurance)

Long strangle: call 620 (premium 25) + put 480 (premium 15); total premium 40; BE 660 / 440				
Expiry price	700	550	400	660 / 440
Net P/L per share	+40	**−40** (max loss)	+40	0

⑨ Binomial — one step (three methods, same answer)

$S = 100$; up 120, down 80; $K = 100$; r_f 10% for the period.

Method	Working	Call
Risk-neutral	$p = (S(1+r) - S_d) \div (S_u - S_d) = (110 - 80)/40 = 0.75$; $C = (0.75 \times 20 + 0.25 \times 0) \div 1.10$	**13.64**
Delta / replicating	$\Delta = (C_u - C_d)/(S_u - S_d) = 20/40 = 0.5$; borrow $B = (\Delta S_d - C_d)/1.1 = 36.36$; $C = \Delta S - B$	**13.64**
Put (parity)	$P = C - S + K/(1 + r) = 13.64 - 100 + 90.91$	**4.55**

Two-step: $u = 1.1$, $d = 0.9$, r 5% per step → $p = 0.75$; terminal call values 21, 0, 0 → $C = 0.75^2 \times 21 \div 1.05^2$ = **10.71**. Work backwards node by node (needed for American puts: compare early exercise at each node).

⑩ Black-Scholes (European, non-dividend)

$$C = S \cdot N(d_1) - K \cdot e^{-rt} \cdot N(d_2)$$

$d_1 = [\ln(S/K) + (r + \sigma^2/2)t] \div \sigma\sqrt{t}$ · $d_2 = d_1 - \sigma\sqrt{t}$

$P = C - S + K \cdot e^{-rt}$ (put-call parity) or $P = K \cdot e^{-rt} N(-d_2) - S \cdot N(-d_1)$

$N(d_1)$ = hedge ratio / delta · **$N(d_2)$ = probability call finishes ITM** · $N(-d) = 1 - N(d)$ · interpolate N() from the table

May 2026 Q6(b): S 120, K 115, 50 days, r 8%, σ 30%, d_1 0.5375, e^{-rt} 0.9891 → $N(d_1)$ **0.7045** (hedge ratio ≈ 70%); $d_2 = 0.5375 - 0.30\sqrt{(50/365)}$ = **0.4264**; $N(d_2)$ **0.6651** (≈ 67%); $C = 84.54 - 75.65$ = **₹8.89**; $P = -120 + 8.89 + 113.75$ = **₹2.63** (ICAI). Verified by code: C 8.89, P 2.64 (rounding).

⑪ BSM assumptions, factors & Greeks (SM)

Assumptions (7): European options · no transaction costs · short-term interest rates known & constant · stocks pay no dividend · stock price follows a random walk · returns normally distributed · variance of return constant over option life.

Factor affecting value (SM)	Effect
(a) Price of underlying	Price ↑ → calls ↑, puts ↓
(b) Time till expiry	Option may become worthless at expiry — time value decays
(c) Volatility	Statistical / historical volatility (SV); higher volatility → higher value
(d) Interest rate	Higher rates → lower PV of exercise price (call ↑, put ↓)

Greek	SM definition
Delta	Degree to which option price moves for a small change in underlying price
Gamma	How fast delta changes — "delta of the delta"; keep small for delta hedging; adjusted only with options
Theta	Change in option price for a **one-day decrease** in time to expiry — time decay
Rho	Change in option price for a **one percentage point** change in risk-free rate
Vega	Sensitivity of option value to change in volatility; used for hedging

REPEAT R13

12 Real options — types (SM)

Option	SM meaning	Like
Growth	Strategic investments: define competitive position, gain knowledge about project profitability, give manufacturing flexibility	European call
Abandonment	Exercised if value from project's assets > PV of continuing the project	American put
Timing	Third choice besides accept / reject: **delay** the decision (when to invest)	American call

Valuation methods = same as financial options (binomial, risk-neutral, BSM). **Expanded NPV = static NPV + option value.**

13 Real vs financial options (SM table — Jan 2026, 4 marks)

Basis	Financial options	Real options
Underlying	Assets normally traded — shares, bonds, commodities	Projects — not traded
Pay-off	Specified in contract → fixed	Estimated from project cash flows → can vary
Exercise period	Short — maximum up to 1 year	Starts from end of 1st year; longer
Approach	Traded → **"Priced"**	Used for decisions → **"Valued"**

14 Risk-neutral real-option numbers (app-computed)

Case	Working
Abandonment: PV now 100; next year 140 or 70; salvage 90; r_f 10%	$p = (110 - 70)/(140 - 70) = 0.5714$; value = 0.4286 × 20 ÷ 1.1 = **₹7.79**
Option to wait: invest 100; PV now 105; next year 130 or 85; r_f 8%	$p = 0.6311$; value = 0.6311 × 30 ÷ 1.08 = **₹17.53** > NPV 5 → **wait**

15 Exotic options (SM 7.5 — 9 types)

Differ from plain vanilla in pay-off & time of exercise; more complex; mostly **OTC**.

Chooser	After a specified period buyer decides whether it is a call or a put
Compound	"Split-fee" / option on an option
Barrier	Activated only if underlying reaches a certain price in a set period
Asian	Pay-off on **average** price over a period
Look-back	Holder chooses most favourable strike from min / max price during life
Basket	Value depends on a portfolio (basket)
Bermuda	Compromise between European & American
Binary / digital	Pay-off on happening of a specific event
Spread	Pay-off on difference between prices of two underlyings

⑥ Index futures (SM 5.2)

- → Based on level of an index (Nifty 50, Sensex); used to **speculate** on the market or **hedge** a portfolio against decline
- → Not based on tangible goods → **all settlements in cash**; traded in number of contracts; value = index × multiplier (SM: 1 contract = 50 units × Nifty)
- → Margin is **continuous** — computed every business day; position squared off at closing price on expiry (SM: last Thursday of the month)
- → Purposes of futures trading (SM): **speculation** and **hedging**. Index futures hedge only **systematic** risk
- → SM TYK theory: reasons index futures became more popular than stock futures; short note on marking to market

⑯ Credit derivatives (SM 8)

- → **CDOs** — backed by pool of bonds, ABS, REITs, other CDOs (securitisation is backed by loans / receivables); covers CBOs & CLOs. **Types**: **Cash-flow CDO** (backed by cash-market debt; relies on collateral's cash to pay SPV securities) · **Synthetic CDO** (credit risk transferred **without** transferring assets; unfunded (CDS only) / fully funded (CLN) / partially funded) · **Arbitrage CDO** (issuer captures spread between collateral return and borrowing cost + management fee)
- → **Risks in CDOs**: default (hits equity / junior tranche first), interest-rate (basis), liquidity, prepayment, reinvestment, foreign-exchange
- → **CDS** = Credit (loan) + Default (non-payment) + Swap (exchange of risk): buyer pays periodic premium; seller pays on default. **Features**: non-standardised private (forward-type) contract; not exchange-traded; ISDA guidelines; can be bought from a third party; bondholder can protect against issuer insolvency; premium rises with risk. **Uses**: hedging, arbitrage, speculation. **Parties**: reference entity, buyer (investor), seller (writer). **Settlement**: physical (deliver bonds, get par) or cash (par − market value)

⑰ Commodity derivatives (SM 10)

- → **Necessary conditions (SM)**: commodity **durable & storable**; units **homogeneous**; **frequent price fluctuations with wide amplitude**; supply & demand **large**; supply flows naturally to market and there are **breakdowns in existing pattern of forward contracting**
- → **Investing**: complements investment in commodity-using companies; invest in a country's consumption & production; no dividends, only price returns. Market advantages: managers prefer derivatives to tangibles; less hassle (delivery); indirect real-asset exposure — inflation hedge
- → **Futures**: easiest & cheapest way; 3 categories — agricultural (soft), energy, metals. **Swaps**: **fixed-floating** and **commodity-for-interest**; valued as a strip of forwards; factors — hedging cost, institutional structure, liquidity, seasonality, futures bid / offer spread, brokerage, credit risk / capital / admin costs

18 Weather & electricity derivatives (SM 11–12)

→ **Weather**: underlying = weather measure (rainfall, temperature, humidity, wind); hedges **volume risk**, not price risk; first deal 1997 (Aquila Energy, OTC). **vs insurance**: insurance covers extreme low-probability events; weather derivatives cover normal-condition uncertainty too. Pricing issues: data reliability, weather forecasting, temperature modelling

→ **Electricity**: **forwards** (fixed quantity at forward price at maturity) · **futures** (standardised trading, settlement) · **swaps** (pay fixed vs floating price; = strip of forwards; locational basis swaps)

19 Derivative mishaps & 6 lessons (SM 13)

Orange County (1994)	Treasurer Robert Citron, no trading background; yield-curve play + ~$2.8 bn inverse floaters; leveraged ~$7.5 bn fund to ~$20 bn via repo; Fed raised rates 6 times in 1994 → loss **~$2 bn**; Citron jailed 1 yr, fined $1,00,000
Barings (1995)	Nick Leeson: arbitrage Nikkei 225 futures Singapore vs Osaka turned speculative long; also ran **back office**, hid losses in error account **'8888'**; loss $827 mn (SM also cites $1.4 bn); 233-year-old bank bankrupt; ING bought it for **£1**. Led to operational-risk capital in Basel II
P&G & Gibson Greetings (1994)	Bankers Trust sold complex **leveraged (ratio / power) swaps**; P&G lost $106 mn; Gibson $3 mn then $16.7 mn; out-of-court settlements; BT taken over by Deutsche Bank (1999)

Lessons: (a) don't buy a derivative you don't understand (value in-house) · (b) due diligence before making Treasury a profit centre (don't link treasurer's pay to profit) · (c) specify risk limits · (d) separate **front, middle & back** offices · (e) a hedger shouldn't become a speculator · (f) stress tests, scenario analysis (VaR + scenarios, 10–20 yrs data for black swans).

20 Correlation — how Ch 9 actually connects

Ch 9 idea	The actual link (app-written cross-reference; the Ch 9 side is SM)
Index-futures beta hedge	**Ch 6**: $\beta_p = \Sigma\, w_i\beta_i$ (weighted average of stock betas) goes straight into contracts = Value × $(\beta_2 - \beta_1)$ ÷ (futures price × lot). Hedge removes only **systematic** risk — unsystematic was already diversified away (Ch 6).
Cost of carry $F = S{\cdot}e^{rt}$	**Ch 10**: interest rate parity $F = S \times (1 + r_h) \div (1 + r_f)$ is the same idea — forward = spot + carry, else arbitrage. Actual F > fair F → buy spot, sell forward/futures (cash-and-carry), in both chapters.
Put-call parity: C + PV(K) = P + S	Used in ICAI answers to get P from C (May 2026: P = −S + C + PV(K)). **Ch 5**: warrant value (MP − E) × n is the **intrinsic value of a call**; a convertible = straight bond + call on shares.
Call / put payoffs	**Ch 10**: importer (payable) buys a **call** on the currency, exporter (receivable) buys a **put** — same max(S − K, 0) / max(K − S, 0) payoff, net of premium. **Ch 12**: interest-rate cap = series of calls on the rate (borrower), floor = series of puts (lender), collar = buy cap + sell floor.
Real options	**Ch 3**: plain NPV ignores managerial flexibility; option to expand / abandon / wait adds value → project worth NPV + option value. A negative-NPV project can still be accepted.
CDO tranches	**Ch 7**: same tranching as securitisation — senior, mezzanine, equity; losses hit the equity tranche first.
Mishaps & 6 lessons	**Ch 2**: risk limits, front/middle/back-office separation, VaR + stress tests = the risk-management framework. **FR (Ind AS 109)**: derivatives are at FVTPL unless designated in hedge accounting.

#2 · 11.6% of marks

Ch 10 — Forex Exposure & Risk Management

Tested every attempt: swap points / broken date, cross rates & arbitrage, receivable / payable hedges, early delivery & cancellation (FEDAI), exposure theory.

① SWIFT & payment gateways (SM)

- → ***SWIFT*** *(Society for Worldwide Interbank Financial Telecommunication) — **purely a messaging system**; assigns **3-letter currency codes** (USD, GBP, CNY, CAD, AUD, HKD, INR, JPY, NZD, SGD, SEK, CHF, EUR)*
- → ***Why important*** *(S): quick settlement (messaging in seconds) besides validation & documentation · catalyst bringing financial agencies to work together · security & reliability reduce **operational risk** · standardised transactions → efficiency & lower cost · full **backup & recovery***
- → ***Payment gateway*** *— virtual equivalent of physical cash transfer; authenticates & routes payment details securely. **Benefits**: 24×7×365; real-time card authorisation; rapid processing; multiple payment options; encryption & verification; real-time reports; refunds; no complex software for merchants; CA-authenticated secure servers; bulk data collection; multiple host interfaces; simple admin control; customer & merchant trust. **Challenges**: customer may lack account with supporting banks; limited banks; reliability, delivery, limited avenues, lack of trust*

② Accounts, positions, quotes (SM)

Term	SM meaning
Nostro	*Bank's FC account kept in a foreign country in that country's currency — "**our account with you**"*
Vostro	*Account maintained by a foreign bank with a local bank — "**your account with us**"*
Loro	*Domestic bank uses the account of a **third-party bank** that holds a Nostro account, to settle forex transactions*
Exchange position	*Total purchase & sale commitments of a bank, delivered or not*
Cash position	*Balance in Nostro account (credited for purchases, debited for sales). Everything affecting cash position affects exchange position — not vice versa*
American / European term	*American: US$ per unit of foreign currency · European: foreign currency per US$*
Direct / indirect quote	*Direct: home-currency price of 1 unit of foreign (₹48 / $) · Indirect: foreign-currency price of 1 unit of home ($0.0208 / ₹)*
Bid / offer / spread	*Bid = dealer buys; offer = dealer sells; spread = difference*
Cross rate	*Rate for a pair in which **neither** currency is the official currency of the country where quoted*
Pip	*Price Interest Point — smallest unit a quote can change*
Merchant rates	*Rates to customers (vs interbank rates). Post-liberalisation banks set own margins; FEDAI rule SA.8: merchant settlements **rounded to nearest whole rupee**; "card rate" for small walk-in deals*

3 Forward points & broken period (SM)

- → Forward rate is quoted as spot + **forward margin (points)**; user loads margin into spot. Points depend on interest rates & length; the currency with the **lower interest rate is always at a premium**
- → **Premium** (direct quote): **added** to both buying & selling rates · **Discount**: **deducted** from both. Points ascending (bid < ask) → add; descending → subtract
- → **Broken period**: forward points assumed to "grow" **uniformly** — interpolate proportionately

SM illustration — spot ₹66.2525 / 67.5945; 2-m points 70 / 90; 3-m 160 / 186; need 2 m 15 d (1 point = 0.0001)		
Points for 15 days = (3-m − 2-m) ÷ 2	45	48
Points for 2 m 15 d	115	138
Forward rate	**66.2640**	**67.6083**
Annual premium on **average** rate = 0.0115 ÷ 66.2583 × 12/2.5	**0.0833%**	0.0980%

Annualised premium = (F − S) ÷ S × 12 ÷ months × 100; when the question says "on average rate", divide by (S + F) ÷ 2 as above.

4 Cross rates & arbitrage

- → Chain like-to-like: A/C bid = A/B bid × B/C bid; ask = ask × ask. Common base: divide bid by ask
- → SM: ₹/$ 62.22 and ¥/$ 102.34 → ₹/¥ = 62.22 ÷ 102.34 = **0.6080**
- → Arbitrage test: buy via one route at ask, sell via the other at bid; gain only if buy-ask < sell-bid

5 Parity theories (SM)

Theory	Formula	SM summary
IRP	$F \div S = (1 + r_D) \div (1 + r_F)$	Forward premium / discount = interest-rate differential; when IRP holds, **covered interest arbitrage** isn't feasible
PPP	$E(S_1) = S_0 \times (1 + I_d) \div (1 + I_f)$	**Absolute** = law of one price; **relative** allows for imperfections (transport, tariffs, quotas)
IFE	Nominal rate = real return + expected inflation	High-interest currencies **depreciate** (high rates reflect expected inflation; also default risk)

SM comparison	Key variables	Summary
IRP	Forward premium vs interest differential	Forward contains premium / discount set by interest differential
PPP	% change in spot vs inflation differential	Spot changes with inflation differential
IFE	% change in spot vs interest differential	Spot changes with interest differential

App-computed checks: IRP — S 83.50, $r_{₹}$ 7%, $r_{\$}$ 5%, 6 m → F = 83.50 × 1.035 ÷ 1.025 = **84.3146**. PPP — 83.50, inflation 6% vs 3% → **85.932**.

6 Forecasting & determination (SM)

→ *Needed for decisions on hedging, short-term financing & investment, capital budgeting, earnings assessment, long-term financing*

→ ***Techniques (4)**: **Technical** (historical data, time-series) · **Fundamental** (economic relationships — subjective, regression, sensitivity) · **Market-based** (current spot / forward rates reflect expectations) · **Mixed** (weighted average of forecasts)*

→ ***Determinants**: major — **inflation** & **interest** rates; others — deficit / surplus on capital / current account, central bank intervention, expectations (**band-wagon** effect), trade barriers, government controls*

7 Forex market & exposures (SM)

→ *Market permits transfer of purchasing power between currencies. **Participants**: non-bank entities, banks, speculators, arbitrageurs, governments*

Exposure	SM meaning
Transaction	*Effect of rate change on obligations existing **before** the change but settled **after** — cash flows from existing contracts*
Translation	*Need to translate foreign subsidiaries' FS into one reporting currency for consolidation (SM example: FC loan retranslated at year-end)*
Economic	*Extent the **economic value** of the firm can decline due to rate changes — alters input costs & output prices → competitive position*

Economic exposure (SM table)	Local revaluation	Devaluation
*Local sales vs foreign competition; exports (local or foreign currency); interest from foreign investments — **inflows***	*Decrease*	*Increase*
*Imports in foreign currency; interest on foreign debt — **outflows***	*Decrease*	*Increase*
Imports denominated in local currency	*Remain the same*	

✍ **Write in exam**

***Transaction vs translation (May 2026, 2 marks):** transaction exposure affects actual cash flows on settlement of existing contracts, while translation exposure only restates FC items on consolidation without a cash flow — hence transaction exposure is more critical.*

8 Internal hedging techniques (SM — 6)

Invoicing in domestic currency	*Seller invoices in own / cost currency; buyer prefers own or stable currency*
Asset & liability management	***Aggressive**: raise inflows in strong / outflows in weak currencies. **Defensive**: match inflows & outflows by currency*
Leading & lagging	*Leading = pay before due date; lagging = delay after due date*
Netting	*Group companies settle inter-affiliate balances for the net amount*
Matching	*Match FC inflows with FC outflows in amount & timing*
Price variation	*Raise selling prices to counter adverse rate change*

9 Money market hedge

SM: agreement to exchange a certain amount of one currency for a fixed amount of another at a particular date. ***Advantages***: fixes future rate (no downside); flexible amount; usable where forwards aren't available. ***Disadvantages***: more complicated than a forward; no benefit from favourable moves.

Receivable	Payable
1. Borrow FC = receivable ÷ (1 + FC borrowing rate × t)	1. Deposit FC = payable ÷ (1 + FC deposit rate × t)
2. Convert at ***spot bid***	2. Buy FC at ***spot ask*** (borrow home currency if needed)
3. Invest home currency at deposit rate	3. Cost = home borrowing × (1 + rate × t)
4. Receivable repays FC loan; compare with forward	4. FC deposit pays supplier; compare with forward

App-computed: receivable USD 1,00,000 in 6 m, USD borrowing 5% p.a., spot bid 83.40, ₹ deposit 7% → borrow USD 97,560.98 → ₹81,36,585 → **₹84,21,366** at 6 m.

10 Forward vs futures (SM currency table)

Feature	Forward	Futures
Amount	Flexible	Standard
Maturity	Any agreed business date	Standard date
Furthest maturity	Open	12 months forward
Currencies	All	Majors
Cross rates	One contract	Usually two contracts
Market-place	Global network	Futures exchanges
Price fluctuation	No daily limit	Daily limit by exchange
Risk	Counterparty	Minimal (margins)
Honouring	Delivery	Mostly reverse transaction
Cash flow	None till maturity	Initial + variation margin
Trading hours	24 hours	4–8 hour sessions

11 Options vs futures (SM)

- → Only the ***writer*** is obliged (options) vs both parties (futures) · premium paid by buyer vs no premium · option buyer's loss restricted, gain unlimited vs unlimited both ways for futures · American option exercisable any time vs futures honoured on the specified date
- → ***Call*** = right to buy FC at a fixed price on / up to a date (importer); ***put*** = right to sell (exporter)

REPEAT R14 · MCQ FAVOURITE

12 Delivery · Cancellation · Extension (SM)

Event	SM treatment
Delivery on due date	At originally agreed rate, irrespective of spot
Early delivery	Bank may accept if customer bears the bank's loss: fixed charges + **swap difference** (loss / gain from offsetting bank's cover) + **interest on outlay of funds** at not less than **prime lending rate** (if inflow, bank may pass interest at term-deposit rate)
Late delivery	Current rate on delivery date — after applying automatic-cancellation provisions
Cancellation on due date	Flat charges + difference between contracted rate and cancellation rate (**reverse** of original) charged / paid
Early cancellation	At the **opposite** rate of the original contract for the date that matches the original maturity (forward rate)
Late cancellation	Automatic-cancellation provisions apply first
Extension on due date	Cancel at spot (as above) and rebook at forward rate for the new period
Extension before due date	Cancel at relevant forward rate and rebook at current forward rate
Automatic cancellation	**FEDAI Rule 6**: overdue contract without instructions on / before due date is automatically cancelled **within 3 working days after maturity**; customer pays the exchange difference but is **not entitled to any profit**

SM Illustration 4 (early delivery, exporter): contract sell US$1,00,000 at 65.40 for 31 Dec; delivered 30 Nov. Bank sells spot 65.22, buys 1-m forward 65.27 + 0.15 = 65.42 → swap loss 0.20 × 1,00,000 = **₹20,000**; interest on outlay 0.18 × 1,00,000 × 18% × 31/365 = **₹275** → net inflow = 65,40,000 − 20,275 = **₹65,19,725**.

⚠ TRAP

Cancellation = the bank's **reverse action**: customer's purchase contract → bank buys back (TT buying); sale contract → bank sells (TT selling). On automatic cancellation the customer bears losses but never gets gains.

13 NDF & rollover (SM)

- **Non-deliverable forward**: cash-settled, short-term forward on a thinly traded / non-convertible currency; P/L = (agreed rate − spot at settlement) × notional
- **Rollover** = spot cancellation + new contract for a later date. **Reasons**: non-receipt of FC from client (exporter); shortage of local currency (importer); non-agreement of payment with clients; non-availability of longer-period forwards

14 Swaps (SM)

- **Currency swap**: spot exchange of principal → continuing exchange of interest → re-exchange of principal on maturity. **Benefits**: considerable cost savings; access currencies that would otherwise command a high premium; diversification of borrowings; treasurers can hedge currency risk
- **Commodity swap**: series of futures-type settlements on a notional amount at predetermined prices / indices — mainly to hedge **oil price** risk
- **Equity swap**: total return on equity / index (dividend + capital) exchanged for fixed or floating interest

15 Popular derivative products (SM)

Forward importer	Counterparty **buys** notional USD against INR at forward rate
Forward exporter	Counterparty **sells** notional USD against INR at forward rate
FX call option	Counterparty (generally importer) has **right to buy** USD at strike
FX put option	Counterparty (generally exporter) has **right to sell** USD at strike
IRS USD floating to fixed	One party pays fixed, other floating (reset periodically) in the same currency on a notional (constant / amortising / accreting)

16 Strategies for exposure management (SM – 4)

Low risk : low reward	**Automatic hedging** in forward market as soon as exposure arises
Low risk : reasonable reward	**Selective hedging** — hedge when forward rates attractive, keep open otherwise
High risk : low reward	Leave all exposures **unhedged** — perhaps the worst
High risk : high reward	**Active trading** — continuous cancellations & re-bookings of forwards

Choice depends on attitude to risk, financial strength, nature of business, vulnerability to adverse movements — no single strategy suits all.

17 Exam method

→ Identify home currency, receivable / payable; exporter uses **bid**, importer **ask**; compare forward / MMH / futures / options / unhedged and **recommend**

18 Correlation — how Ch 10 actually connects

Ch 10 idea	The actual link (app-written cross-reference; the Ch 10 side is SM)
Interest rate parity	**Ch 9**: same no-arbitrage logic as cost of carry. Forward premium/discount % ≈ interest differential; if not → covered interest arbitrage (borrow cheap currency, convert spot, invest, cover forward).
Money market hedge	**Ch 5**: uses money-market borrowing / deposit rates for the exposure period — simple interest, days ÷ 360 or 365 exactly as the question states.
PPP / forecasting	**Ch 11**: foreign project cash flows are converted at forecast rates (PPP / IRP); or discount in foreign currency at foreign rate and convert at spot — same answer only if parity holds.
Currency options / futures	**Ch 9**: identical payoff and margin / MTM mechanics; the 11-row forward vs futures table applies to both chapters.
Currency swaps	**Ch 12**: same comparative-advantage logic as interest rate swaps — each party borrows where it is relatively cheaper and swaps.
Translation of exposure	**FR Ind AS 21**: monetary items at closing rate, difference to P&L; **Ind AS 109**: forward contract is a derivative at fair value.

#3 · 10.7% of marks

Ch 6 — Portfolio Management

Tested every attempt: single-index risk split, Sharpe optimal portfolio & cut-off, CAPM / SML, APT, performance ratios, alternative investments (SGB).

1 Objectives & phases (SM)

Objectives (7)

Security / safety of principal · stability of income · capital growth · marketability · liquidity · diversification · **favourable tax status**

Phases (5)

Security analysis (risk-return of individual securities & correlation) → ***portfolio analysis*** (combine into feasible portfolios) → ***selection*** (efficient → optimal) → ***revision*** (monitor so it stays optimal) → ***evaluation*** (return & risk over the period)

Efficient portfolio: highest return among feasible portfolios with same / lower risk, or lowest risk with same / higher return.

2 Theories (SM)

- → ***Traditional approach***: concerned with the investor — study of age, health, responsibilities, needs for income / capital / liquidity, attitude to risk, tax status; objectives; investment strategy (fixed interest vs equity, high payout vs growth, tax, transaction cost vs switching gains, liquidity); diversification; selection of individual investments (intrinsic value)
- → ***Modern approach (Markowitz)***: risk of an individual asset hardly matters — what matters is its ***contribution to overall risk***; diversify with low / negatively correlated securities; higher return needs higher risk

3 Types of risk (SM)

	Systematic — external, macro, uncontrollable	Unsystematic — internal, micro, controllable
Types	***Interest rate*** (prices move inversely to rates) · ***Purchasing power / inflation*** (worst for bonds & fixed income) · ***Market*** risk	***Business*** risk (variability of operating profit) · ***Financial*** risk (capital structure / leverage, D/E)
Measure	Beta — by ***correlation method*** ($\beta = r_{im}\sigma_i/\sigma_m$) or ***regression method*** ($Y = \alpha + \beta X$)	Residual variance

Diversion of risk: as number of securities ↑ total risk ↓; ultimately only systematic risk remains.

4 Return & risk formulas

$E(R) = \Sigma\, p_i X_i$ · $\sigma^2 = \Sigma\, p_i(X_i - \bar{X})^2$ · $Cov_{AB} = r_{AB}\sigma_A\sigma_B$ · $r_{AB} = Cov_{AB} \div \sigma_A\sigma_B$

Two securities: $\sigma_p^2 = x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2x_1x_2r_{12}\sigma_1\sigma_2$ · ***n securities***: $\sigma_p^2 = \Sigma\Sigma\, x_ix_j\sigma_{ij}$ (variance-covariance matrix)

SM cases: **r = +1** → $\sigma_p = x_1\sigma_1 + x_2\sigma_2$ (no diversification) · **r = −1** → $\sigma_p = |x_1\sigma_1 - x_2\sigma_2|$ (zero-risk mix possible) · **r = 0** → $\sigma_p^2 = x_1^2\sigma_1^2 + x_2^2\sigma_2^2$

App-computed: w 60 / 40, σ 20% / 30%, ρ 0.3 → σ_p = 19.35%; min-variance $w_A = (\sigma_B^2 - Cov) \div (\sigma_A^2 + \sigma_B^2 - 2Cov)$ = 76.6%.

5 Markowitz model (SM)

→ **Assumptions (SM 5.1)**: (i) return adequately summarises outcome · (ii) investors visualise a probability distribution of returns · (iii) risk estimates proportional to variance · (iv) decisions on two criteria — expected return & variance · (v) rational & risk-averse (prefer less risk for equal return) · (vi) yield in different scenarios used as return measure

→ **Efficient frontier** (Mean-Variance Dominance): a portfolio is **not efficient** if another has (a) higher return & lower σ, (b) higher return & same σ, or (c) same return & lower σ. Investor's choice on the frontier depends on risk-return preference

6 Capital market theory — CML

→ Tangent from R_f to market portfolio B → **CML** (R_f B Z). R_f–B = **lending portfolios** (part in risk-free); B–Z = **borrowing portfolios**

→ Slope = $(R_M - R_f) \div \sigma_M$ = reward per unit of risk · $\mathbf{E(R) = R_f + [(R_M - R_f) \div \sigma_M] \times \sigma_P}$

→ CML uses total risk σ (efficient portfolios); SML (CAPM) uses β (all securities)

7 Single index (Sharpe) model

Co-movement between stocks is due to the market index: $\mathbf{R_i = \alpha_i + \beta_i R_m + \in_i}$

$\sigma_i^2 = \beta_i^2\sigma_m^2 + \sigma_{\in i}^2$ · systematic risk = β_i^2 × variance of index · unsystematic = total − systematic · $\mathbf{Cov_{ij} = \beta_i\beta_j\sigma_m^2}$

Portfolio: $\sigma_p^2 = (\Sigma x_i\beta_i)^2\sigma_m^2 + \Sigma x_i^2\sigma_{\in i}^2$ · $R_p = \Sigma x_i(\alpha_i + \beta_i R_m)$ · α_p & β_p = weighted averages

App-computed: β 1.2, σ_m 15%, σ 25% → systematic 0.0324, unsystematic 0.0301. **MTP May 2026** (reverse): σ_p 25.03 → variance 626.50 = $1.73X^2 + 67$ → σ_m = **17.98**.

> ⚠ **TRAP**
> Unsystematic portfolio risk uses **weights squared** ($\Sigma x^2\sigma_{\in}^2$). Keep variance units consistent (15% → 225 or 0.0225).

8 APT (SM)

→ Alternative to CAPM; uses risk premia of several macro factors: $\mathbf{E(R_i) = R_f + \lambda_1\beta_1 + \lambda_2\beta_2 + \ldots + \lambda_n\beta_n}$

→ No surprises → actual = expected. With **unanticipated changes**: $E(R) = R_f + \beta_1(EV_1 - AV_1) + \ldots$ (surprise factor = expected − actual value of factor)

App-computed: R_f 6%, premia 4% & 2%, sensitivities 1.2 & 0.8 → **12.4%**. May 2026: target sensitivities via long / short amounts — two simultaneous equations.

8-MARK FAVOURITE

⑨ Steps (SM)

1. Find "**excess return to beta**" $(R_i - R_f) \div \beta_i$ — i.e. Treynor's ratio — for each stock
2. Rank highest to lowest
3. Compute C_i in ranked order: $C_i = \sigma_m^2 \Sigma[(R_i - R_f)\beta_i/\sigma_{ei}^2] \div [1 + \sigma_m^2 \Sigma(\beta_i^2/\sigma_{ei}^2)]$
4. C^* = highest C_i; select stocks whose excess-return-to-beta is above C^*, reject the rest
5. Weights: $Z_i = (\beta_i/\sigma_{ei}^2) \times [(R_i - R_f)/\beta_i - C^*]$; $X_i = Z_i \div \Sigma Z_i$

App-computed example: R_f 5%, σ_m^2 10.

Sec	R_i	β	σ_e^2	$(R-R_f)/\beta$	Rank	C_i
A	15	1.0	30	10.0	1	2.500
B	17	1.5	40	8.0	2	4.132
D	11	0.8	10	7.5	3	4.982
C	12	1.0	20	7.0	4	**5.314 = C^***
E	8	1.2	35	2.5	5	4.978 (falls → stop)

Sec	Z_i	Weight
A	0.1562	30.27%
B	0.1007	19.52%
D	0.1749	33.88%
C	0.0843	16.33%

⚠ TRAP

Ranking uses **excess return ÷ β**, not ÷ σ. If the question **gives** C^*, go straight to Z_i (May 2026 Q4(a)).

FORMULA CORE

10 CAPM (SM)

- → **$R_j = R_f + \beta(R_m - R_f)$**; risk premium of a stock = β × market risk premium. Graph = **SML** (required return vs β); market β = 1
- → SM: only **one risk-free rate**; if two are given, aggressive = higher, conservative = lower, moderate = average (Illus. 5)
- → **Decisions (SM 8.2)**: CAPM required < actual / expected return → **buy** (under-valued) · required > actual → **sell** (over-valued) · equal → **hold**. Price basis: actual price < CAPM price → under-valued
- → **Advantages**: (i) risk-adjusted return — usable as risk-adjusted discount rate; (ii) cost of equity for a **no-dividend company**
- → **Limitations**: (a) reliability of beta; (b) ignores other (unsystematic) risks; (c) information available — R_f and market return hard to estimate
- → Back-solving (Jan 2026 MCQ): two portfolios' returns & betas → solve R_f and R_m

App-computed: 6 + 1.3 × (14 − 6) = **16.4%**. SM Illus. 2: β 1.40, R_m 14, R_f 10 → 15.6%; premium +2.5 points → 19.1%.

11 Portfolio evaluation (SM)

Measure	Formula	SM meaning
Sharpe ratio	$(R_i - R_f) \div \sigma_i$	Risk premium per unit of **total** risk
Treynor ratio	$(R_i - R_f) \div \beta_i$	Risk premium per unit of **systematic** risk
Jensen's alpha	$R_p - [R_f + \beta(R_m - R_f)]$	Actual return minus CAPM-expected return

App-computed: R_p 15, σ 18, β 1.1, R_f 6, R_m 13 → Sharpe 0.50; Treynor 8.18; Jensen +1.30%.

12 Mixed portfolios (RTP Nov 2026)

Scenario returns × probabilities → E(R), σ, covariance → portfolio variance. Risk-averse → lowest σ for acceptable return; risk-seeker → highest return.

13 Portfolio strategy (SM)

→ **Active (APS) — 4 principles**: **market timing** (business cycle, moving average, advance-decline, econometric models) · **sector rotation** · **security selection** · **use of specialised investment concept**

→ **Passive — 2 guidelines**: (a) well-diversified portfolio at a predetermined risk level; (b) hold relatively unchanged unless inadequately diversified or inconsistent with preferences. Passively managed fund = **index fund** (doesn't remove market risk)

14 Selection of securities (SM)

Bonds — 4 factors

Yield to maturity · **tax shield** · **liquidity** · **risk of default** (credit rating / D/E, interest cover, earning power, industry)

Equity — 3 approaches

Technical (price & volume) · **fundamental** (intrinsic value vs price → buy / hold / sell) · **random selection** (efficient market, fairly priced)

15 Revision & rebalancing (SM)

	Buy & hold ("do nothing")	Constant mix	CPPI
Mechanism	No rebalancing; linear exposure to stocks	Rebalance to fixed % — buy stocks as they fall, sell as they rise	Floor in non-fluctuating assets; cushion in shares
Pay-off	Straight	**Concave**	**Convex**
Down / up market	Definitely poor in down	Not much in down, relatively poor in up	Good in down, well in up
Flat market	Between the other two	**Does well**	**Performs poorly**

16 Asset allocation strategies (SM – 4)

Integrated	Examines market conditions + investor objectives & constraints; allocation incorporating market forecast
Strategic	Optimal mix from historical returns, risk, covariances; periodically restored to target
Tactical	Risk tolerance constant; allocation changed on expected market conditions
Insured	Risk exposure adjusted with wealth — more value, more ability to take risk

17 Fixed income portfolio (SM)

- → **Process (5)**: set objective → investment policy guidelines → strategy (active / passive) → select securities → evaluate vs benchmark. **Return measures**: arithmetic average, time-weighted, rupee-weighted, annualised
- → **Passive**: **buy & hold** (may set duration = benchmark) · **indexation** (replicate bond index) · **immunization** (hybrid; popular with pension funds) · **matching cash flows** (zero-coupon bonds to meet promised payments)
- → **Active**: (1) **forecasting returns & interest rates** — **bullet** (all in one maturity), **barbell** (equal short & long), **ladder** (equal across maturities) · (2) **bond swaps** — **pure yield pickup**, **substitution**, **inter-market (international) spread**, **tax swap** · (3) **interest rate swap**

18 Alternative investments (SM)

- → **Features (6)**: high fees · limited historical data · illiquidity · less transparency · extensive research required · leveraged buying. **Types**: real estate, gold, private equity, REITs, hedge funds, ETFs, MFs, commodities, distressed securities
- → **Real estate** — valuation complex (inefficient market, illiquidity, only approximate comparison, high transaction cost…). **4 approaches**: sales comparison (like P/E), income (perpetual net income ÷ required return), cost (replacement cost + land value, adjusted for location), discounted after-tax cash flow (NPV-like)
- → **Gold forms**: jewellery (making charges) · **gold bars / coins** (storage cost) · **SGBs** — government securities in grams of gold, issued by **RBI on behalf of GoI**, paid & redeemed in cash, quantity of gold protected (market price at redemption), no storage risk / cost, **periodical interest**, held with RBI or in demat · **Gold ETFs** (price based on gold; full transparency; low expenses) · **E-gold** (NSEL, 2010; 1 unit = 1 gram in demat)
- → **Distressed securities**: securities of firms in / near bankruptcy, bought cheap to revive; arbitrage by long debt + short equity. **Risks**: liquidity, event, market, **human** (judge's decision)

SGB scheme specifics used in May 2026 MCQ (2.5% p.a. interest, 8-year tenor, redemption gain exempt for individuals) are **not stated in the SM** — use only the data given in the question.

19 Correlation — how Ch 6 actually connects

Ch 6 idea	The actual link (app-written cross-reference; the Ch 6 side is SM)
CAPM K_e	**Ch 5**: the K_e found here is the discount rate in DDM / FCFE; a stock's equity risk premium = $\beta \times (R_m - R_f)$, not $R_m - R_f$.
$\beta = r_{im}\sigma_i/\sigma_m = \text{Cov} \div \sigma_m^2$	**Ch 9**: β_p (weighted average) decides the number of index futures to sell / buy to change portfolio beta.
Sharpe, Treynor, Jensen	**Ch 8**: same formulas rank mutual funds. Sharpe uses σ (total risk — fund is the whole investment); Treynor uses β (well-diversified); Jensen $\alpha = R_p - [R_f + \beta_p(R_m - R_f)]$.
Fixed-income strategies	**Ch 5**: immunization = duration equal to horizon; bullet / barbell / ladder are duration choices; IRS (Ch 12) changes the duration of the portfolio.
Active vs passive	**Ch 4**: if markets are efficient, passive (index) wins after costs; active needs mispricing. **Ch 8**: index funds / ETFs are the passive vehicle.
Portfolio σ	**Ch 2**: σ_p feeds VaR; lower correlation → lower σ_p → lower VaR.

#4 · 10.5% of marks

Ch 8 — Mutual Funds

Tested every attempt: NAV working, holding-period return (payout vs reinvest), dividend vs bonus plans, back-solving issue price, closed-end premium / discount.

1 Set-up & structure

- → *MF = **trust** pooling savings via **units**; invests in capital & money market; income & gains shared in proportion to units*
- → *Four constituents: **Sponsor** → **Trust (trustees)** → **AMC** → **Custodian**, all SEBI-registered. Trust deed registered under the Registration Act, 1908*
- → *Sponsor: **5 years** in financial services, holds **≥ 40%** of AMC net worth. AMC net worth **₹100 crore** (SM). Trustees ⅔ **independent**; AMC board **50% independent**. AMC removed by majority of trustees or **75%** of unit holders*
- → *Custodian: SEBI-registered, **independent of sponsor & AMC** (DP for demat)*

2 Classification

Basis	Types
Structure	***Open-ended** (buy / redeem any time with fund; no maturity) · **Close-ended** (NFO then exchange; exit on exchange / maturity / repurchase window) · **Interval** (need not list; in-built redemption window; can issue fresh units at NAV in the interval; no fixed maturity)*
Asset	*Equity (**growth** — long-term appreciation; **aggressive** — IPOs, speculative; **income** — high-dividend stocks) · Debt / bond (risks: **I-C-P-R** — interest rate, credit, prepayment, reinvestment) · Money market / liquid · Gilt · Hybrid / balanced*
Special	***Index** · **International** (raises in India, invests abroad) vs **Offshore** (outside India, raises globally to invest in India) · **Capital protection oriented** (protection from portfolio structure, not guarantee; close-ended, listed, structure rated) · **Quant** (rule-based models; no bias, low cost; relies on history) · **Fund of funds** · **ETF** · **Arbitrage** (spot vs futures) · **Sector** vs **thematic***
Ownership	***Public sector** (sponsored by a public-sector company / bank) · **Private sector** · **Foreign** (foreign company operating from India, raises and invests in India)*
Diversified equity (SM 3.2)	***Flexicap** · **Multicap** · **Contra** (out-of-favour / undervalued companies; blends growth + value; needs aggressive risk appetite) · **Index** · **Dividend yield** (yield above the index's; less volatile than growth stocks)*
Others (SM 3)	***Thematic** — broader than sector (multi-sector, international, commodity); market may take long to recognise the theme · **Arbitrage** — open-ended, low-volatility, cash equities + derivatives + debt; spot vs futures differential · **Cash fund** — open-ended liquid, very short-dated debt & money market, growth & dividend options*
Schemes	***Flexicap** (any proportion) vs **Multicap** (≥ 25% each large / mid / small) · **FMP** (no interest-rate risk, still credit risk; close-ended, illiquid) · **ELSS** (80C, 3-yr lock-in; SIP preferred)*
Options	***Growth** (nothing paid) · **Dividend payout** (NAV falls by dividend) · **Dividend reinvestment** (extra units at ex-dividend NAV)*
Plans	***Direct plan** (from 1 Jan 2013) — no distribution expense / trail → lower TER, **higher NAV** than regular*

⚠ TRAP

*Interval ≠ close-ended. CPOF ≠ capital **guaranteed**. FMP is free of **interest-rate** risk, **not** credit risk.*

3 ETF vs index fund; hedge fund vs MF

ETF	Listed; trades all day at real-time prices close to NAV (open-ended fund deals once a day at NAV); **authorised participants** act as market makers; low cost, tax-efficient, no paperwork. Types: index, commodity (gold), bond, currency
Hedge fund vs MF	Lightly regulated private vehicle for selected clients; little public disclosure; **performance fee** (losses borne by investors); uses **leverage, long / short, derivatives**

High-water mark fee (SM: ANP Plan): ₹20 cr; 0.10% fixed + 2% above ₹21 cr. Return 29% → value 25.8 cr → fee ₹2 L + 2% × 4.8 cr = **₹11.60 L**; return 4.5% → 20.9 cr < 21 → **₹2 L only**. Trap: fee on excess over HWM, not on the year's gain.

4 Advantages (12) & drawbacks (4)

Advantages

Professional management · diversification · convenient administration · higher returns · **low cost** (max **2.50%** for equity funds; excess borne by AMC) · liquidity · transparency · other benefits (SIP / SWP / switch) · highly regulated · economies of scale · flexibility · convenience

Drawbacks

1. No guarantee of return (under-performance, entry / exit timing, crashes) · 2. Diversification caps upside · 3. Picking the right fund is hard; past performance no guide · 4. Costs; AMC fees not linked to performance

5 Key terms

Term	Meaning / formula
NAV	Net assets ÷ units outstanding. Net assets = MV of investments + receivables + accrued income + other assets − accrued expenses − payables − other liabilities
Loads	Public offer price = NAV ÷ (1 − entry load) · Redemption = NAV ÷ (1 + exit load) [or NAV × (1 − exit load) where the question says]. Exit load goes back to the **scheme**. No entry load now
Required MF return	$r_2 = r_1 \div (1 - \text{initial expenses}) + \text{recurring expenses}$ (10%, 6%, 2% → **12.64%**)
Expense ratio (TER)	Mgmt & advisory fees, travel, audit, custodian, registrar — **excludes brokerage**; charged daily (1/365) before NAV
Trail commission	0.30–0.75% p.a. on daily average AUM, paid monthly, while money stays invested
Side pocketing	Stressed / downgraded assets moved to a segregated portfolio; main NAV reflects liquid assets; protects continuing holders (IL&FS). Recovery paid only to holders on segregation date
Tracking error	$\sqrt{[\Sigma(d - \bar{d})^2 \div (n - 1)]}$, d = fund − benchmark return. Causes: transaction cost, AMC fees, expenses, cash holdings, sampling. Higher TE = higher risk even if outperforming
Dividend equalisation	New investor pays / exiting investor gets undistributed income per unit: issue = NAV + entry load + income p.u.; repurchase = NAV − exit load + income p.u.

6 Return formulas

HPR = $(NAV_1 - NAV_0 + \text{income distributed} + \text{capital-gain distributed}) \div NAV_0$

Reinvestment: extra units = distribution ÷ reinvestment NAV; return = (total units × closing NAV − cost) ÷ cost

Effective yield p.a. = [(units × closing NAV − investment) + dividends] ÷ investment × **365 ÷ days**

Annualise monthly return × 12 (simple) unless told to compound; **real return** = (1 + nominal) ÷ (1 + inflation) − 1

Pattern	Method
Payout vs reinvest (Sep 2025 Q3(b))	Compute both; reinvest at the given (average) NAV; pick higher return
Entry load + appreciation + distribution reinvested + exit load + tax + inflation (Sep 2025 Q5(b))	Units = amount ÷ POP; end value = units × NAV × (1 − exit load); tax on gain; real return
Dividend % declared	Dividend = units × **face value ₹10** × % (not × NAV)
Find month-end NAV from annual return (PYQ May 2024)	Monthly = 16% ÷ 12 = 1.33%; 0.0133 = (NAV − 78.50 + 0.70) ÷ 78.50 → **₹78.84**
Days invested from yields	Days = period yield × 365 ÷ yield p.a.; count back

7 Dividend vs bonus vs growth plans

→ Build a **units table** per plan to redemption; value = units × closing NAV; deduct STT / tax on short-term units only; annual return = gain ÷ investment × 12 ÷ months

→ **Back-solve issue price** (Jan 2026 Q1(a)): value now = investment × (1 + yield × years); units now = value ÷ NAV; **undo bonuses latest first** (× b ÷ (a + b)); issue price = investment ÷ original units. SM: ₹16 L, 17.5% × 5 → ₹30 L ÷ ₹100 = 30,000 → 24,000 → 20,000 → **₹80**

8 NAV statement (new fund, one month)

1. Cash = amount raised − underwriting commission − marketing / issue costs − investments bought + sale proceeds + dividends received − expenses **paid**
2. Net assets = cash + market value of holdings + receivables (uncollected dividend) − **arrears / outstanding expenses**
3. Distribution = % of **realised** earnings only (sale gains + dividends), deducted from cash
4. NAV per unit = net assets ÷ units (underwritten issue → all units outstanding)

⚠ TRAP

Only expenses **paid** leave cash; arrears are deducted from net assets. Units issued on a cheque = cheque ÷ **unrounded** current NAV; NAV unchanged that day.

HIGH

9 Evaluating a mutual fund

Quantitative (5)

1. Risk-adjusted returns · 2. Benchmark returns (alpha) · 3. Comparison with peers · 4. Returns across economic & market cycles · 5. Financial measures — expense ratio, Sharpe, Treynor, Sortino

Qualitative (3)

1. Quality of portfolio (blue chips, diversification, style; debt: credit quality, maturity, modified duration; low turnover) · 2. Track record & competence of fund manager · 3. Credibility of fund house team

Measure	Formula / note
Sharpe	$(R_p - R_f) \div \sigma_p$ — total risk
Treynor	$(R_p - R_f) \div \beta$ — systematic risk
Sortino	SM: variation of Sharpe using **downside deviation** instead of total σ in the denominator
Jensen α	$R_p - [R_f + \beta(R_m - R_f)]$
M^2 (not in SM)	$R_f + \text{Sharpe}_p \times \sigma_m$ — compare with R_m
Information ratio (not in SM)	$(R_p - R_{benchmark}) \div$ tracking error

10 Reverse-engineering β and NAV

Given	Get
Sharpe & σ → excess return; with Treynor	**β = (Sharpe × σ) ÷ Treynor**
ρ, σ_p, market variance	$\beta = \rho\sigma_p \div \sigma_m$ ("variance 9%" → $\sigma_m = 3$)
Sharpe, Treynor, σ_m	σ = excess ÷ Sharpe; β = excess ÷ Treynor; $\rho = \beta\sigma_m \div \sigma$; **$R^2 = \rho^2$**
Jensen α & Treynor	CAPM return = actual − α; solve R_f and β simultaneously

Expected NAV (SM): NAV ₹70.71, 99% equity, Sharpe 2, σ 11.25, Treynor 15 → β = 22.5 ÷ 15 = 1.5. Market −5% → equity ₹70 × (1 − 7.5%) = 64.75; cash 0.71 − monthly cost 0.25 = 0.46 → **NAV ₹65.21**. Trap: **don't apply β to the cash part**.

11 Closed-ended fund premium / discount

Market price = NAV × (1 ± premium / discount). Investor's return uses **market prices**, so it beats NAV growth if the discount narrows. PYQ May 2026: geometric mean return 3.771% vs NAV growth 2.76%. Geometric mean = $[(1 + r_1)(1 + r_2)\ldots]^{1/n} - 1$.

12 Role of fund manager & FIIs

- → **Active** fund: outperform, generate positive alpha via research & stock picking. **Passive**: match index with minimum tracking error
- → Other roles (**C-M-W-O**): **C**ompliances (SEBI / AMFI, reporting), constant **M**onitoring of performance, **W**ealth creation & protection, control over **O**utsourced work
- → **FIIs** buy units directly from the issuer or through a registered broker on a recognised exchange within SEBI limits; benefits: **better corporate governance**, **more competition**, **capital inflow** (jobs & forex)
- → Fund hedge with index futures: contracts = β × equity value ÷ (futures × lot) — **cash not hedged** (₹90 cr, β 1.1, 4,300 × 50 → 4,605)

13 Formula plans — constant ratio plan

- → Total value = bonds + units × NAV; when NAV move hits the trigger, **reset to the fixed ratio** — buy units after a fall, sell after a rise
- → ICAI checks each NAV against the **previous** one (32 → 38 = +18.75% triggers)
- → Wins in **choppy** markets ending near the start (sells high, buys low); loses in a strong one-way rise (keeps selling the winner)

14 SIP / STP / SWP & vocabulary

- → SIP: fixed amount in; average cost = total ₹ ÷ total units = **harmonic mean** of NAVs ≤ average NAV (₹10,000 at 20, 16, 25, 20 → ₹19.75 vs ₹20.25)
- → SWP: fixed amount out · STP: fixed amount moved scheme to scheme
- → NFO · AUM · folio · **cut-off time** (before → that day's NAV; after → next day's)
- → Gilt fund: rates expected to fall → **lengthen duration**; to rise → shorten

15 Correlation — how Ch 8 actually connects

Ch 8 idea	The actual link (app-written cross-reference; the Ch 8 side is SM)
Sharpe / Treynor / Jensen	**Ch 6**: same formulas; Sharpe for an undiversified investor (σ), Treynor for a diversified one (β), Jensen α positive = fund beat CAPM.
Debt / gilt fund NAV	**Ch 5**: rates ↑ → bond prices ↓ → NAV ↓; % change ≈ −modified duration × Δy, so longer-duration funds swing more.
Hedging a fund	**Ch 9**: sell index futures = Value × β ÷ (futures price × lot) to protect NAV against a market fall.
Index funds / ETFs	**Ch 6** passive strategy and **Ch 4** efficient-market logic — low cost, tracks index.
Units held by a company	**FR Ind AS 109**: MF units generally fail SPPI and are not equity instruments for the OCI election → FVTPL.

> ✍ **Write in exam**
> "NAV per unit = (market value of investments + receivables + accrued income + other assets − accrued expenses − payables − other liabilities) ÷ number of units outstanding."

#5 · 9.6% of marks

Ch 5 — Security Valuation

SM order: return concepts → ERP → required return → discount-rate selection → equity (dividend / earnings / cash-flow models, EV, rights) → preference → bonds → money market → valuers → precautions. TYK theory Qs: **why duration < maturity** (8.8) · **zero coupon bonds** (8.15).

① Return concepts (SM 2)

- → ***Required rate of return*** — *minimum return the investor expects over a period; also called **opportunity cost / cost of capital** (highest expected return forgone elsewhere at similar risk). Often used interchangeably with expected return.*
- → ***Discount rate*** — *rate used to find PV of future cash flows = risk-free rate + risk premium. Each cash-flow stream can carry a different rate (inflation, maturity, default probability → term structure: upward sloping = longer maturity, more inflation / liquidity / default risk).*
- → ***IRR*** — *rate equating PV of future cash flows to market price; average annual return **assuming reinvestment at the IRR**. SM eg: invest 20,000 → 3,000 (yr 1) + 23,000 (yr 2) → IRR **15%**; if the 3,000 is reinvested at only 10% → $(26{,}300/20{,}000)^{1/2} - 1$ = **14.67%**.*

② Equity risk premium vs market risk premium (SM 3.2) ⚠

CAPM: $R_x = R_f + \beta_x(R_m - R_f)$

Market Risk Premium = $R_m - R_f$ · **Equity Risk Premium = $R_x - R_f = \beta_x(R_m - R_f)$**

If x = market (β = 1) → ERP = market premium.

> ⚠ **TRAP**
> Earlier sheet wrongly said "ERP = $R_m - R_f$". Per SM, that is the **market** risk premium; ERP of a stock is **β × MRP**.

③ Required return on equity (SM 4)

Return on share A = R_f + β × Market Risk Premium. CAPM's insight: investors judge risk by the asset's contribution to **systematic** risk (not diversifiable); gives a relatively objective, widely used estimate. SM eg: 5% + 1.5 × 4.5% = **11.75%**. β from covariance (Ch 6): β = $Cov_{im} \div \sigma_m^2$. PYQ May 2024: 30 ÷ 24 = **1.25** → k_e = 12 + 1.25 × 4 = **17%**.

④ Discount rate selection (SM 5)

- → *Cash flows to **equity** → required return on equity; cash flows to **all stakeholders** → cost of capital (WACC)*
- → ***Nominal** CF = not adjusted for inflation; **real** CF = adjusted. Nominal CF ↔ nominal rate; real CF ↔ real rate. Low inflation / short term → nearly same; high inflation → nominal > real.*
- → *Equity valuation uses **only nominal** CF and nominal rate — because corporate tax is stated in nominal terms, so nominal CF reflects taxes accurately. Equity CF → nominal k_e; all-provider CF → **nominal after-tax WACC**.*

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5 Dividend-based models (SM 6.1)

Assumptions: (a) dividend paid annually; (b) first dividend at end of year 1; (c) shares sold at year-end at **ex-dividend** price.

Model	Formula (SM)
One-year holding	$P_0 = (D_1 + P_1) \div (1 + K_e)$ · SM eg: (6 + 36) ÷ 1.20 = **₹35**
Zero growth	$P_0 = D \div K_e$
Constant growth	$P_0 = D_1 \div (K_e - g) = D_0(1 + g) \div (K_e - g)$ — based on Gordon growth model of K_e
Two-stage	$\Sigma\ D_0(1 + g_1)^t \div (1 + K_e)^t$ for n yrs + $P_n \div (1 + K_e)^n$; **$P_n = D_0(1 + g_1)^n(1 + g_2) \div (K_e - g_2)$**
Three-stage	Extraordinary growth → **transition** (advantage depleted by competition) → stable; PV of dividends of 2 periods + PV of price at start of stable period
H-model	$P_0 = D_0(1 + g_n) \div (r - g_n) + D_0 \times H \times (g_c - g_n) \div (r - g_n)$; growth declines **linearly over 2H**; H = half the transition period; = value at normal growth + premium for abnormal growth

SM: one growth rate must be assumed **for infinity**, only then can value be found. These variants also apply to the FCFE model.

Worked (app-computed, re-checked in code)	Result
Constant growth: D_0 3, g 8%, K_e 17%	3.24 ÷ 0.09 = **₹36.00**
MTP Jan 2026: R_f 8, R_m 12, β 1.5 → K_e 14%; D_0 3, g 10%	3.30 ÷ 0.04 = **₹82.50**
H-model: D_0 2, g_c 16%, g_n 6%, transition 10 yrs (H 5), r 12%	(2.12 + 2 × 5 × 0.10) ÷ 0.06 = **₹52.00**
Two-stage: D_0 2, 20% for 3 yrs then 6%, K_e 12%	PV divs 6.90 + PV P_3 43.46 = **₹50.36**
PYQ May 2024: D_0 3, 30% for 5 yrs then 15%, K_e 17%	20.81 + 292.09 = **₹312.90**

> ⚠ **TRAP**
> P_n uses **D_{n+1}** (grow the last dividend once more at g_2) and is discounted by the **year-n** factor. Buy if intrinsic > market; hold if roughly equal.

6 Earnings-based models (SM 6.2)

Investors may forgo cash dividend for higher earnings on retained profits → value on earnings.

(a) Gordon's model — assumes (i) return on retained earnings constant, (ii) retention ratio constant: **$P = EPS_1(1 - b) \div (K_e - br)$** (r = ROE, b = retention)

(b) Walter's approach: **$P = [D + (E - D) \times r/K_e] \div K_e$**

(c) P/E (multiplier) approach: Value = EPS × P/E; P/E estimated from similar company / industry after adjusting for company-specific features

SM TYK 1: BVPS 137.8, ROE 15%, retention 60%, K_e 18% → EPS 20.67, D 8.27, g = 0.6 × 0.15 = 9% → Gordon **₹91.89**; Walter = [8.27 + (12.40) × 0.15/0.18] ÷ 0.18 = **₹103.35**.

REPEAT R11

⑦ Free cash flow models (SM 6.3)

DDM ignores cash needed for long-term capex and short-term working capital → does not reflect true free cash. **FCFF → discount at K_0** (overall firm value); **FCFE → discount at K_e**. **FCFE along with DDM values equity**; FCFF values the firm.

FCFF based on	Formula (SM 6.3.1)
(a) Net income	NI + Interest(1 − t) + Dep ∓ Capex ∓ Δ non-cash NWC
(b) EBIT	EBIT(1 − t) + Dep ∓ Capex ∓ Δ non-cash NWC
(c) EBITDA	EBITDA(1 − t) + **Dep × t** ∓ Capex ∓ Δ non-cash NWC
(d) FCFE	FCFE + Interest(1 − t) + Principal repaid − New debt + Preferred dividend
(e) CFO	CFO + Interest(1 − t) ∓ Capex (US GAAP; don't add interest again if already in CFO)

FCFE (SM 6.3.2) = Net Profit + Dep − ΔNWC − Capex + New Debt − Debt Repayment **+ Net issue of Preference Shares − Preference Dividends**. Can also be used in the multistage growth model.

→ Sep 2025: FCFF with WACC from a **target** structure → equity value per share; then FCFE back-solved

→ May 2026: two-stage FCFE; **implied terminal growth** for a given market price

⑧ DDM vs FCFE (SM 6.4)

DDM	FCFE
Values stock on cash paid as dividend; assumes consistent payout, which can be **less than cash available**. Healthy cash + low payout → **undervaluation**.	Values stock on cash available for distribution **after** capex and incremental WC → **better measure** than DDM.

⑨ Enterprise value (SM 6.5)

EV = Market cap + LT debt + Minority interest − Cash & equivalents (also minus equity investments in affiliates / other companies and LT investments)

Type	Meaning
Total EV	All business activities: mkt cap + interest-bearing debt + minority interest − cash
Operating EV	Total EV − market value of **non-operating** assets (investments, shares in associates)
Core EV	Operating EV − value of **non-core** assets

EV = true economic value; considers debt + equity, **least affected by capital structure** → EV multiples more reliable than equity multiples. **EV/Sales** — firms with negative CF / earnings (cyclical, tech); sales least manipulable. **EV/EBITDA** — capital-intensive firms (heavy D&A), acquisitions; unaffected by depreciation policy & capital structure; inverse = cash return on total investment.

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10 Rights (SM 6.6)

TERP = $(nP_0 + S) \div (n + n_1)$ · **Value of right** = Ex-right price − Subscription price · **Value per shareholding** = (Ex-right − Subscription) ÷ existing shares per right

SM TYK 2: ₹13, 10 lakh shares, raise ₹20 lakh. 1-for-2 → 5 lakh @ ₹4 → TERP **₹10**, right **₹6**. 1-for-4 → 2.5 lakh @ ₹8 → TERP **₹12**, right **₹4**.

11 Preference shares (SM 7)

→ Fixed dividend like debentures. **Non-redeemable** = like perpetual bond: Dividend ÷ required return

→ **Redeemable** = PV of dividends + PV of maturity value at required return. SM eg: FV 10,000, 10%, 3 yrs, 12% → **₹9,519.63**

12 Bond basics, value, theorems (SM 8.1–8.7)

→ **Basics**: par value · coupon rate & frequency (monthly / quarterly / half-yearly / annual) · maturity period · redemption (**bullet** — one-shot at par or premium)

→ **$V = \Sigma\ I \div (1 + k_d)^t + F \div (1 + k_d)^n$** = I(PVIFA) + F(PVIF); k_d = YTM / required return on similar bonds

→ **Semi-annual**: V = I/2 × PVIFA(k_d/2, 2n) + F × PVIF(k_d/2, 2n)

→ **YTM** = discount rate at which PV of the bond's future cash flows = market price

Cause (SM 8.3)	Effect
YTM = coupon rate	sells at **par**
YTM > coupon rate	sells at **discount**
YTM < coupon rate	sells at **premium**
Longer maturity	greater price change for a given change in required return

→ **Price–yield (8.6)**: price varies **inversely** with yield

→ **Price–time (8.7)**: price = par at maturity (no default) → premium bond falls over time, discount bond rises

Worked (app-computed): Face 1,000, coupon 10%, 5 yrs, k_d 12% → **₹927.90**. Exam aid, not SM: approx YTM = [C + (RV − P) ÷ n] ÷ [(RV + P) ÷ 2].

13 Duration (SM 8.8)

Duration = weighted average time within which investor gets back promised principal and promised YTM; measures how quickly a bond repays its true cost; also a measure of **interest-rate risk** (higher duration → higher risk).

Coupon bond's duration is **always less than its maturity** (TYK theory Q1).

Factors: (i) **time to maturity** — shorter → lower duration; (ii) **coupon** — higher → lower; (iii) **YTM** — higher → lower

Macaulay = $\Sigma\ t \times PV(CF_t) \div$ price (years to recover true cost) · **Modified** = Macaulay ÷ (1 + YTM/n), n = compounding periods per year

Same bond (app-computed): D = **4.1355 yrs**; MD = **3.6924** → +1% yield ≈ **−3.69%** price.

13 Immunization (SM 8.9) — contd.

- → **Immunization**: interest-rate risk = **price risk + reinvestment risk**, which move in **opposite** directions. Choose bonds so they offset → portfolio value at end of holding period insensitive to rate changes. **Duration = holding period** → immunized.
- → SM TYK 27: need ₹1 lakh in 2 yrs; Bond X duration 1, Bond Y 3.563 → 2 = x + (1 − x)3.563 → **61% X, 39% Y** (≈ 52 X, 34 Y bonds)

14 Yield curve / term structure (SM 8.10–8.11)

Spot rate = rate for a zero-coupon bond of that maturity (time value of a single future payment). **Yield curve** (term structure) = how YTM relates to term to maturity, for bonds similar in all respects except maturity. **Four patterns** (Saransh): **A. Upward sloping · B. Downward sloping · C. Flat · D. Humped**. Forward rates (rates applicable in future) are derived from spot rates.

Theory	Says (SM)
Expectation	Long-term rates are **unbiased estimators** of future short-term rates → used to forecast them
Liquidity preference	Investors risk-averse; long bonds carry more interest-rate risk and sacrifice liquidity → **premium**; normal curve is **positively sloped**
Preferred habitat (market segmentation)	Each investor has a preferred maturity; rates set by demand/supply in each segment; mismatch → higher rate to pull players out of habitat; curve can be **upward, falling or flat** (Saransh: explains almost any shape)

15 Convexity adjustment (SM 8.12)

Duration assumes a **linear** price–yield link → good only for small Δy. Adjustment = $C^* \times (\Delta y)^2 \times 100$, $C^* = (V_+ + V_- - 2V_0) \div [2V_0(\Delta y)^2]$

16 Convertibles, warrants, ZCB (SM 8.13–8.15)

- → **Convertible debenture** → equity after a period. Shares per debenture = **conversion ratio**; price paid per share = **conversion price**; **conversion value** = share price × shares per debenture
- → **Warrant** = right to subscribe equity during a period at a stated price; issued to **sweeten** a debenture issue. Value = **(MP − E) × n**
- → Warrant vs convertible: (i) warrant's conversion option is **detachable** → traded separately; (ii) warrant exercised for **cash**, convertible needs no cash
- → **Zero coupon bond** (TYK theory Q2): no coupon; issued at **discount**, one lump sum (face value = investment + accrued interest) at maturity; usually long maturity → long-range planning; issued by banks, government, companies — corporate ZCBs riskier (issuer strength, longer maturity) but higher return

MTP May 2025 extra terms (ICAI answer, not SM text): conversion parity = bond price ÷ ratio (5,300 ÷ 20 = 265); conversion premium; straight value (bond floor); downside risk; premium payback.

17 Refunding of bonds (SM 8.16)

- → Refund before maturity when rates **fall** — issue new lower-coupon bonds; hence **call feature**, which costs a **call premium** (repay above face value)
- → A strategic decision evaluated by capital budgeting — generally **NPV**: refund if PV of inflows (**net cash saving**) > PV of outflows (**call premium, interest during transition**, etc.)
- → New bonds usually issued **before** old ones are refunded → interest on **both** during the overlap

Exam method (app-written, not SM text): outlay = call premium (net of tax) + new issue cost + overlap interest (net) − tax saving on unamortised old issue cost written off; savings = interest saved (1 − t) ± issue-cost amortisation shield; discount at after-tax new k_d. SM TYK 25 has this set-up (₹9 m unamortised old cost, 4% call premium, 30% tax).

18 Money market instruments (SM 8.17)

Characteristics (5): short duration · large volume · de-regulated interest rates · highly liquid · safe (issuers' inherent financial strength).

Instrument	Key points (SM)
Call / notice money	Inter-bank; scheduled commercial banks; **call = overnight / 1 day**; **notice = 2–14 days**; manage day-to-day surpluses/deficits
Treasury bills	RBI; **91, 182, 364 days**; issued at discount; **Y = (F − P)/P × 365/M × 100**. TYK 28: 91-day @ 6% → **₹98.53**
Commercial bills	From genuine trade (credit sale); self-liquidating, negotiable; **3–6 months**; recourse to two parties
Certificate of deposit	Negotiable term deposits of banks from bulk depositors at market rates; demat or usance promissory note; issued at discount. E = (FV − SV)/SV × 12/M × 100 (SM eg ₹90 for 6 m → **22.22%**)
Commercial paper	Unsecured negotiable promissory note of high-rated companies for WC; **7 days–1 year**; at discount; India Jan 1990 (**Vaghul Committee**); bypasses banks; **no net increase** in borrowing (replaces WC limits). TYK 30: ₹10 cr, 91 days, 12.04% → net **₹9.70849 cr**
Repo / reverse repo	Sell securities + agree to repurchase ("ready forward"). Differences: (a) repo = RBI lends to banks; reverse = banks lend to RBI (b) repo from seller's (fund-raiser's) view (c) repo meets fund deficiency; reverse cuts excess liquidity (d) repo rate higher (e) repo contains inflation; reverse controls money supply

TYK 31 (repo): dirty = clean + accrued = 99.42 + 10 × 262/360 = **106.70**; 1st leg = 8 cr × 106.70/100 × 98/100 = **₹8,36,52,800**; 2nd leg × (1 + 5.65% × 14/360) = **₹8,38,36,604**.

19 Precautions before accepting (SM 10)

1. Good valuation is **not a precise estimate** — quality ∝ time spent collecting data and understanding the firm
2. More than financial statements — new businesses valued on **narrative** (scalability, replication, growth, cross-sell)
3. Find a **middle ground between numbers and story**
4. **Emotions** move stocks — ignoring them hurts consistency

4-MARK THEORY

20 Role of valuers (SM 9.1) — statutory purposes

1. **M&A / demergers / takeovers** — Income Tax Act, 1961 (tax payable)
2. **Slump / asset / IPR sale** — IBC, 2016 (liquidation; fair or liquidation value)
3. **Conversion of debt / security** — RBI (inbound / outbound foreign investment)
4. **Capital reduction** — SEBI ICDR / LODR / preferential allotment
5. **Strategic financial restructuring** — Companies Act 2013, SARFAESI 2002, Arbitration Act 1996; also Ind AS and court orders

21 Responsibilities (SM 9.2) — Model Code, Rule 12(e), Registered Valuers Rules 2017

1 Integrity & fairness — honest, true info, no disrepute · **2 Competence & due care** — standards, keep skills current, **can't disclaim liability** except for facts given by company · **3 Independence & disclosure** — no bias; no assignment if not independent; no dealing in subject's securities (PIT Regs 2015); no "mandate snatching" / "convenience valuations"; **no success fee**; declare past association · **4 Confidentiality**

5 Information management — contemporaneous records; co-operate in inspections; working papers **3 years** (or till tribunal case disposed) · **6 Gifts & hospitality** — none that affect independence · **7 Remuneration & costs** — transparent; only disclosed fees · **8 Occupation & restrictions** — not too many assignments; no employment unless certificate surrendered

22 Correlation — how Ch 5 actually connects

Ch 5 idea	The actual link (app-written cross-reference; Ch 5 side is SM)
$K_e = R_f + \beta(R_m - R_f)$	**Ch 6**: same CAPM; $\beta = Cov_{im} \div \sigma_m^2$ (variance, not σ). The K_e from Ch 6 is the discount rate in every DDM / FCFE here. Stock's ERP = $\beta \times$ MRP — not MRP.
$g = b \times r$	Gordon earnings model uses the same b and r: $P = EPS_1(1 - b) \div (K_e - br)$ is just $D_1 \div (K_e - g)$ with $D_1 = EPS_1(1 - b)$. If $r > K_e$, more retention raises value (Walter gives the same direction).
FCFF @ WACC, FCFE @ K_e	**Ch 13 / Ch 14**: firm value from FCFF − debt = equity value → ÷ shares = value per share → **swap ratio** = target's value per share ÷ acquirer's. A wrong discount rate (FCFE at WACC) flows straight into a wrong swap ratio.
Nominal CF ↔ nominal rate	**Ch 3 / capital budgeting**: (1 + nominal) = (1 + real)(1 + inflation). Never mix real cash flows with a nominal K_e.
Warrant (MP − E) × n	**Ch 9**: this is the **intrinsic value of a call** with strike E. A convertible = straight bond + embedded call on shares; bond floor = straight value.
Duration, immunization	**Ch 12**: interest-rate risk — price risk vs reinvestment risk are the same two risks; % price change ≈ −MD × Δy is how a 1% rate shock is sized. **Ch 8**: longer-duration debt / gilt funds swing more in NAV when rates move.
Repo, CP, CD, T-bill yields	**Ch 10**: money-market rates feed interest rate parity / money-market hedge (borrow or invest at these rates for the hedge period, days ÷ 360 or 365 as given).

#6 · 8.2% of marks

Ch 14 — Mergers, Acquisitions & Corporate Restructuring

SM TYK theory Qs: **synergy** (para 2) · **reverse merger / takeover by reverse bid** (6) · **equity carve-out vs spin-off** (7.2) · **horizontal vs vertical merger** (3). Numericals: swap ratio, post-merger EPS / MPS, true cost, restructuring.

1 Conceptual framework (SM 1)

- → ***Corporate restructuring*** *= expansion or contraction of operations, or changes in assets, financial or ownership structure.* ***Internal****: new plant, R&D, hiving off non-core, divestment, sell-offs, demerger.* ***External****: M&A, JVs, strategic alliances.*
- → ***Merger*** *= two entities unify into one (legal name in India:* ***amalgamation****).* ***Acquisition*** *= one buys out another; both may remain separate entities.* ***Consolidation*** *= both dissolve, new one created.*
- → *Halsbury: amalgamation = blending of undertakings, shareholders of each becoming substantially shareholders of the amalgamated company. Only one company + rights varied =* ***reconstruction / arrangement****.*
- → *Law:* ***Companies Act 2013, Ch XV (s 230–240)*** *— compromise, arrangements, amalgamations, CDR, demergers, fast-track mergers, cross-border mergers.* ***Takeovers regulated by SEBI****; also FEMA, IT Act (****s 72A*** *revival), Competition Act 2002, IDRA 1951.*
- → *India's first takeover attempt:* ***Swaraj Paul – Escorts****.*

2 Rationale for M&A (SM 2) — TYK Q1 "synergy"

Synergistic operating economics: ***V(AB) > V(A) + V(B)****. Synergy = increase in performance of the combined firm over what the two would achieve independently (Mark Sirower, "The Synergy Trap"). Sources:* ***complementary activities*** *(one has branch network, other efficient production) and* ***economies of scale*** *(shared central services). Economies may be* ***real*** *(less input per unit) or* ***pecuniary*** *(lower prices for bulk inputs).*

- → ***Diversification*** *— unrelated merger cuts business risk → lower discount rate → higher value; more so with* ***independent / negatively correlated*** *income streams*
- → ***Taxation*** *— set off / carry forward of losses; target's losses set off against acquirer's profits*
- → ***Growth*** *— faster than organic: shorter "time to market" (no delays on site, plant, hiring)*
- → ***Consolidation of production capacity & market power*** *— reduced competition*

SM examples: growth / snuff competition — Airtel–Loop, Facebook–WhatsApp · capability — Google–Motorola (patents), Flipkart–Myntra · new market — Airtel–Zain, Cargill–Wipro · funds — Ranbaxy–Sun, Jaypee–UltraTech · tax — Burger King–Tim Hortons, DPL–WBPDCL.

③ Forms (types) of mergers (SM 3) — TYK Q4

Type	SM meaning
Horizontal	Same industry; larger market share; may move towards monopoly / near-monopoly to avoid competition
Vertical	Companies with a **buyer-seller** (or potential buyer-seller) relationship
Conglomerate	Unrelated businesses — neither horizontal nor vertical; pure = no common factors; purpose: use of financial resources, **enlarged debt capacity**, managerial synergy
Congeneric	Related through basic technologies, production processes or markets; extension of product-line / markets / technology; outward move within the industry
Reverse	Private company acquires a **public (shell)** company — bypasses the lengthy process of going public
Acquisition	Purchase of controlling interest via: (i) agreement with majority holder (ii) new shares by private agreement (iii) open-market purchase (open offer) (iv) cash / share issue (v) buyout offer to general body. Acquirer may merge or run target independently. Usually friendly.

Slump sale: business sold as a going concern for a lump sum without values to individual assets; transfer without allotting shares need not go through court (transferor continues).

④ Gains from merger (SM 4.1)

Value of acquirer + stand-alone value of target + **value of synergy** = combined value

Cost of acquisition = premium over market value + integration costs · **Net gain = synergy − premium**

SM eg: V_A 100, V_B 50, V_{AB} 175 → synergy **25**; premium 10 → net gain **15**

Value can be made from **non-synergistic** acquisitions too — **operating improvements** are a big source; managerial talent is the key; tie executive pay / equity to post-merger performance (Anslinger & Copeland, HBR 1996).

⑤ Scheme, evaluation, finance (SM 4.2–4.4)

- **Scheme** = heart of the process; no prescribed form. Must vest **all assets & liabilities** of transferor in transferee (else dispose / settle first — otherwise defective, court won't sanction). Define the **effective date**. Courts won't re-examine valuation if fair — **Hindustan Lever**: SC won't interfere when > 99% shareholders approved.
- **Financial evaluation** asks: (a) **maximum price** to pay (b) principal **risk** areas (c) **cash-flow & balance-sheet** implications (d) best way to **structure** the deal
- **Financing**: cash, stock or both (in Definitive Agreement). Pay in shares even with cash available — keep a **war chest** for future deals, or acquirer's shares are **overpriced**. LBOs: immediate funding by **bridge financing**, long-term funds later.

THEORY · MCQ

Hostile takeover: *acquirer's board approaches target's shareholders directly through a* ***public announcement (tender offer)*** *after the target's board rejects the offer.*

⑥ Takeover strategies (SM 5.1)

Strategy	SM meaning
Street sweep	*Accumulate a large number of target shares* ***before*** *making an open offer → target left with no choice*
Bear hug	*Acquirer* ***threatens*** *an open offer → target's board agrees to a settlement for change of control*
Strategic alliance	*Disarm the acquirer by offering a* ***partnership*** *instead of a buyout; acquirer asserts control from within and takes over*
Brand power	*Alliance with powerful brands to displace target's brands, then buy out the weakened company*

⑦ Defensive tactics (SM 5.2)

Tactic	SM meaning
Divestiture	*Target divests / spins off some businesses into an independent subsidiary → less attractive*
Crown jewels	*Divestiture tactic = selling the crown jewels; not allowed in UK once the deal is known*
Poison pill	*Make itself unattractive, e.g. issue substantial* ***convertible debentures*** *to existing shareholders, convertible when threatened → voting shares rise sharply*
Poison put	*Target issues* ***bonds*** *encouraging holders to cash in at higher prices → cash drain*
Greenmail	*Target's management offers the bidder an incentive (higher-than-market price for* ***its*** *shares) to drop the takeover*
White knight	*Target offers itself to a* ***friendly company*** *to escape the hostile one (keep management)*
White squire	*Like white knight, but shares sold to a company* ***not interested in takeover*** *→ management keeps control*
Golden parachute	*Hefty compensation to managers ousted due to takeover → reduces acquirer's interest*
Pac-man	*Target makes a* ***counter-bid*** *for the acquirer*

Greenmail = target ***pays the bidder*** *to go away. White knight = friendly party* ***takes over.*** *White squire = friendly party buys shares but* ***doesn't take over.*** *SM: hostile takeovers should be avoided as far as possible.*

⑧ Reverse merger (SM 6) — TYK Q2

- → *Smaller company gains control of a larger one; the* ***healthy, prosperous*** *company's undertaking is merged into the* ***sick, non-viable*** *one (sick = net worth eroded). Also "****takeover by reverse bid****" / "****back door listing****".*
- → ***3 tests****: (i) transferor's assets > transferee's (ii) equity issued by transferee > its original issued capital (iii) change of control in transferee through a minority holder / group*
- → ***Benefits****: easy access to capital market · visibility · tax benefit on carry-forward losses of the acquired public company · cheaper, easier route to become public*

⑨ Divestiture (SM 7)

Reasons (4): (i) focus on core (ii) division not contributing enough revenue (iii) firm too big to handle (iv) cash needed urgently for other opportunities. ***Seller's perspective***: selling needs the same skill as buying; know your worth. Sell-side imperatives: competitor pressure; sale inevitable (no new technology, entry barriers, badly positioned, critical mass not reached, poor use of distribution, no new SBUs, not enough capital) · ***window of opportunity*** · focus on core competencies · best interest of shareholders.

Form (SM 7.2)	SM meaning
Sell-off / partial sell-off	Sale of asset, factory, division, product line or subsidiary for cash / securities; unit doesn't fit core strategy; market may undervalue combined business; raises cash (raiders sold subsidiaries to service LBO debt)
Spin-off	Part separated into a new firm; existing shareholders get ***proportionate ownership***; management parted; ***no fresh cash***. Reasons: separate identity · avoid takeover (valuable division spun off) · separate regulated & unregulated lines. Eg Future Group → Ezone
Split-up	Entire firm broken into a series of spin-offs; ***parent ceases to exist***; smaller units more manageable
Equity carve-out	Like spin-off but some shares sold to the public by ***IPO → brings cash***; parent keeps controlling stake; used when the subsidiary grows faster / has higher valuation; fails if carved out as a burden
Demerger of family business	~80% of Indian private companies family-managed; hive off unprofitable units for succession; step 1 identify core / non-core, step 2 cut interest through debt restructuring + sale of surplus assets

> ⚠ TRAP
> ***TYK Q3 — carve-out vs spin-off***: both create a separate company; spin-off shares go to ***existing shareholders*** pro rata with ***no cash***; carve-out sells shares to the ***public (IPO)***, ***raises cash***, parent retains control.

⑩ Financial restructuring (SM 8) = internal reconstruction

- → Internal changes in assets & liabilities ***with stakeholders' consent***, for firms with sizeable losses / eroded or negative net worth → fresh balance sheet free of losses and fictitious assets
- → Sacrifice order: ***equity shareholders most***, then preference holders, debenture holders, lenders, creditors — waivers, lower coupons, ***debt-to-equity conversion***
- → Results: (i) reduction / waiver of claims (ii) assets at real worth by revaluation (iii) gains on appreciation used to write off losses, fictitious assets, provide for doubtful debts

SM Illus 1 (XYZ): benefits = equity reduced 450 + preference 100 + debenture interest waived 26 + creditors 25% waived 75 + land revalued 50 = ***701***; used for P&L 485 + issue cost 5 + preliminary 10 + bad-debt provision 5 + P&M write-down 196 = ***701***. New BS total ***₹825 lakh***; cash = 130 + 150 − 74 = 206.

11 Ownership restructuring (SM 9)

- → **Going private**: listed company converted into private by buying back all outstanding shares (Essar Energy delisted from LSE, 2014); via management buyout or tender offer; usually **heavy debt**
- → **MBO**: company bought by its own management; useful to exit non-core divisions
- → **LBO**: financed entirely or partially **(50% or more) by borrowed funds**; target goes private; secured on acquired firm's physical assets; improve efficiency → cash flow repays debt → may go public again. Targets: market leaders, proven demand, strong management, steady growth. India's first: **Tata Tea – Tetley (2000)**, ₹2,135 cr, ~77% debt; Tata Steel – Corus (7.6 bn)
- → **Equity buyback**: surplus cash buys shares from public (opposite of IPO); shares cease to exist; raises promoter stake

Effects of buyback (SM): (i) controlling holders' % rises (ii) **EPS rises** → price steps up (iii) BS: assets (cash) and equity both fall; cash outflow under **financing activities** and in statement of changes in equity (iv) **ROA, ROE improve**. SM eg: ₹1.5 cr of ₹2 cr cash, 1 L shares bought → assets 5 → 3.5 cr, ROA **4%** → **5.71%**, EPS **2.00** → **2.22**.

12 Unlocking value (SM 10)

Horizontal → optimum size, market share, curb competition, use idle capacity · vertical → save costs, avoid taxes / duties · diversification · use idle funds of another company · merge export / trading co. with industrial co. (cash flow) · merge subsidiary into holding co. (cash flow) · take over a **shell** with licences · nourish a **sick unit** of the group (image). Restructuring → more competitive, survive adverse climate, new direction.

13 Premium & discount (SM 11)

Premiums: **market share, controlling stake, brand value**; discounts: **small player, unlisted company**. Work out scenarios with probabilities. **Timing** matters — economic cycle (GDP), stock market (multiples), global events. Synergy realisation risk (eg Satyam Infoway – India World, 2000). Use several methods; NAV / past-earnings methods may be inadequate for growing or intangible-heavy businesses.

14 M&A failures (SM 12)

5 steps of a successful programme

1 Manage pre-acquisition (keep **secrecy**) · 2 Screen candidates · 3 Eliminate misfits, value the rest · 4 Negotiate · 5 Post-merger integration

Why mergers fail

Acquirers **overpay** · synergy **over-estimated** · poor **post-merger integration** · **psychological** barriers. Evidence: Jensen & Ruback — target holders **~20%**, acquirer **~0%**; McKinsey — 61% of 116 failed, 23% succeeded

To succeed: decide post-merger tasks · choose managers from both companies (and outside) · set performance yardsticks and evaluate · motivate them.

15 Acquisition through shares (SM 13)

- → **Steps**: value acquirer's equity → value target's equity → maximum shares that can be exchanged → pessimistic & optimistic scenarios
- → **Exchange ratio** = acquirer's shares given per target share. SM eg: B's 1,00,000 shares at ₹230; A values itself ₹200 → 230/200 = **1.15** → 1,15,000 shares
- → Share count set by: (1) acquirer compares value per share with / without merger (2) seller compares its value with value of shares received (3) negotiation — reflects **bargaining position**. Fewer shares issued → acquirer better off, target worse off; both bear **synergy risk**
- → **Impact of P/E**: P/E = reciprocal of cost of equity, reflects risk; combined P/E may differ. Value = E × P/E → a **lower combined P/E can offset synergy**; a higher one can add value even without synergy

SM illustration	Result
Illus 2: X 3 L sh, MP 30, EPS 4; Y 2 L sh, MP 20, EPS 2.25	(a) EPS ratio 2.25/4 → 1,12,500 sh; EPS_{XY} = 16.5 L ÷ 4,12,500 = **₹4** (both unchanged) (b) 0.5 : 1 → 4 L sh; EPS **₹4.125**: X +0.125; Y equivalent 4.125 × 0.5 = 2.0625 (−0.1875)
Illus 3: A ₹10 L, 2 L sh, MP 75; B ₹3 L, 50,000 sh, MP 60	(i) MP basis: 50,000 × 60/75 = 40,000 sh (4 for 5); EPS = 13 L ÷ 2.4 L = **₹5.42** (ii) B's EPS 6 unchanged: 3 L ÷ 5 = 60,000 sh → **6 for 5**
Illus 4: Simpson (V 20 L, 50,000 sh) + Wilson (V 10 L); combined 35 L	20,000 sh → Wilson 2/7 = ₹10 L · 25,000 → 1/3 = ₹11.67 L · 30,000 → 3/8 = ₹13.125 L; Simpson gets the rest (25 / 23.33 / 21.875 L)

16 Exam working (ICAI answer patterns; app-written method)

Swap ratio = target's measure ÷ acquirer's (EPS / BV / MP / intrinsic), or weighted average. Banks: CRAR (higher better), NPA (lower better → acquirer ÷ target). **MTP Nov 2026**: 4.0 × 0.40 + 0.8 × 0.25 + 2.0 × 0.35 = **2.5**; promoters 34.50 of 57.50 L = **60%**; EPS 400 ÷ 57.50 = **₹6.956**.

EPS_{AB} = (E_A + E_B + synergy) ÷ (N_A + N_B × ER) · **MPS_{AB}** = EPS_{AB} × P/E_{AB} · **equivalent EPS** of target = EPS_{AB} × ER

Max ER (acquirer not diluted): EPS_{AB} = EPS_A · **Min ER** (target not worse): EPS_{AB} × ER = EPS_B (Illus 3(ii)); same on MPS basis

True cost, share deal = target's % of merged firm × V_{AB} − V_B. PYQ May 2024: 20% × 2,800 − 500 = **₹60 L**; cash deal = cash − V_B.

17 Cross-border M&A (SM 14)

Route for global growth; Tata Steel CFO: organic growth in India too slow for large capital projects, few domestic targets. **Motivating factors (5)**: globalisation of production & distribution · integration of global economies · expansion of international trade & investment · countries reforming laws and giving **investment & tax incentives** · privatisation of state enterprises and banking consolidation.

18 SPACs (SM 15)

- → "**Blank cheque company**": raises funds by IPO to acquire an **unidentified target within a specific time**; no operations / revenue; money in **escrow**; not acquired in time → delisted, money returned; holders may **redeem** if not interested; on approval target becomes public
- → **India**: no specific framework. Companies Act — RoC can strike off a company not commencing operations within **one year**; SPACs take **2–3 years** → enabling provisions needed. Not eligible for listing under SEBI; SEBI planning a framework. **IFSCA** (GIFT City) released a consultation paper (offer size, sponsor holding, minimum application / subscription)
- → **Advantages**: capital even in volatile markets · lower fees, faster listing · negotiate with a **single entity** not thousands of investors · skilled professionals pick the target
- → **Challenges**: complex accounting / reporting; public-company readiness in **3–5 months** of LOI; typically **more expensive than an IPO**; investors' money idle in trust up to two years

19 Correlation — how Ch 14 actually connects

Ch 14 idea	The actual link (app-written cross-reference; the Ch 14 side is SM)
Swap ratio on intrinsic value	**Ch 5 / Ch 13**: value each company first (DDM, FCFE at K_e, FCFF at WACC − debt, P/E × EPS), divide by shares, then ratio. A wrong valuation = wrong ratio.
P/E = 1 ÷ cost of equity	**Ch 5**: with zero growth $P = E \div K_e$, so $P/E = 1/K_e$. A riskier merged firm → higher K_e → lower P/E → can wipe out synergy (SM 13).
Diversification motive	**Ch 6**: negatively correlated income streams cut total risk ($r = -1$ case); but shareholders can diversify themselves — the gain must show up as a lower discount rate / higher debt capacity.
Net gain = synergy − premium	**Ch 3**: this is the acquirer's NPV — accept if positive; synergy is the PV of incremental cash flows.
Reverse merger	**FR Ind AS 103**: reverse acquisition — legal acquirer (listed shell) is the accounting acquiree; tested by who gets control.
Buyback effects	**FR Ind AS 7**: buyback outflow is a financing activity (SM says so); EPS rises because shares fall (Ind AS 33 weighted shares).
Cross-border deal	**Ch 10 / Ch 11**: price in foreign currency → exposure until settlement (hedge); value foreign cash flows via forecast rates or foreign discount rate.

#7 · 7.1% of marks

Ch 11 — International Financial Management

SM TYK theory Qs: **instruments of international finance** · **impact of GDRs on Indian capital market**. Numericals: foreign project NPV (PPP / IRP rates, remittance, double tax), APV, GDR number & cost, ungear / regear beta, netting.

① International capital budgeting — complexities (SM 1.1)

1. Foreign cash flows must be **converted** into parent's currency
2. **Parent** cash flows differ from **project** cash flows
3. Remitted profits taxed in **home and host** country
4. Effect of **forex risk** on parent's cash flow
5. Changing **inflation** shifts the competitive environment
6. Host-country **restrictions on distribution** of cash
7. Initial investment in host country to benefit from release of **blocked funds**
8. **Political risk** reduces chance of expected cash flows
9. Host-country **concessions / benefits** raise profitability
10. **Terminal value** hard to estimate (buyers have divergent views)

② Problems in foreign investment analysis (SM 1.2)

→ **Forex risk**: forecast host inflation; inflate local-currency cash flows; convert at spot × expected depreciation from **PPP**

→ **Restrictions** on profit transfer, depreciation charges, technical differences → project ≠ parent flows; diluted via **internal transfer prices, overhead payments**; adjust blocked funds at their **opportunity cost**

→ **Taxes**: two regimes, royalties / dividends / management fees, withholding tax, treaties, discrimination between remitted vs re-invested profit. Use **higher of home / host tax rate**; if feasible, acceptable under a more favourable regime; if not, look for other tax savings. MNCs cut tax via **transfer pricing**.

③ Project vs parent cash flows (SM 1.3)

→ **Project's own cash flows**: must beat local competitors (else invest in local equity / bonds) — but foreign projects often **replace imports**, so comparison is hard; local evaluation **avoids currency conversion**

→ **Parent's view**: operating + financial flows **actually remitted** are the yardstick (dividends, debt service). Evaluate on **net after-tax operating cash flows**; don't mix financial flows into operating flows.

④ Discount rate vs adjusting cash flows (SM 1.4)

→ Earlier MNCs **raised the discount rate** for political & FX risk. SM: not proper to combine all risks in one rate — political risk relates to future events that can't be foreseen now, so (SM's words) adjusting the rate for it "**penalises early cash flows more than distant cash flows**"; raising it for FX risk ignores **favourable** currency moves.

→ Better: **adjust cash flows** — convert remitted flows into **certainty equivalents** (probability + decision tree), discount at host or parent rate.

NUMERICAL CORE

5 Adjusted Present Value (SM 1.5)

$$APV = -I_0 + \Sigma X_t \div (1 + k^*)^t + \Sigma T_t \div (1 + i_d)^t + \Sigma S_t \div (1 + i_d)^t$$

k^* = **unlevered** cost of capital (operating CF) · T_t = tax saving from financing mix · S_t = before-tax interest **subsidies** · i_d = **before-tax cost of debt** (for the last two terms)

Value-additive: each component discounted at a rate matching its risk. Test basic viability first; if feasible stop, else add the other complexities.

Blocked funds: balances abroad that can't be remitted — **direct** (exchange control) or **indirect** (local / withholding taxes on remittance). If released for the project, they **reduce the investment outlay**.

May 2026 Q3(a) (ICAI): pick cheapest borrowing (forward by IRP), then APV with tax shield discounted at cost of debt.

6 SM scenarios (1.6) — method & answers

Scenario	How SM solves it
Illus 1 — foreign co. (Perfect Inc., US) invests in India	Exchange rates by **PPP**: $E(S_1) = S_0 \times (1 + I_d) \div (1 + I_f)$; inflate price; 55% revenue share + royalty; Indian tax 30% paid next year; only **50% remittable** in year earned, rest next year; convert remittance to $; US tax 35% with DTAA credit (net = 35% − Indian tax), paid next year; discount at **15%** (not WACC 13%, project riskier) → NPV **−$4.15 mn** → reject
Illus 2 — Indian co. invests in Nepal, funds raised there	Use **competitor's** beta: ungear 1.35 × 1850 ÷ [1850 + 510(1 − 0.20)] = **1.106**; regear at 55 : 45 → equity β **1.83**; K_e = 8 + 1.83 × 3 = **13.49%**; WACC = 13.49 × 0.55 + 9 × 0.8 × 0.45 = **10.66%**. Sunk research cost and apportioned overheads ignored → reject
Illus 3 — Indian co. invests in China via **GDR** (Mauritius)	K_e (GDR) = 2.75 ÷ 198 + 10% = **11.39%**; rates by **IRP**: 9.50 × 1.12/1.10 = **9.67**, × (1.12/1.10)² = **9.85**; NPV in CN¥ **−12,025**; in ₹ at 12%: remitted yearly **+₹8,70,008**, remitted at end **+ ₹2,14,257** → accept (₹ weakens vs CN¥)

TRAP

Illus 1: remittance and tax both lag a year → there are year-6 and year-7 cash flows. Illus 3: SM discounts CN¥ flows at 11.39% but ₹ flows at 12%, which is why the two NPVs differ in sign — reproduce SM's working as given.

TYK 1 — foreign currency approach: find the foreign-currency discount rate via IRP, (1.14 × 1.08 ÷ 1.12) − 1, discount $ flows, convert NPV at spot. **Home currency approach**: convert flows at IRP forward rates, discount at 14%.

⑦ FCCBs (SM 2.1)

*Convertible bond in a currency other than issuer's; regular coupon + principal, with option to convert into stock (via warrants activated at a price). Equity feature → **lower coupon**.*

Advantages

*(i) investor flexibility — convert or redeem (normally **3 years**) · (ii) **delayed dilution** of equity / EPS · (iii) easily marketable — conversion upside + minimum fixed interest*

Disadvantages

*(i) **exchange risk** on interest → only low D/E, large forex earners opt · (ii) more debt + forex outgo · (iii) low rate (3–4%) but FX risk on interest & repayment if not converted; benefit only if growth lets shares be issued above current price*

⑧ ADRs & GDRs (SM 2.2–2.3)

- → ***ADR**: issued in USA under stringent **SEC** rules; non-US company's shares deposited with custodian in home country; US$-denominated, traded like US securities; same rights as underlying holders*
- → ***GDR**: negotiable certificate in a currency not native to issuer, representing local-currency equity; mostly USD (some Euro / £); can trade in **more than one currency**, inside and outside US. Shares delivered to custodian → depository (eg Bank of New York) issues DRs. Indian issues took GDR form — marketed globally; fetched higher prices than domestic issues; no other term forex funding available.*

Impact on Indian market (TYK Q2)

*(i) market shifting from **Bombay to Luxembourg** (ii) **arbitrage** possible (iii) market no longer independent of the world (iv) **retail investors sidelined** — free pricing ends discounted issues. Large foreign inflow.*

Markets

*(i) mainly **institutional** investors (ii) emerging-market funds dominate (iii) FIIs switch from shares to GDRs (iv) UK, USA (QIBs), SE Asia (HK, Singapore), France, Switzerland*

***Characteristics**: (i) economic benefits but **no voting rights** (ii) settled via **CEDEL & Euro-clear** (iii) listed on **Luxembourg SE** (iv) OTC trading by market makers (v) freely traded (vi) marketed globally, multi-currency (vii) dividend in issuer currency converted to $ → **FX risk with investor** (viii) cancel after **45-day cooling period**; cost = BSE / NSE price on redemption-advice date.*

***Illus 4**: need $10 m net, 2% float → gross $10.204 m; GDR = 2 sh × 250 × 90% = ₹450 = $7.50 → **1.3605 m GDRs**; K_e = 40 ÷ 441 + 12% = **21.07%**.*

⑨ Euro-convertible bonds & others (SM 2.4–2.5)

- → ***ECB**: convert into predetermined shares (conversion premium), fixed interest, interest & redemption in US$. **Call** (issuer) — redeem early, forces conversion when share price **~130–150%** of conversion price. **Put** (holder) — sell back at set price & date.*
- → ***Euro-convertible zero coupon** (no interest, converts at maturity, ~5 yrs, deferred equity) · **Euro-bonds with equity warrants** (market coupon, detachable warrants) · **Syndicated bank loans** (5–10 yrs, SOFR + spread, covenants) · **Euro-bonds** (currency issued outside its country, eg Yen bond in France; tax / regulation refuge; bearer — fixed, FRN, convertible) · **Foreign bonds** (in lender country's currency, sold there, restricted) · **Euro CP** (< 1 year, discount) · **Credit instruments** (rates differ by speed / risk)*

10 International Financial Centre (SM 3)

IFC caters to customers **outside its own jurisdiction** — hub for funds, products, services on own land under a **different set of regulations**; flexibility in currency trading, insurance, banking.

Benefits (5)

(i) professionals abroad can come and practise here (ii) global opportunities **without leaving homeland** (iii) stops **brain drain** (iv) brings back transactions now done abroad (v) complex derivatives can be traded from India

Constituents (6)

(i) highly developed **infrastructure** (ii) stable **political** environment (iii) **strategic location** (airport, seaport, weather) (iv) **quality of life** (v) rational, fair, transparent **regulation** (vi) **sustainable economy** absorbing shocks

- **GIFT City**: idea 2007 (rival Dubai, Hong Kong) — retain financial business moving out of India. IFSC operationalised at GIFT Multi Services SEZ in **April 2015**; competitive tax regime in **Budget 2016**
- **India INX** (BSE subsidiary) inaugurated **9 Jan 2017**; **NSE** began at GIFT **5 June 2017** (equity, currency, IRF, commodity derivatives)
- Low cost, competitive tax, single-window clearance, relaxed company law, international arbitration centre; moving to unified regulation. **FEMA not applicable** at GIFT City.

11 Sovereign wealth funds (SM 4)

State-owned fund of government money, usually from surplus reserves; legal basis varies (constitutive law, fiscal law, constitution, company law, others).

Sources (5)	Surplus from state natural-resource revenues & trade surpluses · bank reserves from budget excesses · foreign currency operations · privatisation money · government transfer payments
Objectives (8)	Protect & stabilise budget / economy from revenue volatility · diversify from non-renewable commodity exports · beat returns on forex reserves · help monetary authority absorb unwanted liquidity · savings for future generations · fund social & economic development · sustainable long-term growth for target countries · political strategy
Classifications (5)	Stabilisation · savings / future generation · public benefit pension reserve · reserve investment · strategic development SWF (SDSWF)
Sovereign vehicles (4)	SWFs · public pension funds · state-owned enterprises · sovereign wealth enterprises (SWEs)

Risk management ranges from very conservative to high risk tolerance, depending on assets and objectives.

12 Why international WC is complex (SM 5)

(1) wider financing options — local or global (2) interest & tax rate differentials (3) **forex risk** on flows and trade (4) home / host **restrictions** on cash & inventory movement (5) limited knowledge of host conditions (6) only with **full capital convertibility** can funds be positioned freely (not India).

13 Multinational cash management (SM 5.1)

→ **Traditional objectives**: (1) manage & control cash (2) optimum utilisation & conservation of funds

→ **Effective system objectives (5)**: minimise currency exposure · minimise overall cash need · minimise transaction costs · minimise political risk · economies of scale & superior knowledge

→ **Centralised system helps**: (a) minimum cash balance (b) manage centre's liquidity (c) optimal hedging (d) max returns on cash (e) **multinational netting** (f) **transfer pricing** (g) exploit currency correlations — receivables & payables with positive correlation; payables with negative correlation; pooling → **group variance < sum of individual variances**

→ **Two basic objectives**: optimise cash flow movements + invest excess cash. **Six ways**: accelerate inflows (lockboxes, pre-authorised payment) · manage blocked funds (local bank finance, transfer pricing, negotiation, leads / lags) · **leading & lagging** · transfer pricing · **netting** · invest excess cash (Euro-currency market, pooled deposits; weigh conversion costs)

Leading / lagging: adjust timing of intra-group payments on currency expectations — **pay hard-currency payables early (lead)**, **delay soft-currency payables (lag)**. Indian subsidiary paying Japanese one: expects ₹ to fall vs ¥ → **lead**; expects ₹ to strengthen → **lag**. Check host-country restrictions.

14 Netting (SM 5.1.5)

Advantages: fewer cross-border transfers (admin cost) · less FX conversion (transaction cost) · better cash forecasting (net transfers at period end) · accurate reporting and coordinated settlement.

→ **Bilateral**: X buys $20 m from Y, Y buys $30 m from X → Y pays X **$10 m** only (not $50 m)

→ **Multilateral**: each affiliate nets all receipts against all payments; needs centralised communication + disciplined subsidiaries. P → Q → R → P, $50 m each → transfers **fully eliminated**

SM matrix ($000)	Receipts	Payments	Net
France	200	240	pays **40**
Germany	180	130	receives **50**
UK	210	160	receives **50**
Italy	190	250	pays **60**

Gross $780 k → net **$100 k = 87%** reduction.

15 International inventory (SM 5.2)

MNCs hold more than EOQ — **stock piling** — due to import bottlenecks (political) and expected **depreciation** of the importing country's currency. Decide by weighing **cumulative carrying cost vs expected input price rise**; if supply interruption is very likely, stockpile even if costly. Longer lead times + customs → **re-order point much earlier**; reliance on imports largest in a **vertical** set-up.

16 International receivables (SM 5.3)

- → **Inter-firm sales**: exporter wants a **strong** currency, importer a **weak** one; exporter may invoice in a weak currency if it has **debt in that currency** (natural offset). Weak-currency invoice → short credit; liberal credit lets exporter borrow against bills, or is forced by competition (financing importer's inventory) → more sales.
- → **Intra-firm sales**: global allocation of resources; timing doesn't matter to the group — cash-rich unit pays early, cash-short pays late → **leads & lags**.

17 ICAI exam patterns (from papers; app-written summary)

- → **Currency swap inside a project** (May 2026 Q1(a)): principal exchanged at the inception spot rate both ways — no FX risk on principal; other FC flows at forward rates
- → **Uncertain FC flows** (RTP May 2025): CFAT for each exchange-rate scenario → probability-weighted CFAT → perpetuity CFAT(1 + g) ÷ (k − g) − outlay
- → **GDR number & cost** (PYQ May 2024): gross issue = net ÷ (1 − flotation) — same steps as SM Illus 4

18 Correlation — how Ch 11 actually connects

Ch 11 idea	The actual link (app-written cross-reference; the Ch 11 side is SM)
PPP / IRP forecast rates	**Ch 10**: same formulas — PPP $E(S_1) = S_0(1 + I_d)/(1 + I_f)$ (Illus 1); IRP $F = S(1 + r_h)/(1 + r_f)$ (Illus 3). Direct quote: home inflation / rate on top.
Ungear / regear beta	**Ch 6 / Ch 3**: $\beta_{asset} = \beta_{equity} \times E \div [E + D(1 - t)]$ (debt $\beta = 0$); use a **proxy** firm in the same business; regear at the project's own D/E; then CAPM → WACC.
APV	**Ch 3**: base-case NPV at unlevered k + PV of financing side-effects; tax shields at cost of debt — identical method, just with FX flows.
GDR cost $K_e = D_1 \div$ net proceeds + g	**Ch 5**: Gordon growth model; flotation cost lowers net proceeds and raises K_e.
FCCB / ECB call & put	**Ch 9 / Ch 5**: convertible = straight bond + call on shares; issuer's call forces conversion once share price is well above conversion price.
Leading & lagging, netting	**Ch 10**: internal hedging techniques — same definitions; netting cuts transaction exposure before any external hedge.
Pooling cash	**Ch 6**: less-than-perfectly correlated flows → group variance < sum of individual variances (diversification).

#8 · 5.9% of marks

Ch 12 — Interest Rate Risk Management

Numerical every attempt, different tool each time: FRA (Sep 2025), cap & floor (Jan 2026), gap / Earnings at Risk + zero-cost collar (May 2026). SM TYK theory Qs: **Forward Rate Agreements** · **swaptions and their uses**.

① Interest rates & benchmarks (SM 1–2)

→ *Interest = cost of borrowing + compensation for service and **risk** of lending (default) + protection against **inflation**.*
__Factors__ (macro): (a) demand / supply of money (b) inflation (c) government — biggest borrower; RBI via note issue / __OMO__ changes CRR, SLR, bank rate

→ ***Benchmark (reference) rate**: basis for other rates — loans, overdrafts, mortgages, forwards, futures, options and especially **swaps**; floating loans = benchmark + bps by credit rating; decided by an independent body; also used to value bonds*

→ ***LIBOR**: manipulation news 2012 → decided 2017 to end → from **1 Jan 2022** use **Alternative Reference Rates (ARRs)**. ARRs differ: (i) based on **actual overnight transactions** (secured or unsecured), LIBOR unsecured and relied on panel banks' judgment (ii) **near risk-free, no term premium***

Region	ARR	Regulator	Nature
USA	**SOFR** — Secured Overnight Financing Rate	Federal Reserve Bank of New York	Secured
UK	**SONIA** — Sterling Overnight Index Average	Bank of England	Unsecured
Europe	**€STER** — Euro Short-Term Rate	European Central Bank	Unsecured
Japan	**TONAR** — Tokyo Overnight Average Rate	Bank of Japan	Unsecured
Switzerland	**SARON** — Swiss Average Rate Overnight	SIX Swiss Exchange	Secured

***India**: repo, PLR, MCLR, but most common are **MIBOR** (rate bank charges a borrower) and **MIBID** (rate at which bank would borrow from another bank) — used in IRS, FRA, floating-rate debentures.*

Secured ARRs = SOFR and SARON (both start with S). The other three are unsecured.

② Interest rate risk (SM 3)

*Change in bond prices from rate changes + impact on interest income → **price and reinvestment** risks both matter.*

THEORY + GAP NUMERICAL

③ Types of interest rate risk (SM 3.1)

Type	SM meaning
Gap / mismatch	Assets, liabilities, off-BS items with different principal, maturity or **re-pricing dates**. **Positive gap** = RSA > RSL → rate **rise raises NII**. **Negative gap** = RSL > RSA → NII rises when rates **fall**. **Earnings at Risk (EaR) = gap × assumed rate change**; bank fixes an EaR limit (based on last / current income) and a trigger for on- / off-BS hedging.
Basis	Rates on different assets / liabilities change by different magnitudes (eg assets fixed, liabilities floating); high for banks funding composite assets from composite liabilities; NII expands = favourable basis shift
Embedded option	Prepayment of loans, call / put on bonds, premature withdrawal of deposits; faster and bigger rate moves → bigger risk; adjust gap statements, charge penalties based on opportunity cost
Yield curve	Yield-curve movements over business cycles affect portfolio values and income
Price	Assets sold before maturity; prices and yields inverse; tied to **trading book** → limit size, holding period, duration, defeasance period, stop-loss, MTM
Reinvestment	Uncertainty about rate at which future cash flows can be reinvested; mismatches vary NII

Gap report — 5 limitations (SM)

(1) only timing difference of re-pricing; ignores **basis and embedded option** risk (2) assumes all items in a time band re-price together; ignores effect on **economic / market value** (3) ignores timing differences in payments caused by rate changes (4) assumes **parallel shift** of yield curve — seldom happens (5) ignores variability in **non-interest** revenue & expenses

④ Measuring interest rate risk (SM 3.2)

- → Must identify and quantify first; system should cover gap, basis, embedded option, yield curve, price, reinvestment and net interest position risks
- → Techniques: **maturity gap analysis** (sensitivity of **earnings**) · **duration** (sensitivity of **capital**) · **simulation** · **VaR** — often used in combination / hybrid

May 2026 Q5(b) (ICAI): re-pricing gap = RSA − RSL (negative); EaR = gap × rate change × ¼ (quarter) vs limit of 8% of NII → hedge if EaR exceeds the limit. Rates up hurts a negative gap; rates down helps.

5 Hedging methods (SM 4)

Traditional: (i) Asset Liability Management (ALM) (ii) Forward Rate Agreement (FRA). **Modern**: (i) Interest Rate Futures (ii) Interest Rate Options (iii) Interest Rate Swaps (+ swaptions).

6 ALM (SM 4.1.1)

Evolved by **RBI**; introduced in Indian banks w.e.f. **1 April 1999**. Comprehensive, dynamic framework to measure, monitor and manage **liquidity, interest rate, forex, equity and commodity price** risks, integrated with business strategy; structure the balance sheet so **net interest earnings are maximised** within risk preference. Extends to liquidity, market and trading risk, funding & capital planning, profit planning, growth projection. First step in long-term strategic planning — an **intermediate-term planning** function.

7 Forward Rate Agreement (SM 4.1.2) — TYK Q1

- → Borrower / lender protects against adverse rate changes; **OTC** (not exchange-traded); used by banks to fix cost of future deposits / revenue of variable loans indexed to SOFR, €STER, MIBOR
- → **Off-balance-sheet**; **no transfer of principal** (notional); **cash-settled** for the difference
- → RR > FR → **seller pays buyer** (buyer = borrower protected); RR < FR → buyer pays seller
- → Settled at the **start** of the period → difference **discounted at the actual (RR) rate**

Settlement = N × (RR − FR) × (dtm/DY) ÷ [1 + RR × (dtm/DY)] (DY = 360 or 365)

SM eg: **1 × 4** FRA (deposit placed after 1 month, matures 4 months from contract) at 5%, $1 m, 90 days; SOFR 6% → 1 bp = $25 → $2,500 ÷ [1 + 0.06 × 90/360] = **$2,463.05** to buyer.

TYK 1 (app-computed): ₹60 cr, 3 months in 6 months, FRA 9.30% → at 9.60% company **receives ₹4,39,453**; at 8.80% company **pays ₹7,33,855**. TYK 2: 6 × 9 FRA ₹100 cr at 5.25%, 3-month MIBOR 5.50% → **+25 bp**, settlement ≈ **₹6,16,523**.

8 Interest rate futures (SM 4.2.1)

- → Futures on an **interest-bearing asset** — bonds are the underlying, not the rate. Borrower fearing a **rise sells** futures (prices fall when rates rise → profit on buying back)
- → NSE: futures on **6, 10 and 13-year GoI securities** and **91-day T-bill**. Settlement: **daily MTM** + **physical delivery** any day in expiry month; final settlement only on expiry date
- → **Conversion factor** (published by NSE; introduced by RBI): CF × futures price = delivery price of a given bond
- → **Cheapest to deliver**: seller's profit = futures price × CF − quoted spot price; CTD = bond that **maximises profit / minimises loss**

NUMERICAL CORE

⑨ Interest rate options (SM 4.2.2)

Also called **Interest Rate Guarantee (IRG)**: a right, not obligation — insurance against adverse moves while keeping favourable ones. An IRO is basically a **series of FRAs** exercisable at set rates each period.

Option	Who / what	Payoff (SM)
Cap (call)	Buyer (**borrower**) pays premium; receives difference when benchmark > cap rate; no obligation if below. Evolved from IRGs on floating-rate loans (1985); LBO boom — lenders required caps. Features: notional, benchmark, cap rate (= strike), period with reset / payment dates	**$N \times \max(0, r - r_c) \times d_t/\text{days}$**
Floor (put)	Buyer (**lender / depositor**) protected from **fall** in rates; = right to **sell an FRA** → European put on an FRA; multi-period floor = bundle of puts on a sequence of FRAs	**$N \times \max(0, r_f - r) \times d_t/\text{days}$**
Collar	**Buy a cap + sell a floor** → rate bounded on both sides; cheaper than a cap alone (floor premium offsets); **zero-cost collar** = floor premium exactly offsets cap premium; gives up benefit of a big fall	**$N \times [\max(0, r - r_c) - \max(0, r_f - r)] \times d_t/\text{days}$**

SM cap eg: 1-yr cap, $1 m, 6-month SOFR cap 5%, reset 15 July (first period excluded — known at inception); SOFR 5.5% → paid next 15 Jan (184 days): 1,000,000 × 0.005 × 184/360 = **$2,555.56**.

⑩ ICAI patterns

- **Jan 2026 Q4(a)**: borrower **buys cap**, depositor **buys floor**. Lump-sum premium converted to per-period: **$P = A \times r_p \div \text{PVAF}(i, n)$** ($i$ = fixed rate per period). AB Ltd ₹5 m, 0.4%, PVAF(4%, 10) 8.1108 → **₹2,466**; gain 2% × ½ × 5 m = 50,000 → net **₹47,534**. XY Ltd ₹2 m, 0.5%, PVAF(4%, 6) 5.2421 → **₹1,908**; net **₹18,092**. ICAI also accepted $n - 1$ resets.
- **Cap to peg at 9%** (MTP): pay actual rate, receive excess over 9% → net cost never above 9%; years at or below 9% → option lapses. Cap preferred to a collar here — selling the floor at 9% causes outflow when rates fall below 9%.
- **May 2026 Q4(c)**: zero-cost collar — value of cap = value of floor (PV of expected payoffs) → solve the floor rate.

> ⚠ **TRAP**
> Cap / floor is per **reset period** — multiply by the fraction of the year (½ for 6-month resets). The first period is usually excluded (rate already known).

11 Interest rate swaps (SM 4.2.3)

- → Counterparties exchange payments indexed to two different rates on a **notional** principal that is **never exchanged**. Early swap dealers took a brokerage fee only; contract was between the end users
- → **Fixed / floating swap**: fixed rate · floating rate reset periodically · notional · term with reset and payment dates. Settlement **2 days after trade date**. Fixed rate quoted as **All-In-Cost (AIC)** against a flat floating index
- → Fixed payment (semi-annual) = $N \times AIC \times 180/360$ (30-day months, 360-day year); floating receipt = $N \times R \times d_t/360$

Type	SM meaning
Plain vanilla (generic / coupon)	Fixed ↔ floating on notional; floating = SOFR, €STER, SONIA, MIBOR, PLR
Basis rate (non-generic)	Both legs floating on different benchmarks (eg 1-month vs 3-month SOFR)
Asset swap	Fixed-rate investments (bonds) swapped for floating-rate investments (an index)
Amortising	Notional **declines** over the life — for issuers of redeemable bonds / debentures (hedges the redemption profile)

12 Swaptions (SM 4.2.4) — TYK Q2

- → Option on an IRS — right, not obligation, to enter a swap at a future date, fixed rate and term. **Payer (call) swaption**: pay fixed, receive floating. **Receiver (put) swaption**: receive fixed, pay floating
- → **Features**: option on a **forward-start IRS** with terms fixed at inception · "**3-month into 5-year**" = option to enter a 5-yr IRS in 3 months · option period = trade date to expiry · premium in **basis points** · can be **cash-settled** (MTM off the forward curve). Pricing: model of distribution of the forward zero-coupon curve
- → **Uses**: (a) traders & treasurers (b) speculate / hedge swap books (c) hedge **embedded optionality** (d) borrowers targeting an acceptable rate (e) businesses **tendering for contracts** (f) protection on **callable / puttable bonds**

13 Correlation — how Ch 12 actually connects

Ch 12 idea	The actual link (app-written cross-reference; the Ch 12 side is SM)
Price & reinvestment risk	**Ch 5**: same two risks that **immunization** offsets (duration = holding period); duration measures sensitivity of capital, gap measures sensitivity of earnings.
Cap, floor, collar	**Ch 9**: cap = series of calls, floor = series of puts on the rate; collar = long call + short put — same as a zero-cost option collar on shares.
FRA settlement discounted at RR	**Ch 10**: same idea as early settlement of a forward — money paid at the start of the period is worth less, so discount at the actual rate.
IRS / swaptions	**Ch 10**: currency swaps use the same comparative-advantage logic; **Ch 6**: IRS changes the duration of a bond portfolio (active fixed-income strategy).
IRF conversion factor, CTD	**Ch 5**: bond price = PV of coupons + redemption — CF makes bonds with different coupons / maturities comparable.
SOFR / SONIA / MIBOR	**Ch 11**: syndicated loans are priced at SOFR + spread; **FR Ind AS 109**: IBOR reform and hedge accounting.

#9 · 4.0% of marks

Ch 2 — Risk Management

Short chapter — learn every list. SM TYK theory Qs: **significance of VaR** (3.2) · **financial risk from different perspectives** (2). Numerical: VaR for a single share / two-share portfolio.

1 Types of risk faced by an organisation (SM 1)

Risk	SM meaning & example
Strategic	*Company's strategy becomes less effective and it struggles to reach its goal — technology change, new competitor, shift in demand, input-cost rise.* ***Kodak*** *(built a digital camera in 1975, saw it as a threat, left behind) ·* ***Nokia*** *(late to touch screens → Samsung led) ·* ***Xerox*** *(positive — moved into laser printing)*
Compliance	*Complying with Companies Act 2013, SEBI etc.; non-compliance → fine / imprisonment. Real problem when entering a* ***new business line or geography*** *(cement company entering sugar in another state)*
Operational	***Internal****; failure to cope with day-to-day problems; relates to* ***people and process*** *(paying ₹1,00,000 instead of ₹10,000 → second checker or system flag)*
Financial	*Unexpected changes in prices, exchange rates, credit rating, interest rates; political risk included as it leads to country risk*

2 Categories of financial risk (SM 1.4)

- → ***Counter-party****: non-honouring of obligations (goods not delivered after payment, loan / interest not repaid) — includes* ***credit risk***
- → ***Political****: faced by overseas investors — (i) confiscation / destruction of property (ii) rationing of remittance (iii) restriction on currency conversion (iv) restriction on borrowings (v) invalidation of patents (vi) price control*
- → ***Interest rate****: rate changes alter assets & liabilities; most important for* ***banks*** *(earn the spread). Inherent in both fixed and floating rates — floating borrower suffers when rates rise; fixed borrower pays comparatively more when floating rates fall*
- → ***Currency****: cash flows change with exchange rates, adversely or favourably — ₹ depreciation helps an exporter (****Infosys****), hurts an importer (****IOC****)*
- → ***Liquidity****: inability to meet liabilities when due — inadequate cash or mismatch in timing; common in banks (maturity mismatch)*

3 Evaluation of financial risk — 3 views (SM 2) — TYK Q2

View	SM
Stakeholders	*Equity holders see* ***financial gearing*** *as risk (last in winding up); lenders see high gearing as risk of default on interest & principal*
Company	*Excessive borrowing, or lending to someone who defaults, can force* ***liquidation***
Government	*Failure of a bank (****Lehman Brothers****) or downgrade of a financial institution spreads distrust; includes* ***wilful defaulters****; extends to* ***sovereign debt crisis***

TYK THEORY Q1 + NUMERICAL

④ Value at Risk (SM 3)

*Measure of investment risk: under normal market conditions over a set period (say one day), **how much an investment might lose** — portfolio, capital investment or forex. Answers: (i) what is the **worst case**? (ii) what will be the **loss**? First applied **1922 at NYSE**; entered finance in the **1990s**; now the world's most widely used measure of financial risk.*

Features (SM 3.1)

*(i) **Components**: time period · confidence level (generally **95% / 99%**) · loss in % or amount (ii) **statistical** tool based on standard deviation (iii) any **time horizon** — day, week, month (iv) **probability** of maximum loss, assuming normal distribution (v) **risk control** by setting maximum-loss limits (vi) **Z score** = number of σ from the mean; Z × σ = VaR*

Application / significance (SM 3.2) — TYK Q1

*(a) measure **maximum possible loss** on a portfolio or trading position (b) **benchmark** for performance measurement of operations / trading (c) fix **limits for front-office** dealers in treasury (d) help management decide **trading strategies** (e) tool for **ALM**, especially in banks*

VaR = Z × σ (in ₹) × √t · Z = **2.33** at 99%, **1.65** at 95% (one-tailed) · $\sigma_{t\ days} = \sigma_{1\ day} \times \sqrt{t}$

SM working	Answer
SM 3.3: ₹2 cr in X Ltd, daily σ 2%, 252 trading days, 99%	σ = ₹4 lakh → 1-day VaR = 4 × 2.33 = **₹9.32 lakh**; 10-day = √10 × 9.32 = **₹29.47 lakh**
TYK 1: ₹200 lakh each in XYZ & ABC, daily σ 1% each, r = 0.3, 10-day 99%	Daily σ each ₹2 lakh; variance = $2^2 + 2^2 + 2 \times 0.3 \times 2 \times 2 = 10.4$ → σ = **₹3.22 lakh** (or 0.80623% of ₹400 lakh); 10-day σ = 3.22 × √10 = **₹10.18 lakh**; VaR = 2.33 × 10.18 = **₹23.72 lakh**

⚠ TRAP

*Scale σ by **√t**, never by t. Use portfolio σ with the correlation term — two 1% shares with r = 0.3 give 0.806%, not 1%. 252 days is only needed to annualise.*

⑤ Identifying & managing financial risk (SM 4)

Counter-party risk (4.1)

***Hints (5)**: (a) failure to obtain resources to complete the project (b) regulatory restrictions (c) hostile action of a foreign government (d) let down by a third party (e) insolvency. **Management (6)**: (1) **due diligence** before dealing (2) don't over-commit to one entity / group (3) know exposure limits (4) review credit limits & approval procedure regularly (5) rapid action on likely default (6) performance guarantees, insurance, other instruments.*

Political risk (4.2)

Identify (4): insistence on resident investors / labour · restriction on currency conversion · **expropriation** of foreign assets · price fixation

Assess country risk (4): political rankings in business magazines · macro-economic conditions · popularity & stability of government · advice from home-country **embassies**

Mitigate (4): **local sourcing** of materials & labour · **joint ventures** · **local financing** · **prior negotiations**

Interest rate risk (4.3) — identified from (6)

Monetary policy · government action (eg **demonetisation**) · economic growth · release of industrial data · foreign investment · stock-market changes. Management → Ch 12.

Currency risk (4.4) — parameters (6)

(1) **Government action** (Brexit → £ lowest since the 1980s) (2) **nominal interest rate** (IRP) (3) **inflation** (PPP) (4) natural calamities (5) war, coup, rebellion (6) change of government and its attitude to foreign investment. Management → Ch 10.

Mnemonic for currency parameters: **G-I-I-N-W-C** — Government, Interest, Inflation, Natural calamity, War, Change of government.

⑥ Correlation — how Ch 2 actually connects

Ch 2 idea	The actual link (app-written cross-reference; the Ch 2 side is SM)
VaR of a 2-share portfolio	**Ch 6**: $\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2r\sigma_1\sigma_2$ — the same formula; lower r → lower σ_p → lower VaR. That is diversification expressed in rupees.
Interest rate risk	**Ch 12**: identified here, managed there (gap, FRA, caps, swaps); **Ch 5**: price vs reinvestment risk, duration.
Currency risk — IRP, PPP	**Ch 10**: the parity conditions forecast the rate; the hedges (forward, MMH, options) manage it; **Ch 11**: political-risk list = complexity list in international capital budgeting.
Political risk mitigation	**Ch 11**: local financing and JVs reduce blocked-fund and expropriation risk; certainty-equivalent cash flows rather than a higher discount rate.
Operational risk, limits	**Ch 9**: Barings (back office run by the trader), Orange County — lessons = separate front / middle / back office, risk limits, VaR + stress tests.
Financial gearing as risk	**FM / Ch 14**: high gearing raises K_e; LBO targets carry the most gearing risk; **AAA**: SA 570 going-concern indicators (defaults, negative net worth).

#10 · 3.4% of marks

Ch 1 — Financial Policy & Corporate Strategy

Pure theory + one numerical (SGR / EFR). SM TYK theory Qs: **interface of financial policy and strategic management** (para 5) · **balancing financial goals vis-à-vis sustainable growth** (para 6).

① Advanced role of the CFO incl. value creation (SM 1)

*Traditional: shareholder **wealth maximisation**, financial health, financial controls; governance, compliance, ethics. Now (globalisation, information growth, pandemic) also **strategic & operational decision-making**. Post-pandemic areas:*

- → ***(a) Risk management** — oversee the whole risk-management framework*
- → ***(b) Supply chain** — financial viability of SCM for sustainable growth*
- → ***(c) M&A and corporate restructuring** — strategic; an error can collapse the business*
- → ***(d) ESG financing** — shift from traditional to **sustainability financing***
- → *CFOs now lead **value creation**, on a sustainable basis over the long term.*

② Strategic financial decision-making framework (SM 2)

- → *Capital investment = **springboard for wealth creation**; capital is the limiting factor → strategic allocation of limited funds*
- → *3 essentials: clear, realistic **strategy** · financial resources, controls & systems (**finance**) · right **management** team & processes → **Strategy + Finance + Management = Fundamentals of Business***
- → ***Strategy** = long-term direction and scope to achieve competitive advantage through configuration of resources in a changing environment, fulfilling stakeholders' aspirations. Investors care about profit **and its likelihood** — risk-return trade-off*
- → ***Strategic financial management** = application of FM techniques to strategic decisions to achieve the decision-maker's objectives; combines **backward-looking** accounting with **forward-looking** FM; identifies strategies that **maximise market value**; allocates scarce capital; implements and monitors*

Functions (SM 2.2) — 5

Continual search for best opportunities · selection of best profitable ones · optimal mix of funds · systems of internal control · analysis of results for future decisions. Strategy = long term (senior mgmt) · tactics = intermediate (middle) · operations = short term (line).

Key decisions — 4

***Financing** (equity–debt mix) · **Investment** (long-term projects, risk vs return) · **Dividend** (payout vs reinvestment) · **Portfolio** (judge each investment by its contribution to the **aggregate** performance of the corporation, not in isolation)*

③ Strategy at different hierarchy levels (SM 3)

Level	SM
Corporate	Which businesses to compete in; develop & coordinate the portfolio. Answers 3 questions: **Suitability** (will it achieve the common objective) · **Feasibility** (kind and number of resources needed) · **Acceptability** (stakeholders' satisfaction — financial and non-financial)
Business unit (SBU)	Any profit centre planned independently; coordination of operating units; building & sustaining **competitive advantage**
Functional	Departments — R&D, operations, marketing, finance, HR; business processes & value chain; give input upward, then translate higher strategy into action plans

Finance has the highest importance in top-down / bottom-up planning: corporate strategy = **deployment** of resources; financial strategy = **mobilisation and effective use of money** — the most critical resource; others can be mobilised with an adequate monetary base.

④ Financial planning (SM 4)

→ Backbone of business & corporate planning; defines the feasible area of operation. **Financial Planning = Financial Resources + Financial Tools + Financial Goals**

→ Outcomes: **financial objectives** (set first, consistent with mission) · **financial decision-making** · **financial measures** to evaluate performance (ratios, cash-flow analysis)

⑤ Interface of financial policy & strategic management (SM 5) — TYK Q1

Starting point and end point of an organisation is money. No business can run or expand without an internally and / or externally mobilised financial base.

1. **Sources of finance & capital structure** — the most important dimensions of a strategic plan. Ownership capital (equity, preference), borrowed (debentures), public deposits (short / medium term; may pay more than banks), overdraft, cash credit, bill discounting, bank loans, trade credit. D/E norms: **1 : 1 public sector**, **2 : 1 private sector**; higher for capital-intensive, high-cost priority-sector and backward-region projects
2. **Investment & fund allocation** — policies for fixed and current assets. Proposals of 3 types: **new product** · **more capacity** for an existing product · **cost reduction** / efficiency. Compare within group by capital budgeting; **project evaluation & selection** are the key jobs; best allocation under resource constraints
3. **Dividend policy** — payout vs retention; dividend = earnings that cannot be profitably used. Stable dividend → positive price impact; constant % of earnings → flexibility, signals lower risk; minimum DPS + extra in good years; weigh investment needs, shareholder preference, cash vs stock dividend

Financial policy can't be framed in isolation — investors constantly evaluate it; frame it **during corporate planning**, not later. Interdependence: sometimes **strategy is the cause and financial policy the effect**, sometimes the reverse.

⑥ Balancing financial goals vs sustainable growth (SM 6) — TYK Q2

- → Forces managers to consider the financial consequences of sales increases; set growth goals consistent with operating & financial policies — conflict if they aren't
- → Right distribution of resources for **current and future** stakeholders: **"save fuel" campaign** = demarketing, a **stable growth strategy** of the oil industry under resource constraints. Variants: **incremental growth, profit, pause** strategies
- → Growth **too fast or too slow** both hurt — keep actual growth close to the sustainable growth ratio

Financially sustainable — 7 (CIVICUS)

More than one **source** of income · more than one **way** of generating income · regular strategic, action & financial planning · adequate financial systems · good public image · value clarity · **financial autonomy**

Sustainable organisation — 6 (CIVICUS)

Clear strategic direction · scan environment for opportunities · attract, manage, retain competent staff · adequate admin & financial infrastructure · demonstrate effectiveness & impact to leverage resources · community support & involvement

⑦ Sustainable growth rate (SM 6.1)

SGR (Robert C. Higgins) = maximum growth in sales achievable given profitability, asset utilisation, desired payout and debt ratios — how fast a firm can grow **without borrowing more** (beyond it, external funds). **SGR = ROE × (1 − dividend payout ratio)**. Variables: net margin, asset turnover, assets-to-equity, retention rate.

- → **Assumes** the firm wants to: (1) keep a target capital structure **without issuing new equity** (2) keep a target payout (3) grow sales as fast as the market allows. If willing to issue equity → no financial constraint on growth
- → Twin cornerstones: **growth strategy** and **growth capability** — neglect either and long-term growth fails
- → Sustainability: **very weak** — total stock of capital assets constant · **weak** — preserve critical resources · **strong** — preserve resources under primacy of **ecosystem** functioning
- → Useful when a borrower seeks more loans → too much debt: raise equity or slow growth to the sustainable level
- → **Mature firms** (actual < SGR): use surplus cash — higher dividends, buybacks, repay debt, hold liquid assets (each **lowers** SGR) — or **acquire** fast-growing companies
- → Growth from **volume and inflation**; inflationary asset growth must be financed like real growth → more external finance, higher historical-cost D/E → if creditors fix D/E, **inflation lowers SGR**
- → **Mitsubishi Corporation**: dropped fixed 3-year midterm plans for a ~2020 growth vision — "future pull" to double the business, clear portfolio strategy (proactive reshaping), growth with financial discipline

NUMERICAL

⑧ External Funding Requirement (SM 6.2)

To grow, both current and non-current assets must rise; with constraints on D/E or current ratio, find how much external funding is needed and of what type. ***Analytical approach****: project each balance-sheet item against sales under the conditions.*

Formula approach*:*

$EFR = (TA \div S) \times \Delta S - (\text{spontaneous payables \& liabilities} \div S) \times \Delta S - m \times S_1 \times (1 - d)$

m = net profit margin · S_1 = revised sales · d = payout. Then split between short-term, long-term debt and equity using the given conditions.

SM Illustration — Sundry Ltd (₹ lakh)	Working
TA 14,000; payables 2,000; sales 6,000 +20%; margin 4%; payout 50%	Retained = 7,200 × 4% × 50% = **144**; extra assets = (14,000 − 2,000) × 20% = 2,400 → **EFR = 2,256**
Sales : (ST loan + payables) = 4 : 3	New = ¾ × 7,200 = 5,400; existing 2,400 + 3,000 = 5,400 → **short-term nil**
FA : LT loan = 1.5	7,200 ÷ 1.5 = 4,800 − 4,000 → **long-term 800**
Balance from equity	2,256 − 0 − 800 = **1,456**; check D/E = 4,800 ÷ (3,000 + 1,456 + 2,000 + 144) = **0.727** ≤ 1.5 ✓

⚠ TRAP

Deduct only ***spontaneous*** *liabilities (payables & provisions) — not loans. Retained earnings use* ***revised*** *sales. Always verify the stated D/E or ratio condition at the end.*

⑨ Correlation — how Ch 1 actually connects

Ch 1 idea	The actual link (app-written cross-reference; the Ch 1 side is SM)
SGR = ROE × (1 − payout)	**Ch 5**: the same g = b × r drives Gordon's model $P = D_1 \div (K_e - g)$ and Gordon's earnings model $EPS_1(1 - b) \div (K_e - br)$. If $r > K_e$, retaining more raises both growth and value.
Mature firm's surplus cash	**Ch 14**: buybacks raise EPS / ROE (SM 9.4 effects) and lower SGR; acquiring fast-growing firms raises actual growth.
Portfolio decisions	**Ch 6**: judge an investment by its contribution to total risk-return, not alone — the Markowitz idea applied to a corporation's projects.
Investment decisions under risk	**Ch 3**: NPV, sensitivity, real options; capital rationing = "allocation of limited funds".
CFO roles: risk, ESG	**Ch 2**: the risk framework the CFO now owns — strategic, compliance, operational, financial risk; VaR limits for treasury dealers.
Inflation lowers SGR	**Ch 5 / Ch 3**: nominal vs real — inflated asset values need real financing even with no real growth.

#11 · 3.4% of marks

Ch 15 – Startup Finance

Pure theory – 4-mark questions. SM TYK theory Qs: **innovative sources of funding a start-up** (para 2) · **pitch presentation** (para 3). Also asked: unicorn, VC characteristics / stages / process, Startup India definition, succession planning.

① Basics of startup financing (SM 1)

→ *Initial infusion of money to turn an idea into reality. **Banks aren't interested** – no assets to rely on, cash flows / repayment capacity can't be forecast*

→ *Left with: sell assets, borrow against home, family & friends – risk of bankruptcy and strained relationships → keep family loans **strictly business-like***

→ *Speed up operations so outside investors can see the venture and the founder's own risk; **bootstrapping** (personal finances or the venture's operating revenue) avoids investors dictating terms; need a strong **business plan** and advice from experienced entrepreneurs*

② Innovative ways to finance a startup (SM 2) – TYK Q1

*Banks are the least likely benefactors → **maximise non-bank financing**.*

Source	SM point
(i) **Personal financing**	Investors won't put money in if the founder hasn't contributed own savings
(ii) **Personal credit lines**	Based on personal credit (eg credit cards); banks cautious – only if the business has cash flow to repay
(iii) **Family & friends**	Believe in you; keep the loan **in writing** (promissory note)
(iv) **Peer-to-peer lending**	Group of people lend to each other; small / ethnic groups of similar faith or interest support each other
(v) **Crowdfunding**	Small amounts from a **large number** of individuals via social media & crowdfunding websites
(vi) **Microloans**	Small loans by individuals at **lower interest**; one individual or aggregated across many
(vii) **Vendor financing**	Company lends to its customer to buy its products; or suppliers **defer payment** until goods are sold (30 → 45 / 60 days) – depends on creditworthiness, may cost more
(viii) **Purchase order financing**	Startups can't fund a large new order → PO financier **pays the supplier directly**; transaction completes, profit flows to the startup
(ix) **Factoring receivables**	Factor pays most (say **80–90%**) upfront; balance on due date after collecting, less interest

③ Pitch presentation (SM 3) — TYK Q2

Short presentation (**not more than 20 minutes**) to investors on the company's prospects and why they should invest; usually a PowerPoint **pitch deck** giving a quick overview of the business plan; face-to-face or online, to investors, customers, partners, co-founders. **9 points**:

1. **Introduction** — who you are, what you do; short; interesting facts, milestones
2. **Team** — investors invest in the team, not only the idea; promoter's background; past joint results
3. **Problem** — the problem solved. SM eg: **Facebook (2004) vs Orkut** — profile visible only if you are added (privacy)
4. **Solution** — how it is solved. SM eg: **Flipkart (2007)** — cash on delivery (cards rare) + own supply chain for timely delivery
5. **Marketing / sales** — market size, target customers, how to attract them, growth & revenue forecasts
6. **Projections / milestones** — educated guess if no history: **income statement** (monthly in year 1, quarterly in year 2, annual after), **cash flow** (cash burn / generated, closing cash), **balance sheet** (annual)
7. **Competition** — how you differ; details of competitors acquired (names, prices)
8. **Business model** — how the company earns revenue and profit; gross profit compares models. SM eg: two movie-rental firms, revenue ₹5 cr; A goes online → COGS ₹4 cr → ₹2 cr → GP margin **20%** → **60%**. Show revenue streams & timeline, pricing vs competitors, **customer acquisition cost, lifetime value**, retention
9. **Financing** — money raised, from whom, how used; if none, what was achieved with minimum funding; how much is sought and its use; investors like founders who invested own money

④ Unicorn (SM 4)

- → **Privately held** start-up valued at **US$1 billion**; term coined by VC **Aileen Lee in 2013**; mythical animal = statistical **rarity** of success. Features: private · US$1 bn valuation · rarity · new ideas, disruptive innovation, consumer focus, high technology
- → Status can be **lost** if valuation slips below US$1 bn
- → India's first: **InMobi** (Sept 2011; SoftBank US$200 m; founded 2007). Fastest: **Udaan** (2018, just over 2 years). Then Flipkart 2012, Mu Sigma 2013, Ola & Snapdeal 2014, Paytm / Quikr / Zomato 2015 …
- → India = **3rd largest** start-up ecosystem (after US, China); 59,000+ DPIIT-recognised start-ups; as of 3 Oct 2023, **111 unicorns** worth US$349.67 bn. Next milestone: **Decacorn** (> US$10 bn)

5 Bootstrapping (SM 5.1.1)

Start and build from personal finances or the company's operating revenue. More money at inception → complacency and waste; own savings → caution, curbs waste.

- → **(a) Trade credit** — suppliers reluctant; show a well-crafted financial plan to the owner / CFO, get a reference; maybe half on credit; **sell before you pay**; cuts working capital (esp. retail)
- → **(b) Factoring** — receivables sold to a commercial finance company, which collects and does the paperwork; can be on a **non-notification** basis; cuts bookkeeping, collection, credit-check costs; frees cash (esp. B2B / government with long delays)
- → **(c) Leasing** — lease the photocopier (₹5,000 p.m. vs ₹1,00,000) or van (vs ₹5,00,000). Lessor: depreciation tax benefit, capital appreciation, profit. Lessee: smaller payments, can walk away at end, lease rentals tax-deductible

6 Angel investors (SM 5.1.2)

- → Invest in small startups, often from family & friends; one-time or ongoing; **more favourable terms** — invest in the **entrepreneur** rather than the viability of the business; focus on first steps, not profit
- → Also called informal investors, angel funders, private, seed investors, business angels; affluent individuals; take **equity or convertible debt**; may use crowdfunding platforms / angel networks
- → Use **own money** (VCs manage pooled funds); vehicle may be an LLC, business, trust or fund. Early failure = total loss → look for a defined **exit** (acquisition, IPO)

7 Venture capital (SM 5.1.3)

Investment vehicle managing investors' funds for startups with **exceptional growth potential**; professionals invest alongside management in young, rapidly growing companies. VCs: finance new & rapidly growing companies · buy **equity** · help develop new products · add value through **active participation**.

Characteristics (4)

(i) **Long horizon** — minimum **3**, maximum **10 years** (ii) **Lack of liquidity** — adds an illiquidity premium (iii) **High risk**, high return (iv) **Equity participation** — board seat, supervise decisions

Advantages (7)

Long-term equity → solid capital base · partner sharing risk & reward · practical advice from experience · network of contacts · further funding rounds · experienced in preparing an **IPO** (incl. NASDAQ) · can facilitate a **trade sale**

THEORY

⑧ Stages of VC funding & risk matrix (SM 5.1.3.4)

Stage	Years locked	Risk	Activity financed
Seed money	7–10	Extreme	Prove a concept / idea, R&D; low-level financing
Start-up	5–9	Very high	Prototypes, product development & marketing
First stage	3–7	High	Commercial production & marketing (early sales)
Second stage	3–5	Sufficiently high	Expanding market, working capital — selling but ***not yet profitable***
Third stage	1–3	Medium	Expansion, acquisition, product development for a ***profit-making*** company — ***Mezzanine financing***
Fourth stage	1–3	Low	Going public — ***Bridge financing***

⑨ VC investment process (SM 5.2) — 6 steps

1. ***Deal origination*** — directly or via intermediaries (mainly practising CAs); VC states sector, stage, promoter, turnover focus; company gives ***Investment Memorandum*** (business model, financial plan, exit plan, tentative valuation)
2. ***Screening*** — committee of senior VC people
3. ***Due diligence*** — verify documents; usually external consultants; fee paid by VC or shared
4. ***Deal structuring*** — both win; convertible structure lets promoter buy back; ***tag-along clause*** — promoter must sell part of stake along with VC
5. ***Post-investment*** — VC nominee on board; strong MIS, budgeting, governance, covenants, milestone reporting; professional management
6. ***Exit plan*** — (a) sell to third parties (IPO or private placement to other VCs) (b) promoter ***buy-back*** at pre-agreed rate (IRR generally ***18–25%***), often as right of first refusal

⑩ VC funds in India (SM 6)

→ ***Evolution***: committee in the 1970s; first all-India VC funding by ***IDBI, ICICI, IFCI*** about ten years later; CCI guidelines Nov 1988 (narrow, restrictive); World Bank seminar → TDICICI, GVFL, Canbank, Pathfinder; CCI abolished; foreign finance cos allowed ***1995***; re-regulation 1996 favoured foreign players; boost from IT revolution ***1997***; dotcom bust → shift to later-stage funding

→ ***Formation***: trust · company · LLP. ***Investors***: FIs, banks, pension funds, corporations, HNIs

→ ***Structures***: ***domestic*** — pooling vehicle (trust or company; ***trust preferred*** for flexibility; India doesn't recognise limited partnerships) + separate investment adviser · ***offshore*** — LLC / LP abroad invests directly, managed offshore, Indian adviser does diligence · ***unified*** — offshore vehicle invests in a local trust into which domestic investors contribute directly

11 Startup India (SM 7)

Launched **16 Jan 2016**. Per **GSR 127(E), 19 Feb 2019**, an entity is a startup: (i) up to **10 years** from incorporation, as a **private limited company, registered partnership firm or LLP** in India (ii) turnover **≤ ₹100 crore** in every year since incorporation (iii) working on innovation / improvement of products, processes or services, or a scalable model with high employment or wealth creation. **Not** if formed by splitting up or reconstructing an existing business.

- → Startup India & StandUp India (2016); SIDBI-run **Electronic Development Fund** — government an LP for the first time; Mudra; **100% tax holiday u/s 80-IAC**; exemption from angel tax
- → **Startup India Seed Fund Scheme (SISFS)**, DPIIT, Jan 2021: **₹945 crore**; ~3,600 entrepreneurs via 300 incubators over 4 years; DPIIT-recognised, incorporated **≤ 2 years** before applying; up to **₹50 lakh**; proof of concept, prototype, trials, market entry, commercialisation
- → Why India thrives: (i) talent pool (ii) cost-effective workforce (10 m+ join yearly) (iii) internet use (2nd largest base) (iv) technology (AI, blockchain) (v) variety of funding; 100% automatic FDI route in most sectors

12 Succession planning (SM 8)

- → Identify critical positions and develop people to fill them; vital for family businesses (often non-family leader, family ownership); can be a **liquidity event** transferring ownership to rising employees
- → **Need (5)**: risk mitigation (search takes 6–9 months) · cause removal (fraud, gross negligence, barred leader) · talent pipeline · conflict resolution · aligning culture, vision, values
- → **Strategy (4 steps)**: evaluate key leadership positions → map competencies needed → identify competencies of current workforce (train / hire outside) → **bridge leader** (outsider who prepares young family members)
- → **Challenges in startups (3)**: founder mindset ≠ corporate mindset · premature (processes slow growth) · founders are the face of the startup

13 Correlation — how Ch 15 actually connects

Ch 15 idea	The actual link (app-written cross-reference; the Ch 15 side is SM)
VC illiquidity premium, buy-back at 18–25% IRR	**Ch 5 / Ch 13**: required return = risk-free + premiums; the buy-back rate is the VC's target IRR — discount exit value at it to price the entry stake.
Exit by IPO / trade sale	**Ch 14**: trade sale = acquisition; VC exit is one reason companies are sold; SPAC is another route to listing.
Factoring, vendor & PO financing	**FM (WC)**: working-capital finance; **Ch 11**: accelerating cash inflows and receivables management.
Unicorn valuation	**Ch 13 / Ch 5**: startups valued on narrative (scalability, growth) — the SM Ch 5 valuer's precaution "numbers vs story".
Business-model gross margin	**Ch 1**: strategy + finance + management; pitch "projections" = financial planning (FR + FT + FG).
Startup definition, 80-IAC	**DT / FR**: tax holiday. Convertible debt from angels: the **issuer** splits it under Ind AS 32 if it is a compound instrument; the **holder** carries it at FVTPL under Ind AS 109.

#12 · 3.4% of marks

Ch 3 — Advanced Capital Budgeting Decisions

SM TYK theory Qs: **certainty equivalent approach · sensitivity analysis · project appraisal under inflation · steps in simulation analysis.** Numericals: inflation NPV, ENPV / σ / CV, RADR, CE, sensitivity, scenario, decision tree, replacement cycle (EAC), APV.

① Current trends — factors (SM 1)

*Intermediate assumed certain cash flows; in practice projects face (i) **inflation** (ii) **change in technology** (iii) **change in government policies**.*

② Inflation in capital budgeting (SM 1.1) — TYK Q3

→ *Inflation raises revenues and costs; net revenues after adjusting for inflation = net revenues in current terms. Two distortions:*

→ ***(1) Depreciation on historical cost**: after-tax CF = (R − C)(1 − T) + DT — (R − C)(1 − T) moves with inflation, **DT does not** → inflation **reduces the real rate of return**. SM eg: ₹80,000, 4 yrs, revenue 60,000, costs 20,000, tax 50% → CF 30,000 p.a.; with 10% inflation nominal CF 32,000 → 39,282 but **real** CF 29,091 → 26,830 (falls)*

→ ***(2) Cost of capital contains an inflation premium**: $R_N = R_R + P$ (SM's formula). If the rate has an inflation premium, reflect inflation in cash flows; no inflation → discount at the real rate*

***Illus 1**: outlay 40,000, real CF 15,000 × 4 yrs, cost of capital 12% incl. 10% inflation premium. Discounting real CF at 12% is **inconsistent**. Either (i) nominal CF = 15,000 × 1.10^t at 12% → NPV **₹17,369**, or (ii) real rate = 1.12 ÷ 1.10 − 1 = **1.8%** on real CF → **₹17,367** — same answer.*

***Different inflation for revenue and cost**: $NPV = \Sigma\ [\{R_t\ \Pi(1 + i_r) - C_t\ \Pi(1 + i_c)\}(1 - T) + D_tT] \div (1 + k)^t - I_0$. **Illus 2**: ₹8 lakh, revenue inflation 10/9/8/7%, cost 12/10/9/8% (compounded), tax 60%, dep 2 lakh (shield 1.2 lakh, not inflated), k 10% → NPV **₹1,06,744**.*

⚠ TRAP

*Inflate cumulatively: year-3 revenue = R × 1.10 × 1.09 × 1.08. The depreciation tax shield is **never** inflated.*

③ Technology change (SM 1.2)

***Why analyse**: alters production process · benefits (quality, delivery, flexibility) · may lower cost of capital · better cash inflows · may need extra capex · shorter product life cycle. **How**: (1) at decision time use sensitivity / scenario / simulation (2) after launch, revise estimates (3) if unviable, evaluate **abandonment** (4) adjust the discount rate. Review continuously.*

4 Government policies (SM 1.3)

- → Affect the firm directly or via suppliers, buyers, service providers; positive or negative; revise estimates, consider abandonment. Some change rarely (New Industrial Policy 1991), others within a year: **fiscal policy** (spending, taxation, borrowing) and **monetary policy** (OMO, reserve requirements, selective credit control, currency supply, repo; MPC meets at least **4 times a year**)
- → **Domestic**: (a) interest-rate changes affect cost of debt / YTM — little effect on long-term-funded capex but big on **working capital** (overdraft); small changes ignored, ~**100 bps** calls for revision (b) fiscal policy changes tax and depreciation rates → revise cash flows
- → **International**: (a) bank rate and money supply move exchange rates (b) tax on foreign income / **DTAA** changes

5 Risk in investment decisions (SM 2)

- → Decisions under (i) **certainty** (ii) **risk** — probabilities can be assigned (iii) **uncertainty** — they can't
- → **Risk** = variability of actual vs estimated returns; measured by σ and CV. Risk: distribution known; uncertainty: not known (terms used interchangeably)
- → **Why adjust for risk**: (1) opportunity cost — are returns proportionate to risk vs other options? (2) to know the real value of inflows — higher risk, higher premium and return

6 Internal & external risk factors (SM 3)

Factor	SM meaning & example
Project-specific (internal)	Timely completion, estimation errors in resources & cash flows — nuclear vs hydel project
Company-specific (internal)	Rating downgrade, KMP changes, IPR / legal violations, labour disputes — two banks with different default exposure
Industry-specific	Regulatory restrictions, technology — leather, breweries
Market	Substitutes, demand, access to resources — coal supply to a thermal plant
Competition	Rival entry, product dynamism, change in tastes
Economic conditions	Monetary & fiscal changes (new taxes, cess), inflation, GDP, savings, disposable income
International	Trade / market-access restrictions, recession, bilateral agreements, political & geographical — limits on outsourcing jobs abroad

7 Techniques of risk analysis (SM 4)

Statistical

Probability · variance / σ · coefficient of variation

Conventional

Risk-adjusted discount rate · certainty equivalent

Others

Sensitivity · scenario · simulation · decision tree

TYK THEORY Q1

⑧ Statistical techniques (SM 4.1)

ENCF = $\Sigma\ NCF_i \times P_i$ · **ENPV** = $\Sigma\ ENCF_t \div (1 + k)^t$ (incl. outflows) · $\sigma = \sqrt{\Sigma\ P_i(NCF_i - ENCF)^2}$ · **CV = σ ÷ expected return / cash flow**

CV = risk per unit of expected return — compare projects of different size; **lower CV preferred**. SM eg: 3,00,000 × 0.3 + 2,00,000 × 0.6 + 1,20,000 × 0.1 = **₹2,22,000**.

⑨ Risk-adjusted discount rate (SM 4.2.1)

→ Investors demand more from risky projects: rate = compensation for delaying consumption + inflation (= **risk-free**) + **risk premium** (depends on perceived risk & risk aversion). **RADR = R_f + risk premium**

→ **Illus 7**: ₹100 lakh, CF 25/60/75/80/65 → at 7% NPV **144.34**; at RADR 14% NPV **99.79**

→ **Advantages**: (1) easy to understand (2) builds the premium into the discount rate. **Limitations**: (1) hard to find the premium / RADR (2) NPV yes, but **σ of the project can't be computed**

⑩ Certainty equivalent (SM 4.2.2) — TYK Q1

CIMA: expressing risky future cash flows as the certain cash flow the decision-maker would consider equivalent — **indifferent** between the risky amount and the (lower) riskless amount. Lets the decision-maker build in his **utility function**; a set of riskless cash flows replaces the original.

1. **Step 1**: α_t = certain CF ÷ risky (expected) CF; multiply each CF by α_t. Coin toss: win ₹10,000 or nothing; indifferent at ₹3,000 certain → **CE ₹3,000, α = 0.3**
2. **Step 2**: discount at the **risk-free rate** — cost of capital would **double-count** risk
3. **Step 3**: normal capital budgeting; for **IRR compare with risk-free rate**, not the required return

α between 0 and 1; α = 1 → certain or risk-neutral; industry usually risk-averse. **NPV = $\Sigma\ \alpha_t\ NCF_t \div (1 + k_f)^t - 1$**. **Illus 8**: ₹45 lakh; 10/15/20/25 lakh × 0.90/0.85/0.82/0.78 at 5% → **₹5,34,570**.

Advantages

(1) simple to understand and apply (2) easily set for different risk levels in different years

Disadvantages

(1) no objective / mathematical way to estimate — subjective (2) reflects **management's** risk perception, ignores **shareholders'** → rarely used

RADR vs CE: CE is **superior** — it doesn't assume risk grows with time at a constant rate (each year's α fits its risk). But RADR is preferred in practice — easier to adjust a rate than specify a series of α.

11 Sensitivity analysis (SM 4.3.1) — TYK Q2

- → **CIMA**: modelling and risk-assessment procedure in which changes are made to significant variables to find their effect on the planned outcome; special attention to variables of special significance
- → Change **one variable at a time**; the more sensitive the NPV (IRR), the more **critical** the variable. **Steps**: (1) find variables influencing NPV / IRR (2) establish mathematical relationships (3) analyse effect of change in each
- → SM Illus 9: 2.5% adverse change in each variable → **selling price** has the maximum effect (NPV −24.82%)
- → **Advantages**: (1) identifies **critical issues** (2) **simple**. **Disadvantages**: (1) assumes variables are **independent** (2) **ignores probability** of changes

12 Scenario analysis (SM 4.3.2)

Extends sensitivity: brings in **probabilities** and changes **several variables at once**. Base (most likely) case → worst case (low sales & price, high variable cost) → best case; or mixed. Variability of outcomes across scenarios (eg boom / normal / recession) measures risk. **More complex** than sensitivity — all inputs change simultaneously.

13 Simulation — Monte Carlo (SM 4.3.3) — TYK Q4

Exact replica of the actual situation via a model; infinite calculations for possible outcomes and probabilities; ties together sensitivities and probability distributions; named after casino gambling (from first nuclear-bomb work); gives a **probability distribution of NPV**, not a point estimate.

1. **Model the project** — NPV vs **parameters** (set by decision-maker, constant across runs) and **exogenous variables** (stochastic, outside control)
2. Specify parameter values and probability distributions of exogenous variables
3. Select a value **at random** from each exogenous distribution
4. Compute NPV for those values
5. **Repeat** 3 & 4 many times
6. Plot the NPV distribution; compute **mean and σ**

Advantages — "strength lies in variability"

(1) predict bad market situations beforehand (2) handles many exogenous variables with any distribution, complex inter-relationships beyond analytical methods, forces explicit consideration of inter-dependencies

Shortcomings (4)

(1) hard to model and specify distributions (2) inherently **imprecise** — misleading when a tail matters (3) built by experts, decision-maker may not understand / use it (4) NPV at **risk-free** rate (to avoid pre-judging risk) → hard to interpret

14 Decision tree (SM 4.3.4)

- → Graphic display of the link between a present decision and future events, decisions and consequences — for **sequential** decisions. **Decision** (manager controls; acts rationally) vs **event** (no control; assign probabilities)
- → **Steps**: (1) define investment (launch local / national / international; extend factory vs outsource; dig for oil) (2) identify alternatives (3) draw the tree (4) evaluate — from the **last decision (right) backwards**; at each decision point keep highest NPV, truncate others; compute expected NPV at chance points
- → Rules: decision node = **rectangle**; chance node = **circle**; alternatives as straight lines; drawn **left to right**; number nodes; add values & probabilities

15 Replacement decisions (SM 5)

→ **Replace existing machine** (newer version or a different machine with the same function), mostly to maintain existing levels of operation. Method (app-written): incremental NPV — cost of new less sale of old, change in operating cash flows and depreciation tax shield, terminal values

→ **Optimum replacement cycle** (continuous replacement): NPVs of different cycle lengths aren't comparable → **EAC = PV of costs ÷ PVAF for the cycle**; choose the **lowest EAC**. SM taxi: EAC 1 yr 3,40,044 · 2 yr **3,15,331** · 3 yr 3,17,639 → replace every **2 years**. ICAI MTP machine: 4-year cycle lowest at ₹58,656

16 Adjusted Present Value (SM 6)

APV = Base-case NPV + PV of tax benefit on interest. Base case at the **unlevered** cost of equity (all-equity firm; unlevered K_e from Ch 13). **Separates investment and financing decisions**; credits the project for the debt used to finance it.

17 Correlation — how Ch 3 actually connects

Ch 3 idea	The actual link (app-written cross-reference; the Ch 3 side is SM)
Real vs nominal (1.12 ÷ 1.10 − 1)	**Ch 5**: equity is valued on **nominal** CF at nominal rates; **Ch 11**: PPP uses the same inflation ratio to forecast exchange rates.
RADR = R_f + premium	**Ch 6**: CAPM gives the premium objectively — $\beta(R_m - R_f)$; **Ch 11**: SM says don't load political / FX risk into the rate — use certainty equivalents.
Certainty equivalent at R_f	**Ch 9**: risk-neutral valuation also discounts at R_f after adjusting the cash flows / probabilities — same "adjust numerator, not denominator" logic.
σ and CV of NPV	**Ch 6**: σ and CV of returns; **Ch 2**: σ × 2 = VaR.
Abandonment, decision tree	**Ch 9**: real options (abandon, expand, wait) give value that static NPV misses; decision trees value them.
APV	**Ch 11**: SM's APV for foreign projects adds interest subsidies; **Ch 13**: unlevered K_e via asset beta.
Replacement cycle / EAC	**FM**: EAC from Intermediate; equal-life comparison logic as in lease vs buy.

#13 · 3.4% of marks

Ch 4 — Security Analysis

Theory-heavy. SM TYK theory Qs: **EMH and its misconception · Dow theory · Elliot wave theory · indicators of economic performance.** Numericals: moving averages / EMA, RSI (ICAI paper), dividend-growth value.

① Fundamental analysis (SM 1)

- → *Share price depends on **expected future dividends**; PV of dividends = **intrinsic value**. Price below intrinsic → **buy**; above → **sell***
- → *Constant dividend: $P_0 = D \div k$. Constant growth ($g < k$): **$P_0 = D_1 \div (k - g) = D_0(1 + g) \div (k - g)$** — **Gordon's dividend growth model** (Myron J. Gordon); invalid if $g \geq k$ (infinite price) — not restrictive, long-term g unlikely to exceed k. $g = 0$ → back to $D \div k$*
- → ***Link to P/E (SM 1.1)**: with payout fraction **b**, $D_1 = bE_1$ → **$P_0 = bE_1 \div (k - g)$**; so $P/E = b \div (k - g)$. Price / P/E above fundamental → overpriced → sell; below → buy*
- → *Three concentric circles: **economy** → **industry** → **company** analysis*

> ⚠ **TRAP**
> ***Symbol trap**: in SM Ch 4, **b = payout** ratio. In SM Ch 5 (Gordon earnings model) and Ch 1 (SGR), **b = retention**. Read the definition given in the question.*

② Economic analysis (SM 1.2)

Past / present / expected performance of the economy and sectors; income & consumption trends drive profits, dividends, prices.

Factors (4)

*(a) growth of **national income** — nominal GDP vs **real** growth (b) growth of **industrial sector** (by demand) (c) **inflation** — WPI / CPI; consumer industries fine-tune pricing (d) **monsoon** — forward & backward linkages*

Techniques (3) — TYK Q4

*(a) **Anticipatory surveys** — expert opinion on construction, capex, inventory, consumer spending; drawbacks: intentions may not materialise; not forecasts per se (b) **Barometer / indicator approach** (c) **Economic model building** (GNP model / sectoral analysis)*

- → ***Indicators**: **leading** (turn before the economy) · **roughly coincidental** (turn with it) · **lagging** (turn after it). Show direction, not magnitude; may conflict → **diffusion / composite index** (hard to compute; doesn't remove irregular movements). **Money supply** ↑ → inflation; stocks a hedge → prices rise*
- → ***GNP model steps**: (i) hypothesise total demand (GNP) from political stability, inflation, economic levels (ii) forecast components — consumption, gross private domestic investment, government purchases, net exports (iii) add them (iv) compare with an independent agency's forecast and test consistency*

THEORY

③ Industry analysis (SM 1.3)

Industries don't grow at the same rate → assess demand, cost structure, government constraints.

Factor (7)	SM
(a) Product life cycle	High profit in initial & growth stages, medium but steady in maturity, sharp decline at the end
(b) Demand-supply gap	Excess supply cuts profit (lower price); shortage raises it
(c) Barriers to entry	Innate to product / technology, or created by existing firms
(d) Government attitude	Crucial determinant of prospects
(e) State of competition	Leaders, homogeneous vs differentiated products, demand by customer / area, growth / cyclical / defensive / decline industry
(f) Cost conditions & profitability	Cost allocation & controllability, price, capacity (installed / idle / operating), capex needed; ratios — GP, OPM, ROE, return on capital
(g) Technology & research	Nature of technology, expected changes, capex-to-sales over time, R&D spend, sales & profit in short / immediate / long run

Techniques: (a) **regression & correlation** on demand drivers (GNP, disposable income, per-capita consumption, price elasticity) (b) **input-output analysis** — flow of goods through the economy to spot growing / declining industries.

④ Company analysis (SM 1.4) — 13 points

- **(a) Net worth & book value** — equity + preference + free reserves − intangibles − losses; BV may not reflect worth: price reflects **future earnings**; BV on **historical cost**
- **(b) Sources & uses (funds flow)** — short-term funds financing long-term assets → liquidity crunch
- **(c) Cross-sectional & time series** — common-size statements, ratio analysis
- **(d) Size & ranking** — capital employed, profit, ROI, sales vs peers; know-how, R&D, price leadership
- **(e) Growth record — P/E**, % growth in EPS, % growth in **net block** (capacity). SM: A ₹150 / EPS 25 = P/E 6 vs B ₹250 / 25 = 10 → A preferable if all else equal. "**Growth is the single most important factor**"
- **(f) Financial analysis** — key: EPS, P/E, yield, book value, intrinsic value (10 years, vs industry); solvency (current, liquid, D/E)
- **(g) Competitive advantage** (Apple's brand, Samsung in mobiles) · **(h) Quality of management** — experience, skills, integrity; earnings calls, MD&A, insider selling, 5-year record · **(i) Corporate governance** — LODR 2015 compliance, stakeholder policies, disclosure quality & timeliness, independent directors · **(j) Regulation** — Companies Act, SCRA, SEBI Act; FEMA / RBI · **(k) Location & labour relations** · **(l) Pattern of shareholding** (PNB nationalisation — institutions raised compensation) · **(m) Marketability** — public holding, speculative interest, exchange, volume

Techniques: correlation & regression (simple / multiple) · trend analysis · decision tree (multiply probabilities along each sequence, then add). Also **group analysis** — SEBI disclosure of group companies; RBI group exposure norms.

TYK THEORY Q2 & Q3

5 Technical analysis (SM 2.1–2.3)

- → Studies price charts assuming trends **repeat** (investor psychology follows patterns); studies investor and price behaviour. Asks: (i) is there a trend? (ii) will it reverse? Methods: **visual** (charts) and **statistical**
- → **Assumptions (4)**: (i) value depends on **supply & demand** (ii) S & D governed by rational and irrational factors (iii) prices move in **trends** lasting a substantial time (iv) relies on charts, not financial statements
- → **Principles (3)**: (a) **market discounts everything** (b) **price moves in trends** (c) **history repeats itself**

6 Dow theory (SM 2.4.1) — TYK Q2

Charles Dow (founder of Dow Jones), late 19th century; measures relative strength of the market; a business barometer. Based on **DJIA** and **DJTA**. Three simultaneous movements: **primary** (main bull / bear trend, 1 year to 36 months or longer) · **secondary** (opposite direction, 2 weeks to a month or more) · **daily** fluctuations (not interpreted, but studied). Tells **where the market is going, not how far**. Successively higher highs & higher lows → bull; lower highs & lows → bear.

Three moves up: (1) accumulation by far-sighted investors (2) first reports of good earnings (3) widespread reports of well-being + rampant speculation. Then far-sighted investors sell → first move down; non-sustainability confirmed → second; **distress selling** → third.

7 Elliot wave theory (SM 2.4.2) — TYK Q3

Ralph Elliot, 1934, inspired by Dow theory and nature; 75 years of price data. Wave = price movement from one change in direction to the next in the same direction, from buying / selling impulses. **Impulsive (basic) waves**: 3 or 5 waves in the main direction (bull or bear). **Corrective (reaction) waves**: 3 waves against it. Waves 1, 3, 5 directional; 2 & 4 corrective. **Complete cycle = 5-wave impulse + 3-wave correction (a, b, c) = 8 waves**, bullish and bearish phases; then a fresh cycle.

8 Random walk theory (SM 2.4.3)

Prices are statistical ups and downs, **unpredictable**; no link between present and future prices; successive changes independent. **M. G. Kendall** (SM: "Kendell") — like a roulette wheel; "drunk walking in a blind lane". (a) prices can't be predicted (b) trends are a statistical expression of past data (c) no connection between successive peaks and troughs.

9 Charts (SM 2.5)

Line (closing prices joined) · **Bar** (vertical high-low bar; ticks for open & close) · **Japanese candlestick** (white = close > open = bullish; black = close < open = bearish; **Doji** = open ≈ close) · **Point & figure** (detects reversals; choose **box size** — smaller = more sensitive — and **reversal criterion**; X rising, O falling).

THEORY + NUMERICAL

10 Market indicators (SM 2.6)

- → **Breadth index** = net advances or declines ÷ issues traded; supports Dow averages = technical strength; contradicts = weakness (reversal)
- → **Volume**: rising price + rising volume → buy (unsatisfied demand); falling + rising volume → bear. SM also: rising market with decreasing volume indicates a bull market; falling with dwindling volume a bear market. Use with Dow theory
- → **Confidence index** = high-grade bond yields ÷ low-grade bond yields; rising precedes a rising market; usually (not always) a **leading** indicator
- → **Relative strength**: some securities rise faster in bull / fall slower in bear and keep that strength — measured by historical returns or ratio to industry / market
- → **Odd-lot theory**: **contrary-opinion** — average person usually wrong; predicts tops in bull markets and reversals

11 Support, resistance & patterns (SM 2.7–2.8)

Peak after a fall = **resistance** (selling pressure); trough after a rebound = **support** (buying pressure); breach = new levels. SM eg: A ₹100 / ₹125; B ₹140 / ₹160.

Pattern	Signal
Channel	Uniform tops & bottoms; up-sloping = rising, down-sloping = falling
Wedge	Tops and bottoms move in opposite directions, or same direction at different rates
Head & shoulders (top)	Most important **reversal** pattern; **bearish** — below neckline → fall ≈ head-to-neckline distance → **sell**
Inverse head & shoulders	**Bullish** — above neckline → **buy**
Triangle / coil	Uncertainty; breakout direction hard to predict
Flags & pennants	Previous trend likely to **continue**
Double top / double bottom	Bearish / bullish
Gap	Open vs previous close; wider gap = stronger continuation signal

12 Moving averages (SM 2.9)

$AMA_{n,t}$ = $(P_t + \dots + P_{t-(n-1)}) \div n$. Trend: **200-day** (or 30-week) = long term · **60-day** = intermediate · **10-day** = short term

$EMA_t = a \times P_t + (1 - a) \times EMA_{t-1}$, $a = 2 \div (n + 1)$ — latest price weighted most

Buy signal	Sell signal
(a) price rises through a flattening MA (b) price falls below a **rising** MA (c) price above MA falls but turns up before reaching it	(a) price falls through a flattening MA (b) price rises above a **falling** MA (c) price below MA rises but turns down before reaching it

ICAI paper (RSI): RSI = 100 − 100 ÷ (1 + RS), RS = avg gain ÷ avg loss; update averages as (previous × 13 + today) ÷ 14. >70 overbought · 50–70 bullish · 30–50 bearish · <30 oversold (ICAI answer: 34.64 → bearish).

TYK THEORY Q1

13 Evaluation of technical analysis (SM 2.10) & FA vs TA (SM 3)

For (3)

(a) crowd psychology makes trends persist; spotted early (b) demand-supply shifts are gradual (c) information is assimilated over time, so price keeps moving one way

Against (4)

(a) no convincing explanation of tools (b) evidence for random walk (c) by the time a trend is signalled it has happened (d) **self-defeating** — more users, less value

Efficient market → TA may not work; real markets are imperfect → useful **with** fundamental analysis. **FA vs TA**: method — macro (GDP, inflation, rates) + micro (sales, profitability, solvency) vs historical market data (price, volume, open interest) · rule — "prices discount everything" vs "price captures everything" · use — **long-term** vs **short-term** investing.

14 Efficient market theory (SM 4) — TYK Q1

- **Eugene Fama** (Chicago), 1960s: all price-sensitive information is fully reflected in prices → no one can consistently beat the market; higher return only by taking **more risk**
- **Why random**: information free & instant to all · keen competition makes prices reflect intrinsic values · prices change only on **new, unpredictable** information
- **Misconceptions (SM 4.2)**: (1) EMH doesn't mean prices always equal fair value — the future is uncertain, surprises keep prices fluctuating (2) managers' failure to outperform doesn't mean incompetence — markets are efficient **because** managers compete well (3) random movement ≠ irrational market — **rational, competitive investors make price changes random**
- **Levels**: **weak** — past prices & volumes · **semi-strong** — + all public information · **strong** — all information, public and private
- **Weak-form tests**: **serial correlation** (none found, short-term) · **run test** (runs vs a random series — supports random walk) · **filter rule** (buy after N% rise, sell after N% fall — doesn't beat buy-and-hold after commissions)
- **Semi-strong**: Fama, Fisher, Jensen & Roll — 940 NYSE splits 1957–59; Ball & Brown (SM: "Boll") — earnings announcements. **Anomalies**: slow adjustment to quarterly earnings surprises · small firms beat large · low P/E beats high P/E · **Monday** returns lower
- **Strong**: insiders and exchange specialists earn superior returns; **mutual fund managers do not** (Burton Malkiel, *A Random Walk Down Wall Street*)
- **Challenges (4)**: (a) information inadequacy / misinformation (b) limited processing capacity (Herbert Simon; David Dreman) (c) irrational behaviour (Keynes; **L. C. Gupta** — "blind man firing a gun") (d) monopolistic influence of big institutions

Equity research (SM 5): buy / sell / hold calls, also used in M&A swap ratios. Tools: Bloomberg Terminal (1981), Benzinga Pro, Refinitiv Eikon, MarketXLS, Stockopedia, Koyfin, Finbox.io, GuruFocus, Business Quant, YCharts.

15 Correlation — how Ch 4 actually connects

Ch 4 idea	The actual link (app-written cross-reference; the Ch 4 side is SM)
Intrinsic value, Gordon	**Ch 5**: same DDM; P/E = payout ÷ (k − g). **Ch 14**: equity research feeds the swap ratio.
EMH levels	**Ch 6 / Ch 8**: efficient markets favour **passive** (index / ETF) strategies; MF managers not beating the market = strong-form evidence; Jensen's α measures any "beating".
Leading indicators, money supply, inflation	**Ch 2**: interest-rate & currency risk identifiers; **Ch 3**: government-policy impact on capital budgeting.
Odd-lot, confidence index	**Ch 5**: confidence index uses bond yields — high-grade vs low-grade = credit spread.

#14 · 3.4% of marks

Ch 7 — Securitization

Pure theory — every list matters. SM TYK theory Qs: **PTS vs PTC** (para 7) · **problems in securitisation, especially in India** (para 6).

① Why & what (SM 1–2)

→ *Firms with large receivables still face a liquidity crunch; borrowing changes D/E and adds financial risk → **securitisation***

→ *Creation of a pool of assets from illiquid financial assets (receivables, loans) that are marketable — **repackaging / rebundling illiquid assets into marketable securities** (auto loans, credit-card receivables, residential mortgages, future receivables)*

→ ***RBI**: transactions where **credit risks in assets are redistributed** by repackaging them into tradable securities*

Features (6)

*(i) **Creation of financial instruments** backed by collateral (ii) **Bundling** (pooling) & **unbundling** (fixed-denomination instruments) (iii) **Risk-management tool** — on non-recourse basis, default risk shifts from originator to investor (iv) **Structured finance** — tailored to investors' risk-return profile (v) **Tranching** — pool split by risk & return (vi) **Homogeneity** — securities in a tranche are homogeneous, small denominations for small investors*

② Benefits (SM 3)

Originator (4)

*(i) **Off-balance-sheet financing** → frees capital, better liquidity, expansion (ii) **specialisation** in core business — servicing moves to SPV; non-recourse shifts default burden (iii) better **financial ratios** — banks / FIs manage capital-to-risk-weighted-asset ratio (iv) **lower borrowing cost** — rated papers (credit enhancement) issued at lower rates → originator earns a spread*

Investor (3)

*(1) **Diversification** — backed by different asset types (2) **Regulatory requirement** — eg buying micro-industry-backed assets meets RBI **Priority Sector Lending** (3) **Protection against default** — recourse to originator; insurance*

THEORY

③ Participants (SM 4)

Primary	Role
Originator	*Initiator / securitizer; sells assets on its books, receives funds; transfers **legal and beneficial interest** to SPV*
SPV	*Created only to execute the deal (company, firm, society or trust); holds legal title; **ring-fences** assets from originator's balance sheet; pays originator upfront; issues **ABS / MBS***
Investors	*Individuals, MFs, PFs, insurers, FIs; receive interest and principal*

Secondary	Role
Obligors	*Owe money to the originator — "**main root**" of the process; their credit standing is paramount*
Rating agency	*Assesses: strength of cash flow · mechanism for timely interest & principal · credit quality of obligors · liquidity support · strength of legal framework*
Receiving & paying agent *(servicer)*	*Collects from obligors, passes to SPV, follows up defaulters / legal action; usually the originator or affiliate*
Agent / trustee	*Oversees that all parties perform; protects investors*
Credit enhancer	*Originator (over-collateralisation, cash collateral) or third party / bank (letter of credit, surety bond)*
Structurer	*Investment banker / arranger; brings parties together; ensures legal, regulatory, accounting, tax compliance*

④ Mechanism (SM 5) — 7 steps

1. ***Create pool*** — *segregate similar assets (for MBS: interest rate, risk, maturity, concentration)*
2. ***Transfer to SPV***
3. ***Sale of securitised papers*** — *SPV designs instruments by interest, risk, tenure, pool (pass-through / pay-through)*
4. ***Administration*** — *sub-contracted back to the originator, who collects and passes to SPV for a fee (servicer / conduit)*
5. ***Recourse to originator*** — *unless specified otherwise, defaulted receivables go back to the originator*
6. ***Repayment*** — *SPV pays investors interest & principal from the pool*
7. ***Credit rating*** *of instruments before sale*

5 Problems, especially in India (SM 6) — TYK Q2

Problem	SM
Stamp duty	Transfer of Property Act 1882 — on mortgage debt up to **12%** in some states. **PTCs exempt** (evidence only a receivable, not debt); some states reduced duty
Taxation	No specific IT Act provision — SPV taxed as representative trustee, or investors on their share? capital-gain implications unclear
Accounting	Off-balance-sheet in name, but receivables leave the books — problem esp. on transfer **without recourse**
Lack of standardisation	Each originator uses own documentation & administration
Inadequate debt market	No developed secondary market for ABS
Ineffective foreclosure laws	Lenders struggle to transfer property on default → MBS less attractive

6 Instruments (SM 7) — TYK Q1

Basis	Pass Through Certificate (PTC)	Pay Through Security (PTS)
Cash flows	**All** receipts (interest + principal) passed on **immediately**	SPV **debt** securities backed by the assets — servicing **desynchronised** from asset cash flows
Investor's claim	Direct, **proportional beneficial interest** in assets held in trust	Claim on SPV's debt securities
Prepayment	Distributed proportionately → **skewed** early cash flows	Surplus can be **reinvested short term** by SPV
Maturity	**Single** — all terminate together on final payment	**Several tranches** with varying maturities

Stripped securities: cash flows split into **Interest Only (IO)** and **Principal Only (PO)**; highly volatile → less preferred. Yields **rise** → borrowers postpone repayment of cheaper loans → **PO price falls**; IO value **rises** (more interest). Rates **fall** → borrowers prepay → IO falls. Prices driven mainly by investor perception.

7 Pricing (SM 8)

- → **Originator's angle**: price at a rate whose outflow can be amortised over time by investing the amount raised
- → **Investor's angle**: discount best estimate of expected cash flows at the **YTM of a comparable security** (credit quality, average life); or use the yield curve + **spread** for the credit quality

8 Risks (SM 9)

- → **Credit / counter-party** — prime risk; bankruptcy and non-performance of the **servicer**
- → **Legal** — recent concept in India; no conclusive precedent or explicit statute → disputes over ownership create pay-out uncertainty
- → **Market**: (a) **macro-economic** — eg fall in industrial production hits commercial-vehicle operators → CV loan repayments; fall in CV prices → wilful default and repossession (b) **prepayment** — rates fall → retail prepayments → **reinvestment risk** (c) **interest rate** — floating pool vs fixed pay-outs (or vice versa) → shortfall even at 100% collection; hedge with **interest rate swaps**

9 Blockchain & tokenization (SM 10)

- → **Blockchain / DLT**: shared, peer-to-peer, decentralised, open ledger with no trusted third party; **append-only** — entries permanent, transactions irreversible (changes recorded as new transactions); not controlled by any bank, corporation or government. Analogy: a **shared Google Doc**
- → Flow: transaction initiated → broadcast → network validates using **cryptography** → represented as a block → block added to chain → complete

Applications (5)

(a) **Financial services** — automated trade lifecycle / transaction log of assets (b) **Healthcare** — secure, private, interoperable data (c) **Government** — land & vehicle registration, e-voting (d) **Travel** — payments, passports / IDs, reservations, insurance, loyalty (e) **Economic forecasts** — decentralised prediction markets, voting, trading

Risks (4)

(i) members' different **risk appetites**; who manages risk / accountability with no one in charge (ii) reliability depends on technology — tampered **consensus mechanism** makes data unreliable (iii) no central authority → hard to maintain **process controls**, understand general IT controls (iv) **information overload**; hard to find competent monitors

Tokenization = converting tangible and intangible assets into **blockchain tokens** (real estate, artwork). **Similar to securitisation** in: (i) **liquidity** for illiquid assets (ii) **diversification** (iii) **tradable** → wealth (iv) **new fee opportunities** for institutions.

10 Securitisation in India (SM 11)

Pioneered by **Citi Bank** (auto loans); then sales-tax deferrals, aircraft receivables. **SARFAESI Act 2002** — tackle NPAs without court. Now key funding for MFIs and NBFCs; CMBS emerging; dominated by few players (ICICI Bank, HDFC Bank, NHB). CRISIL: volumes may return near pre-pandemic high of **₹1.9 trillion** (FY19 & FY20). SEBI allows **FPIs** into securitised debt of unlisted companies up to a limit.

LINKS

11 Correlation — how Ch 7 actually connects

Ch 7 idea	The actual link (app-written cross-reference; the Ch 7 side is SM)
Tranching	***Ch 9**: CDOs use the same senior / mezzanine / equity tranches — losses hit the lowest tranche first; CDS transfers the credit risk of the tranche.*
Pricing at YTM of comparable + spread	***Ch 5**: bond value = PV of expected cash flows at YTM; spread over the yield curve for credit quality.*
IO / PO price behaviour; prepayment risk	***Ch 5**: price-yield relationship and reinvestment risk; **Ch 12**: embedded option risk (prepayment) and reinvestment risk in banks' gap analysis.*
Floating pool vs fixed pay-outs	***Ch 12**: hedge the mismatch with an interest rate swap (pay floating received from the pool, receive fixed).*
Off-balance-sheet, non-recourse	***FR Ind AS 109**: derecognition only if substantially all risks & rewards are transferred — **with recourse**, the receivables usually stay on balance sheet (the SM's "accounting problem").*
Investor diversification	***Ch 6**: a pool of many obligors diversifies unsystematic (individual default) risk; systematic (macro) risk remains — the SM's "macro-economic risk".*
SARFAESI, foreclosure	***Ch 14**: SARFAESI also requires valuations in restructuring; **AAA**: NPA / bank audit context.*
Blockchain risks	***AAA**: general IT controls, consensus-mechanism reliability = audit of automated environments.*

12 Write-in-exam lines

> ✍ **Write in exam**
> *"Securitisation is the process of repackaging illiquid financial assets such as receivables and loans into marketable securities. The RBI defines it as transactions where credit risks in assets are redistributed by repackaging them into tradable securities."*

> ✍ **Write in exam**
> *"In a pass-through certificate all cash flows from the pool are passed to investors immediately and all certificates mature together; a pay-through security is a debt security of the SPV backed by the assets, which permits tranches of different maturities and short-term reinvestment of surplus cash."*

#15 · 2.8% of marks

Ch 13 — Business Valuation

SM TYK theory Qs: **EVA vs MVA · relative valuation is a "relative" value by comparative analysis — elaborate.** Numericals: NAV / NRV, P/E & capitalisation, DCF + TV, asset beta, relative (EBITDA multiple, EV), chop-shop, EVA, SVA, LBO, distressed firms, startup methods.

① Conceptual framework & terms (SM 1–2)

- → *First corporation valued / IPO: **Dutch East India Company** (United East India Company). Valuation = science + art; part of strategic FM*
- → ***Purposes (5)**: (a) information for internal stakeholders (b) comparison with similar enterprises — management efficiency (c) future **public listing** (d) strategic planning — value drivers, deploying surplus cash (e) **ball-park price** for acquisition*
- → ***PV** (₹1,000 in a year at 10% ≈ ₹909) · **IRR** (rate making NPV zero; higher IRR more likely selected) · **ROI**: (1,400 − 1,000) ÷ 1,000 = 40%; with additions use **average investment**: 1,000 + 200 during year, value 2,000 → 800 ÷ 1,100 = **72%**; historical ratio — course-correct only*
- → ***Gordon perpetual growth**: little used by investors — more parameters needed; dividends rarely grow steadily; "darling of academicians". **Terminal value** = "horizon value" at exit, usually by the Gordon model*

② Asset-based approach (SM 3.1)

*For M&A: unlisted → negotiated price; listed → market cap is a guide, but depends on market efficiency and may reflect only a small quoted portion. Three approaches: **asset, earnings, cash flow** (+ others). Asset approach values income-**generating assets**, not income; least useful for **IT companies** (IPR, people).*

Method	SM
***Net asset value** (book value)*	*Fixed assets + net current assets − long-term debt. Simple (historical cost) but balance sheet isn't a valuation device → **lower limit**; ignores intangibles (brand, IPR)*
***Net realisable value** (liquidation / adjusted book value)*	*Realisable value of assets − liquidation expenses − liabilities (expenses ignored if bought as going concern); buyer offers lowest prices; implies break-up; useful when acquirer will sell one part and integrate the rest. SM eg: NRV 1,05,000 ÷ 5,000 shares = **₹21***
Replacement value	*Cost of equivalent assets & liabilities in the open market; usually slightly above NRV; seen as the **maximum** an acquirer would pay; ignores staff loyalty*

*Conclusion: shows net worth fairly on going-concern basis but **ignores future revenue-generating ability** and market dynamics.*

FORMULA CORE

③ Income-based approach (SM 3.2)

For acquirers continuing the business; include extra earnings from the acquisition.

→ ***P/E (earning-yield multiplier)***: *price = EPS × P/E; for listed cos; captures expected growth & market expectations; for unlisted use peer / industry P/E. Gives the* ***minimum acceptable price*** *to target's holders. Steps: (i) P/E of equivalent quoted company (ii) adjust* ***down*** *for non-listing risk (iii) future* ***maintainable*** *EPS (iv) multiply*

→ ***Capitalisation of earnings***: *value = expected annual* ***maintainable profit*** *(weighted average of past profits, adjusted for synergy / scale) ÷ capitalisation rate (required earning yield = EPS ÷ price = 1 ÷ P/E). Forward-looking; but future profit hard to estimate, extraordinary items treated differently*

④ Cash-flow (DCF) approach (SM 3.3)

Free cash discounted to match risk; avoids over-reliance on history. ***5 steps***: *(a) free cash flows (b) forecast (c) discount rate from cost of capital (d) terminal value (e) PV of FCF + TV, interpret.*

SM eg: *EAT 600 − one-time income 200 + one-time expense 100 + depreciation 100 = FCF 600; grow 5%, less ΔWC & capex → 500 / 550 / 651.50; at 8% PV 463 / 471.35 / 517.29;* ***TV = 517.29 × 1.03 ÷ (0.08 − 0.03) = 10,656.17****; total* ***12,107.81****. Note: SM grows the year-3* ***PV****, so this TV figure is already a present value. TV is often the major part of value → validate g and WACC.*

⑤ Cost of equity (SM 4)

→ ***CAPM***: $R = r_f + \beta(r_m - r_f)$ — *popular but* ***over-simplifies*** *risk into two buckets*

→ ***APT*** *(Stephen Ross, 1970s): multiple macro factors, each with its own β:* $\mathbf{R_f + \beta_1 RP_1 + \beta_2 RP_2 + \ldots + \beta_n RP_n}$*; APT doesn't name the factors — analysts choose them*

⑥ Asset beta & unlisted firms (SM 4.3)

Private firms (often little debt) and a listed firm entering a ***new line of business*** *shouldn't use their own WACC → use* ***asset (ungeared) beta*** *= business risk only.*

$\beta_a = \beta_e \times E \div [E + D(1-t)] + \beta_d \times D(1-t) \div [E + D(1-t)]$*; with* $\beta_d = 0$: $\beta_a = \beta_e \times E \div [E + D(1-t)]$*; regear:* $\beta_e = \beta_a \times [E + D(1-t)] \div E$*.* $\beta_e \geq \beta_a$ *(equal if debt-free)*

Steps: *(1) find* ***pure-play*** *(proxy) firms and their* β_e *(2) de-gear each (3) one proxy → use it; several →* ***average*** *(or most appropriate) (4) re-gear at* ***own*** *capital structure (5) CAPM → required return / value*

Illus 1 *(X Pvt Ltd):* β_a *1.8, D:E 40:60, t 30% →* $\beta_e = 1.8[1 + 0.7 \times 40/60] =$ ***2.64***; $K_e = 5 + 2.64 \times 6 =$ ***20.84%***; *WACC = 20.84 × 0.6 + 12 × 0.7 × 0.4 =* ***15.864%***; *FCF 100/120/150 → value* ***272.15****. EBITDA 90 − extraordinary gain 10 − pending write-off 20 = 60 × 5 = EV* ***300****.*

⑦ Relative valuation (SM 5) — TYK Q2

- → Asset / earnings / cash-flow approaches give **intrinsic** value; relative valuation gives a **"relative" value by comparison with peers** — "valuation by multiples"; increasingly used to **validate** intrinsic value; aims at a value near market price (eg bond YTM close to the government index is more credible)
- → **4 steps**: (1) find the **drivers** for the multiple (2) compute via financial ratios (3) find **comparable firms** & compare (4) **iterate** to smooth deviations
- → **Step 1**: **EV multiples** — EV/EBITDA (popular), **EV/invested capital** (capital-intensive), **EV/sales** (cash-rich, big order book, organic growth) · **equity multiples** — P/E (celebrated), **PEG** = P/E ÷ growth (high-growth / sunrise industries)
- → **Step 2**: right ratio — exporter's value swings with FX (P/E misses Brexit-type shocks); EV/invested capital misfits asset-light firms or those with investment property
- → **Step 3** (hardest): no two firms alike — X (FTE pricing) vs Y (unit transfer pricing) differ in risk; niche firms → look across industries or **regression**; "take benchmarks with a pinch of salt"
- → **Assumptions (3)**: (i) market is **efficient** (ii) fundamentals–multiples link is **linear** (iii) comparables similar in structure, risk & growth. EV via entity value adjusted for debt, or balance-sheet approach

SM illustration	Answer
Illus 2: GP 10 L − indirect 4 L = EBITDA 6 L; 1 L shares; debt 3 L; surplus funds 5 L; r_f 4.5, r_m 12, β 0.9	(a) rate 4.5 + 0.9 × 7.5 = 11.25% → 600 ÷ 0.1125 = 5,333.33 thousand → **₹53.33** per share (b) 600 × 5 = 3,000 − 300 + 500 = equity **₹32 lakh**
Illus 3: H K Ltd 70,000 sh at ₹12; LT debt 2 L; cash 5 L	8.40 L + 2 L − 5 L = EV **₹5,40,000**

⑧ Contemporary approaches (SM 6.1)

- → Industry-specific metrics: **price per page visited** (internet), **price per subscriber** (online store) — still check cash-to-sales and DCF
- → **Goodwill-based**: asset value + goodwill as a multiple of sales or **footfall** (retail giant buying a mom-and-pop store)
- → **PER**: EPS 40, average price 50 → 1.25 (relative figure; use sector median)
- → **LBO by PE firms**: buy with **~70 : 30 debt : equity**, sell in 3–5 years. SM eg: EBITDA 100 → … Y3 EBDAT 112 × **exit multiple 7** = 784 − debt 100 = equity **684** (SM shows Y3 EBITDA 178; 20% growth on 144 would give 172.8). Risk: cyclical slowdown leaves heavy debt

9 Chop-shop / break-up (SM 6.2)

Find **undervalued multi-industry** firms worth more split up — buy assets below replacement value. **Steps**: (1) identify segments; average capitalisation ratios of pure-play firms (2) theoretical value per ratio (3) **average** the theoretical values. **Illus 4** (Cornett, market €4 m): cap/sales €4,005,000 · cap/assets €2,880,000 · cap/operating income €5,650,000 → average **€4,178,000** > €4 m.

10 Economic Value Added (SM 6.3)

- → Holistic: valuation and economic contribution; value is created only if returns **exceed the cost of capital**; separates operations from financing. Negative EVA = destroying value
- → **Why EVA**: others are number-crunching; EVA makes **management accountable** for decisions (eg rejecting a positive-IRR but slower-payback project shows up)

EVA = NOPAT − (Invested capital × WACC) = NOPAT − capital charge. NOPAT = EBIT − tax (depreciation stays as operating expense) + **non-cash** items (eg bad-debt provision). Invested capital = total assets − **non-interest-bearing current liabilities** (creditors, advance billing) + non-cash adjustment = equity + long-term debt, generally at **start** of year.

Illus 5: EBIT 410 − tax 123 + provision 20 = NOPAT **307**; IC = 1,300 − 400 + 20 = **920**; WACC = 800/900 × 8.45% + 100/900 × 12% × 0.7 = **8.44%**; charge 77.65 → EVA **229.35**.

11 Market Value Added (SM 6.4)

MVA = current market value of the firm − invested capital (eg 1,000 − 920 = 80). Addresses EVA issues (ignoring value drivers, book value); no basis for share value but gauges efficiency from **market-cap** view — value perceived by the market. EVA = **derived** value for discerning investors. High-MVA firms become market darlings / pricey; EVA and MVA may be briefly negatively correlated (eg ex-dividend) — gap closes.

TYK Q1	EVA	MVA
Measures	NOPAT − WACC × invested capital — internal, period performance	Market value − invested capital — market's view of future value creation
Based on	Accounting data adjusted (NOPAT, non-cash items)	Market capitalisation
Use	Management accountability; returns above hurdle rate	Perceived value added; not a basis for share value

12 Shareholder Value Analysis (SM 6.5)

NOPAT is historical and misses future investment opportunities → SVA uses **value drivers** (sales potential, investments, cost of incremental capital). **Steps**: (a) FCF from value drivers (b) discount at WACC (c) add TV (d) add market value of non-core assets & marketable investments (e) less debt = equity. SM: PVs 57.54 + 43.46 + 44.86 + TV (40.45 × 1/0.12 = 336.95) = 482.81 + investment property 35 − debt 19 = **498.81**.

13 Fair value (SM 7)

Buy at fair price — "no less no more". CA: arm's length; analyst: PV in cash; speculator: arbitrage. Booms defy fair value (dot-com). No single correct method → give a **range** (seller's minimum, buyer's maximum); deal by negotiation. **Exam approach**: (i) value by as many methods as data allows (ii) comment on each (iii) add any further information to the conclusion.

14 Going vs non-going concern (SM 8)

Going concern assumed (else different basis, disclosed). **Non-going concern = liquidation value** (assets sold, liabilities paid). Going-concern value = **total value** — usually higher: includes future profits, intangibles, goodwill (premium); liquidation also means lay-offs and reputation damage. Use non-going concern only when the firm has no value as a going concern.

15 Distressed companies (SM 9)

Distress = unable / struggling to pay creditors (high fixed costs, illiquid assets, cyclical revenue); can spread to operations; still may have value. **Why conventional methods fail**: TV assumes infinite life and growing CF (negative CF here) · negative, declining revenue — CF hard to estimate, DCF values a going concern, may give negative equity · discount rates assume sound firms → adjust for failure probability.

Method	SM
Modified DCF	Probability distribution of CF across outcomes, adjusted for default; K_e from updated D/E + unlevered β; K_d from updated default risk. Short-cut: **Expected CF_t = CF_t × (1 − probability of distress$_t$)**
DCF + distress value	(i) value as going concern on the path back to health (ii) probability of distress (iii) distress sale value (% of book or DCF value). **Equity = DCF equity × (1 − p) + distress sale value × p**
APV	**Firm value = unlevered value + (tax benefits of debt − expected bankruptcy cost)**; unlevered = FCFF at unlevered K_e; **expected bankruptcy cost = (unlevered value − distress sale value) × p**
Relative	Revenue & EBITDA multiples (P/E, P/BV often unusable); adjust down subjectively — telecom avg 2× revenue → **1.25×** for a distressed firm

16 Startups (SM 10)

Standard approaches (income, asset, market) presuppose an established, cash-generating business; startups have **no history**, value rests on untested growth and founders ("walking in the dark … night-vision goggles"). **Why traditional methods fail**: income — no / minimal positive CF · asset — mostly IP / intangibles, and going concern so not just realisable value · market — **disruptors** without established competitors (some market elements used in later rounds).

Value drivers: **product** (ready / prototype vs idea; market testing) · **management** (credentials, balance — over half of Indian unicorns have IIT / IIM founders) · **traction** · **revenue** streams · **industry attractiveness** (tourism amid lockdowns) · **demand-supply** of investors · **competitiveness** (first-mover; Ola after Uber).

Method	SM
Berkus (Dave Berkus)	5 success factors — basic value · technology · execution · strategic relationships · production & sales; caps **$2 m pre-revenue**, **$2.5 m post-revenue**; simple, ignores market factors
Cost-to-duplicate	All costs to build the startup / product incl. physical assets; criticised — ignores future revenue
Comparable transactions	"How much were similar startups valued at?" — XYZ ₹560 cr ÷ 24 cr users ≈ ₹23 / user × ABC's 1.75 cr ≈ **₹40 cr**; adjust for differentiators
Scorecard	Pre-revenue: average pre-money of peers × Σ(weight × score). Weights: team 0–30% · opportunity 0–25% · product 0–15% · competition 0–10% · marketing / channels 0–10% · need for more investment 0–5% · others 0–5%. Eg marketing 10% × 150% = 0.15
First Chicago	DCF + market: worst / normal / best case values × probabilities → weighted value
Venture capital	Investor's target multiple (10×, 20×, 30×) or IRR; discount exit value to get **post-money** value

17 Digital platforms (SM 11)

- → Software infrastructure for many-to-many interactions: **marketplace** (Booking.com, Uber, Amazon) · **search** (Google, Bing, Baidu) · **repository** (Spotify, YouTube, GitHub) · **communication** (WhatsApp, Teams, Telegram, Slack) · **community** (Facebook, LinkedIn) · **payments** (Paytm, GPay)
- → **Income**: **top-down** — TAM → SAM → SOM, time to reach share (ambitious for nascent firms) · **bottom-up** — from limited resources. Cash burn early. FCFF at WACC, FCFE at K_e; beta hard (few comparables) → add **company-specific risk premium (CSRP / alpha)**, higher for nascent platforms
- → **Market**: comparables scarce, profits / EBITDA negative, capital-light (low BV) → compare on **revenue drivers** (bookings, users, merchants, views per user, search speed) · **Cost**: hours of coding — ignores revenue capacity

Professional / consultancy firms (SM 12): compare historical data with industry KPIs and competitors (audited statements, tax returns); factor risk in projected growth (TV); **normalise** income and cash flows (add back non-cash and firm-specific items).

18 Impact of ESG on valuation (SM 13)

- → **E**: climate change, water, waste, emissions, biodiversity · **S**: employee development, diversity & inclusion, community, health & safety, customer · **G**: board independence & diversity, anti-bribery, tax transparency, ethical conduct
- → Developments: ESG funds > $50 bn in 2020; ESG assets > $35 trillion · green bonds crossed **$1 trillion** (2020) · sustainability taxonomies (EU) · IFRS single global ESG standard · **SEBI** proposed ESG disclosure framework (Feb 2023)
- → High ESG → **cheaper debt**, access to green / social / sustainability-linked bonds; ESG moved from "good to have" to "must have"
- → **In valuation**: adjust the discount rate (practical, less explicit) or **cash flows** (more explicit): **E** — 2-degree scenario analysis, carbon points · **S** — cost of labour conditions, CSR, welfare · **G** — penalties, fines, taxes from poor governance

19 Case studies

Vodafone–Idea: market value, comparable multiples, NAV (31 Dec 2016); **no DCF** (no projections provided); share-exchange basis; Idea fell up to 14.57% — "**perceived value**" isn't quantifiable. **Facebook–WhatsApp**: $21.8 bn; 450 m users (70% active), +1 m a day; ~**$55 per user** incl. $4 bn retention payout.

20 Correlation — how Ch 13 actually connects

Ch 13 idea	The actual link (app-written cross-reference; the Ch 13 side is SM)
Asset beta, re-gearing	**Ch 11**: SM Nepal illustration uses the competitor's beta the same way; **Ch 3**: APV needs the unlevered K_e from here.
DCF, FCFF / FCFE, EV	**Ch 5**: the FCFF / FCFE formulas and EV types live there; TV = Gordon model.
Valuation range	**Ch 14**: seller's minimum and buyer's maximum set the swap-ratio range; chop-shop is the M&A "break-up" logic.
APT, CAPM	**Ch 6**: APT surprise formula, CAPM / SML — the K_e inputs.
EVA	**FR Ind AS 103 / 36**: value in use and goodwill impairment use the same discounted-cash-flow idea; EVA is the management-performance view.
Startup & VC methods	**Ch 15**: VC buy-back IRR of 18–25% = the VC method's target rate.
Distressed APV	**Ch 3 / Ch 11**: same APV, with expected bankruptcy cost subtracted.